



GE VERNOVA

**ADMS**

16.25.1

Substation Editor User Guide

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# 1. Introduction

The Substation Editor (SSE) is a stand-alone application module that is supplied as part of the standard ADMS product suite. It is a modeling tool used to produce both displays, and portions of the SCADA model and the ADMS network topology model (also referred to as the DNOM model).

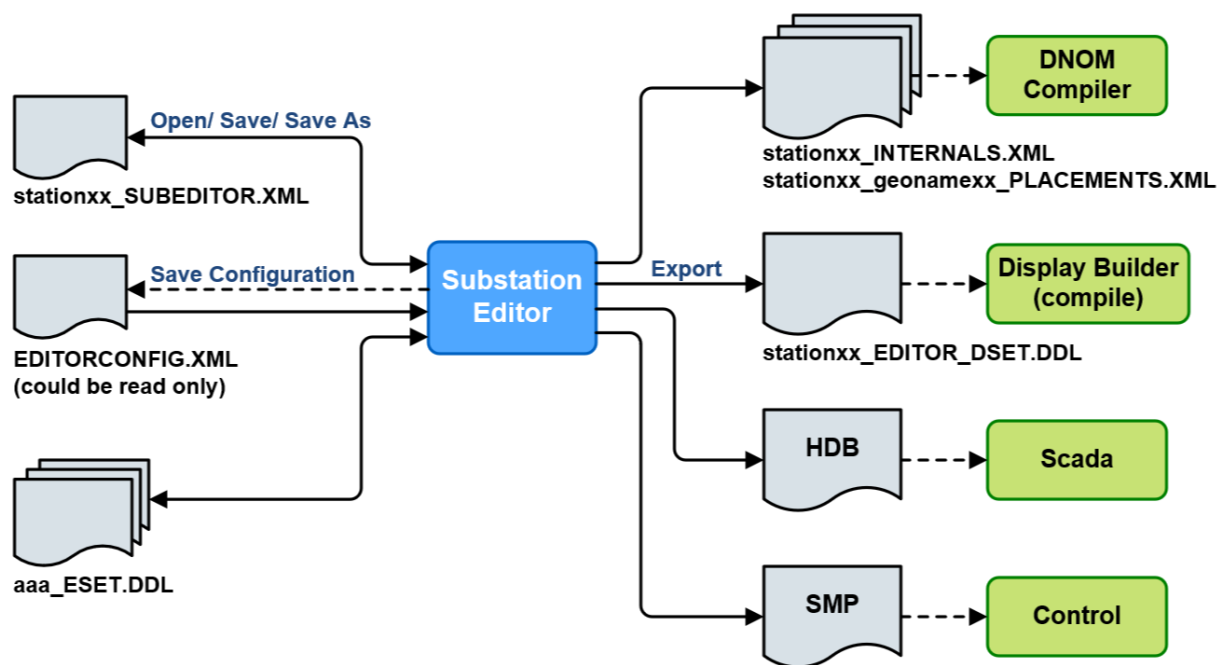


Figure 1. Substation Editor - Conceptual Overview

The production of network and SCADA objects is inseparable from the production of visual elements. This interdependent modeling is supported by the Substation Editor constructs described in [Modeling Constructs](#).

## 1.1. Overview of Substation Editor Input Files

**station\_SUBEDITOR.XML:** This file preserves the device, measurement, and graphic information used by Substation Editor to produce the INTERNALS and PLACEMENTS files described in [Overview of Substation Editor Output Files](#). In addition, it preserves the modeling support constructs and configuration values that vary between stations or even between views.

**substation\_editor\_configuration.XML:** This file preserves the modeling support constructs and configuration values that are common to all stations and views. For more details about configuring the Substation Editor, see [Using the Substation Editor](#).

**UserPreferences.xml:** This file is serialized to and deserialized from the %UserProfile% directory. It preserves the settings described in [Set User Preferences](#).

**station.xml** or **station\_SUB.csv:** This is the station externals file obtained from the asset management system and formatted as required by the Converter tool. For more information, refer

to the *ADMS Modeling Reference Guide*.

## 1.2. Overview of Substation Editor Output Files

As output, Substation Editor:

- Creates a file for the Model Manager that describes the newly exported station internals and station placements files (Station.issg).
- Creates network models for the internals of substations (station\_INTERNALS.XML).
- Creates schematic displays for the internals of substations (station\_PLACEMENTS\_view.XML).
- Creates schematic or geo-schematic views of feeders and equipment groups (station\_PLACEMENTS\_view.XML).
- Creates Scada database import files (HDB file, processed by Scada's hdbimport).
- Creates Scada oneline displays (station\_DSET.DDL processed by Browser's Display Builder).
- Creates Control database import files (SMP file, processed by the SCADA Server).
- Creates a file that contains the Associated Models defined using Substation Editor (global\_internals.xml).

Although both the station\_SUBEDITOR.XML file and the substation\_editor\_configuration.XML file are output by Substation Editor, they are serialized purely for persistence and are not input to any other application.

The HDB, SMP, and DSET.DDL files are fed into the Platform model building tools, which are described in the respective product's documentation.

The primary source of information for the ADMS model is the station externals files (named station.XML or station\_SUB.csv), which has almost always been exported from the asset management system (usually a GIS)<sup>[1]</sup>.

The station\_INTERNALS.XML and station\_PLACEMENTS\_view.XML files are input to CONVERTER, along with the station externals file and various supplementary configuration files, to produce the ADMS model file named station.mod. Only the following files and their contents are covered in this document:

- **station\_INTERNALS.XML:** This file defines the equipment "inside the fence" of the substation. It also defines as many of the SCADA measurements as the modeling user wants to represent on any of the station's views. Only one station\_INTERNALS.XML file is produced for each station.
- **station\_PLACEMENTS\_view.xml:** This file defines the graphic representation of the given view. One PLACEMENTS file is produced for each view modeled in the SUBEDITOR file.
- **GLOBAL\_INTERNALS.XML:** This file defines values for certain properties of some types of equipment. One GLOBAL\_INTERNALS.XML file is produced each time the Global Internals tab of the Substation Editor Configuration is changed. The contents of this file are discussed in the context of the "Global Internals" section of the *Substation Editor Installation Guide*.

[1] It is possible to construct the station externals file using Substation Editor, but that task is not within the scope of this document.

## 2. Modeling Constructs

Substation Editor supports several modeling constructs. Some of them are intended to make repetitive modeling tasks easier; others are useful for maintaining uniformity across views and across stations; and still others support graphic editing.

### 2.1. Placements and Groups

The terms "placements" and "groups" carry multiple meanings. Some of them are described in the *ADMS Modeling Overview and Converter User Guide*.

For the Substation Editor, a "placement" is the instantiation of a graphic element on a view. For example, when you move a picture from the Elements tab onto a view, you have created one or more placements (<EditorPointPlacement>, <EditorPolyPlacement>, etc.).

#### Note

When a picture that contains nested pictures (picture-in-picture) or symbols is instantiated, its individual elements are instantiated as individual placements. These individual placements are gathered into an <EditorGroup>, which is a special type of placement.

Substation Editor uses the term "group" in several slightly different, but related ways:

- In the same way as Microsoft Visio, when shapes are grouped as a single object. Substation Editor allows you to group placements (via the Group Items  toolbar button) for relocation, deletion, and duplication.
- In a way similar to the FG Display Builder concept of nested pictures. [Elements](#) describes modeling an equipment group by copying an ESET element from the Elements tab onto a view. That ESET element was created in FG Display Builder as a picture containing picture-in-picture elements (as well as symbols and other graphic elements). Each picture-in-picture, symbol, polyline, etc. within the definition of the Feeder Bay picture is instantiated as a separate placement, and Substation Editor groups () them as it creates them in the view.

#### Note

Information about which Feeder Bay picture (for example, FEEDER\_23KV\_W\_REGULATOR\_BYPASS) was used to create the group is lost once the group has been instantiated in the view. However, the names of the nested pictures are shown as the Element Names (for example, ALG\_LIM\_NS\_AAMP) on the Properties tab.

- As a way to create a Postage Stamp Display (for example, from external Capacitors or Topology Device Groups). As in the previous point, Substation Editor instantiates and groups the placements on the view from the elements contained in the circuit pattern picture.

#### Note

"Topology Device Group" is just a label for a tab in a Create Postage Stamp Displays pop-up display. It is not a special kind of "group" within the Substation Editor. It is not a Substation Editor modeling construct. It is, however, one of the CONVERTER DNOM objects (Topology Device Group = <UIContainer> and others to be supported in the future).

## 2.2. Associations

Substation Editor uses multiple mechanisms to associate a DNOM object, and default values for its properties, with a graphic element. For more information, refer to the "Associations" section of the *Substation Editor Installation Guide*.

For the full list of object associations that have been defined for the demonstration system network model, including information about how the parameters can be tuned, refer to [Network Object Associations References](#).

## 2.3. Templates and Proxies

The term "template" is used for several different constructs. Sometimes it is used generically to indicate a graphic element that is used to create a placement (for example, an ESET element that appears on the Elements tab, which is used as the template for instantiating a placement on a view). In other situations, "template" refers to one of the following modeling constructs.

**Global Template Names** may be configured on the Template Names tab of the Substation Editor Configuration pop-up, as described in the "Template Names" section in the *Substation Editor Installation Guide*.

**Object Template Names** (or "object templates") are created by being referenced within the value of a placement's properties. This creation can occur:

- On the Properties tab of a placement
- On the Associations tab of the Substation Editor Configuration pop-up for a given element
- From Display Builder in the Picture Placement or Symbol Placement properties, as described in [Modeling Equipment Groups](#).

**Group Template Names** behave just like Object Template Names except that their scope is the placement group: change the value for one placement in the group, and the new value is used for all placements in the same group.

**Postage Stamp Proxies** are a special set of Group Template Names. They and their values are defined by the Create Postage Stamp Displays function.

Each Global Template Name, Object Template Name, and Group Template Name is also known as a **proxy**.

### Note

Unless a proxy is actually used in setting the value of at least one property of at least one placement, it is not preserved in the SUBEDITOR file. In a later editing session, if you need to use one of the Postage Stamp Proxies, for example, you must create it and define it just as though it were a Group Template Name.

**View Templates** contain both graphic and DNOM definitions, and are intended to serve as a starting point for modeling a new substation. They may use proxies, but are not themselves proxies. For more information, see [View Templates](#).

## 2.4. Postage Stamp Displays

The Create Postage Stamp Displays function is used to produce SCADA displays and DNOM measurement objects for "outside the fence" equipment, which is modeled in the station externals file. In general, that function is described in [Postage Stamp Displays](#). This section describes some of constructs used by the Create Postage Stamp Display functionality.

### 2.4.1. Circuit Pattern

The standardized description of the electrical topology in a Postage Stamp Display is known as its Type (on the Create Postage Stamp Displays pop-up) or less generically as its Circuit Pattern. The Circuit Pattern is determined in two steps, which are described in more detail in [<UIContainer> Postage Stamp Display](#).

1. Starting with the To node of the primary device, the sequence of line segments and supported devices is traced node-to-node. The abbreviation of each equipment type is added to the unsorted pattern, resulting in something that might look like:

CLL or SLSLXLX

2. Without changing the node-to-node order of the letters, the pattern is sorted into its alphabetically lowest arrangement:

CLL remains CLL

SLSLXLX becomes LSLXLXS

However, the simplest way to determine the Circuit Pattern of a UI Container is to examine the SubstationEditorTrace log. For details, see [Substation Editor Configuration and ESETs](#).

### 2.4.2. Postage Stamp Proxies

Six proxies are generated for each device (including line segments) in the Circuit Pattern, formatted as:

|                       |             |            |
|-----------------------|-------------|------------|
| %/device/number/%     | %LINE3%     | 114639483  |
| %/device/number/NAME% | %LINE3NAME% | 1075-32998 |



|                          |                |             |
|--------------------------|----------------|-------------|
| %/device/number/STATION% | %LINE3STATION% | PEBBLEBEACH |
| %/device/number/FRNODE%  | %LINE3FRNODE%  | 84N32SE     |
| %/device/number/TONODE%  | %LINE3TONODE%  | 16SW499E    |
| %/device/number/PHASE%   | %LINE3PHASE%   | ABC         |

The numbers are assigned sequentially for each equipment type in the sorted Circuit Pattern starting from the left:

| CLL |            | LSLXLXS |                    |
|-----|------------|---------|--------------------|
| C   | CAPACITOR1 | L       | LINE1              |
| L   | LINE1      | S       | SWITCH1            |
| L   | LINE2      | L       | LINE2              |
|     |            | X       | (X=external) LINE3 |
|     |            | L       | LINE4              |
|     |            | X       | (X=external) LINE5 |
|     |            | S       | SWITCH2            |

For more details, see [How Proxies Can Be Used to Set Properties](#).

## 2.5. Measurement Points

In order to attach measurements (Analog and Statuses) to the correct node, Substation Editor expects the person modeling to create a Measurement Point at that node, and then to attach the measurements to the Measurement Point.

The attachment is applied either by hand after the new view has been created (using the Ctrl+A technique, described in [Creating SCADA Measurement Mapping to Measurement Point](#)), or automatically by designating the measurement point's ESET element as the Default Association for the measurement's ESET element. For example:

**Substation Editor Configuration**

☒ Associated Element:

☐ Add Association:

**Measurement Info**

☒ MEAS Type:

MEAS Group:

☒ Default Assoc:

Default CAM:

Create Meas: ☒

Associations | General | External Station Configurations | Topology | Linker | Global Internals | Template N.

**Note**

When you have multiple instances of the same MEAS Type that attach to different nodes, you must have a separate Element-to-Default-Association pairing. The following example is for a second Volts on Phase A measurement in the same view as the example above:

**Substation Editor Configuration**

☒ Associated Element:

☐ Add Association:

**Measurement Info**

☒ MEAS Type:

MEAS Group:

☒ Default Assoc:

Default CAM:

Create Meas: ☒

Associations | General | External Station Configurations | Topology | Linker | Global Internals | Template N.

## 3. Substation Editor User Interface Overview

When the Substation Editor starts, it opens the program with no model files or displays loaded. The layout of the editor is shown in the image below.

There are two main display areas, the Main Pane and the Explorer Pane. It is important to note that different items can be displayed on both of these panes by selecting the tabs at the bottom of the screen.

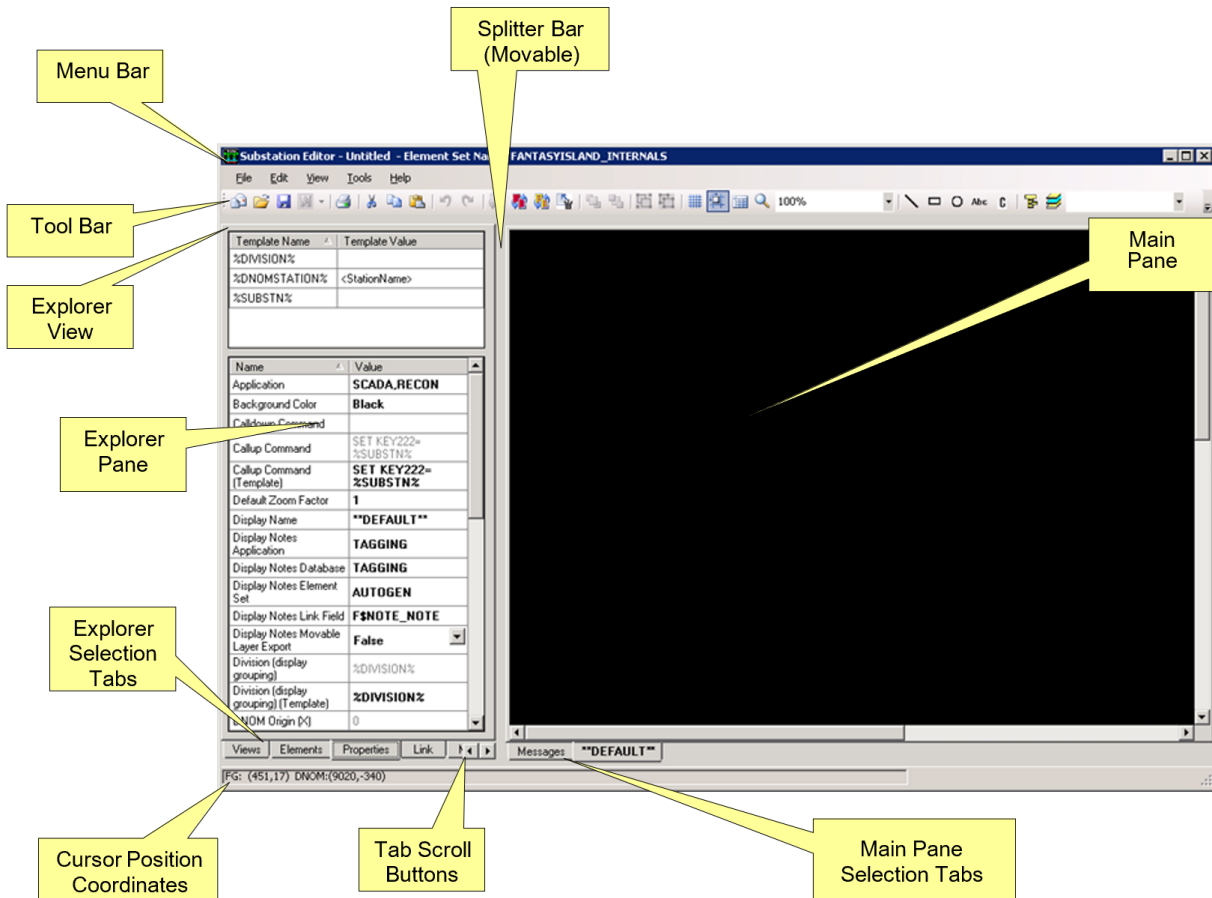


Figure 2. User Interface Layout

### 3.1. Menus

A set of pull-down menus (the Menu bar) along the top of the screen controls the basic operations of the Substation Editor, as shown in the following image.

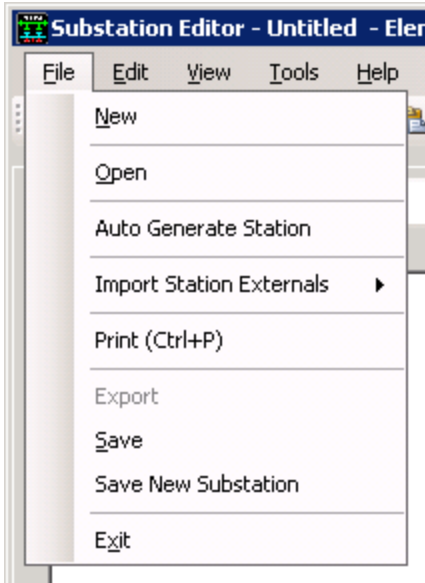


Figure 3. Menu Bar - Pull-Down Menu

The main entries on the Menu bar are:

- **File:** Controls the importing and saving of model files and element sets, and the importing and exporting of display files.
- **Edit:** Standard Undo/Redo and Cut/Copy Paste functions.
- **View:** Selects the required view from the currently loaded model.
- **Tools:** Opens the Substation Configuration Editor, and displays the properties of the currently loaded substation model.
- **Help:** Displays the online help pages.

For a detailed explanation of each of the pull-down menu entries, refer to [Menu Guide](#).

## 3.2. Toolbar

The toolbar provides quick access to the main functions for manipulating the display in the Main Pane. The toolbar is shown in the following three images. More-detailed descriptions of the functions of each button on the toolbar are given in [Toolbar Guide](#).

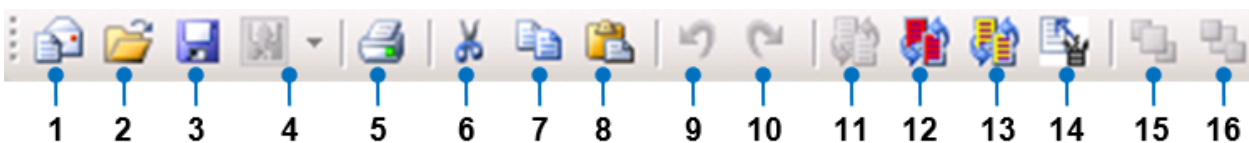


Figure 4. Substation Editor Toolbar - Left Section

| SSE Toolbar Left Section Key |                 |         |
|------------------------------|-----------------|---------|
| 1.                           | New             | 3. Save |
| 2.                           | Open a New File |         |

#### SSE Toolbar Left Section Key

|                              |                                           |                                    |
|------------------------------|-------------------------------------------|------------------------------------|
| 4. Export                    | 5. Print                                  | 6. Cut                             |
| 7. Copy                      | 8. Paste                                  | 9. Undo                            |
| 10. Redo                     | 11. Re-Apply Associations for Placement   | 12. Re-Apply Associations for View |
| 13. Re-apply Template Values | 14. Correct DNOM Station for Measurements | 15. Bring to Front                 |
| 16. Send to Back             |                                           |                                    |



Figure 5. Substation Editor Toolbar - Center Section

#### SSE Toolbar Center Section Key

|                       |                      |                 |
|-----------------------|----------------------|-----------------|
| 1. Group Items        | 2. Ungroup Items     | 3. Show Grid    |
| 4. Snap to Grid       | 5. Configure Grid    | 6. Zoom         |
| 7. Current Zoom Level | 8. Select Zoom Level | 9. Draw a Line  |
| 10. Draw a Rectangle  | 11. Draw a Circle    | 12. Insert Text |
| 13. Add Place Holder  | 14. Menu Properties  |                 |



Figure 6. Substation Editor Toolbar - Right Section

#### SSE Toolbar Right Section Key

|                                       |                                          |                            |
|---------------------------------------|------------------------------------------|----------------------------|
| 1. Layers                             | 2. Current Active Layer                  | 3. Select the Active Layer |
| 4. Show/Hide Network Layer            | 5. Align Left                            | 6. Align Vertically        |
| 7. Align Connection Points Vertically | 8. Align Right                           | 9. Align Top               |
| 10. Align Horizontally                | 11. Align Connection Points Horizontally | 12. Align Bottom           |
| 13. Show/Hide Laterals                | 14. Show/Hide Labels                     |                            |

The number of toolbar items available depends on the current size of the Substation Editor window. In Maximized mode, all of the toolbar items are available. However, if there is not enough space to display all of the toolbar items, some of them will be hidden and can be reached by clicking the button in the right corner of the toolbar.

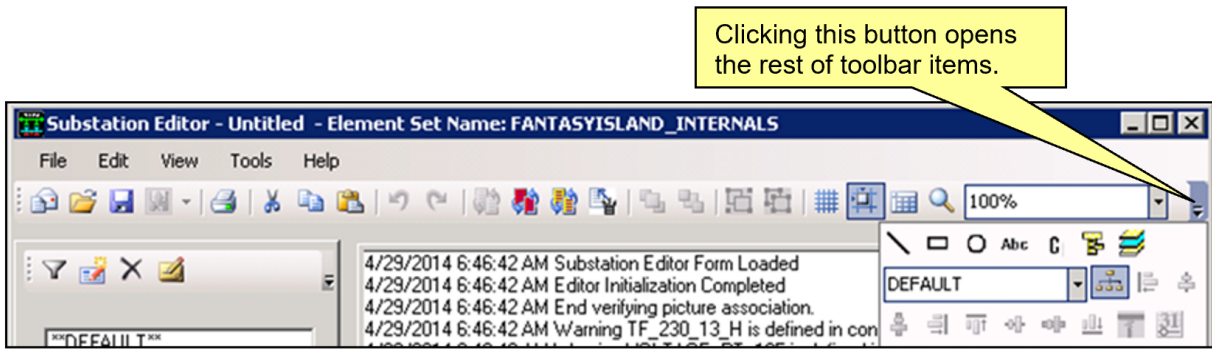


Figure 7. Substation Editor Hidden Toolbar Items

## 3.3. Substation Editor Tabs Overview

A number of different data views are provided through the Explorer Pane and Main Pane. These views are selected through the tabs at the bottom of the display, as shown in the following image.

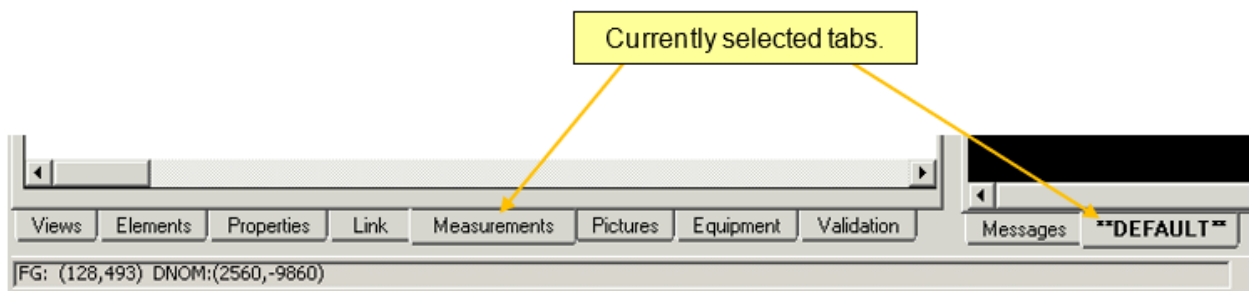


Figure 8. Substation Editor Tabs

### 3.3.1. Messages Tab

When the Substation Editor is first started, it displays a blank Default display in the Main Pane.

Optionally, you can select the Messages tab (refer to the image below) to view the Message log at any time during the editing session. The Messages tab contains a real-time log of actions that are performed during the editing session, such as opening/saving displays, changing the configuration of the Substation Editor, creating/deleting views, etc.

Actions performed with network objects and their properties are not written in the Message log. Undo/Redo actions are also not written in the Message log. Errors and other diagnostic messages will be displayed on the message log, which can be viewed at any time during the editing process by clicking the Messages tab. All messages in the Message log have a time stamp and a brief description of the action performed.

If problems are experienced while using the Substation Editor, it can be useful to check the Message log for information about the likely cause.

The Messages tab can be moved to the Explorer Pane (and back to the Main Pane) at any time, as

described in Arranging the Tab Views).

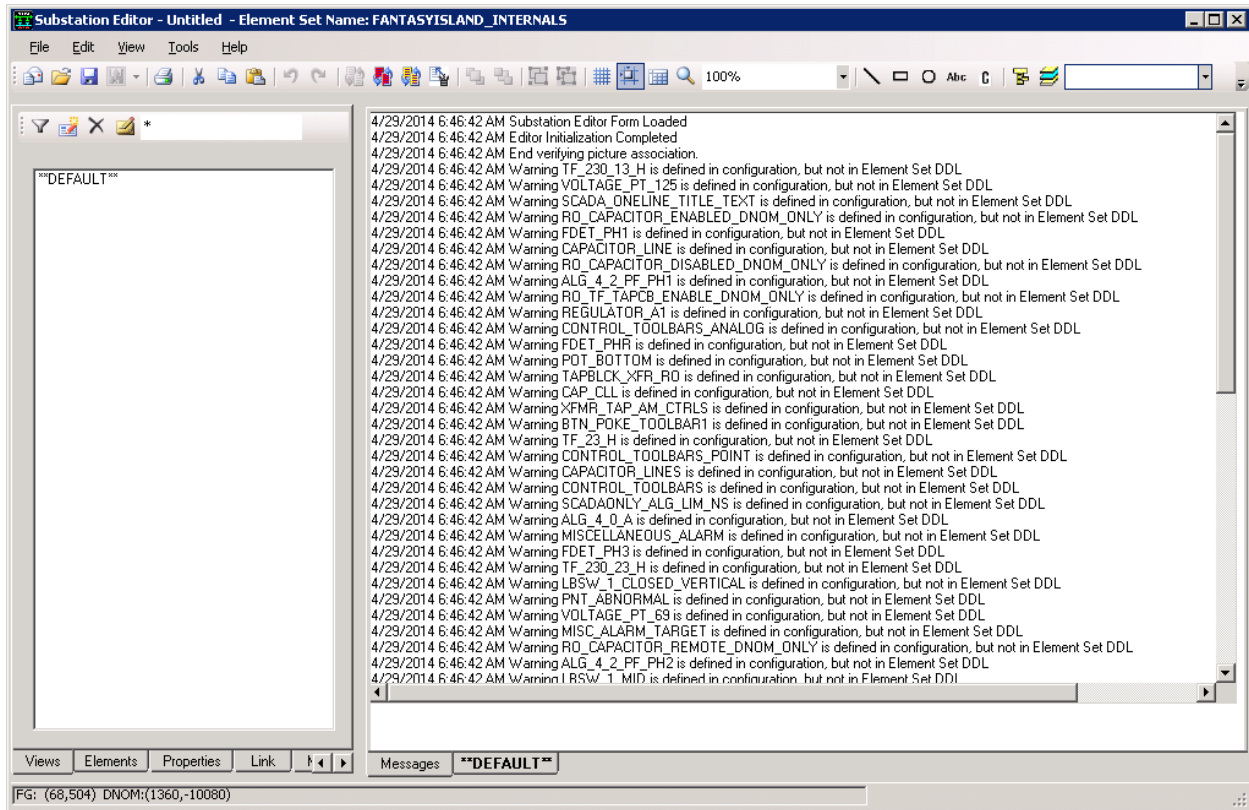


Figure 9. Message Log Display

### 3.3.2. Views

The Views tab contains a list of displays that have been created within the current network model. Clicking the name of the view in the list will cause the view to be displayed in the Main Pane; this view is now active and can be edited.

A new view can be created by clicking the Add button (refer to the following image). This displays a pop-up requesting the name of the new view.

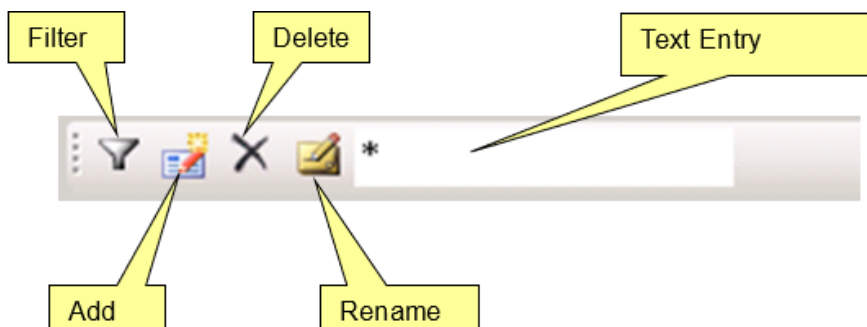


Figure 10. Explorer View Toolbar

The list can be filtered by entering some text in the text entry field and then clicking the Filter button.

An existing view can be deleted by selecting it on the list and then clicking the Delete button.

The name of a view displayed in the Explorer Pane can be changed by clicking the Rename button. This displays a pop-up where the new name can be entered.

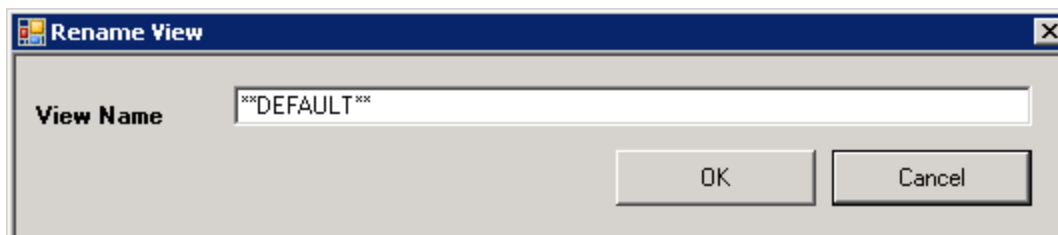


Figure 11. Changing the Name of a View

Note that the preferred way to change the View name is to change the Display Name property on the Properties tab. By doing this, the display name and the view name are consistent, which makes navigation more efficient.

### 3.3.3. Elements

Selecting the Elements tab displays the complete element set for the display drawing. Any element can be dragged onto the display.

The toolbar (refer to the image below) allows you to filter the element set in a number of ways to make finding the required object easier. Clicking any of the buttons toggles the display of that object type on or off. The colored background of a button indicates that the object type is currently displayed. By default, all object types are displayed when the Substation Editor is started.

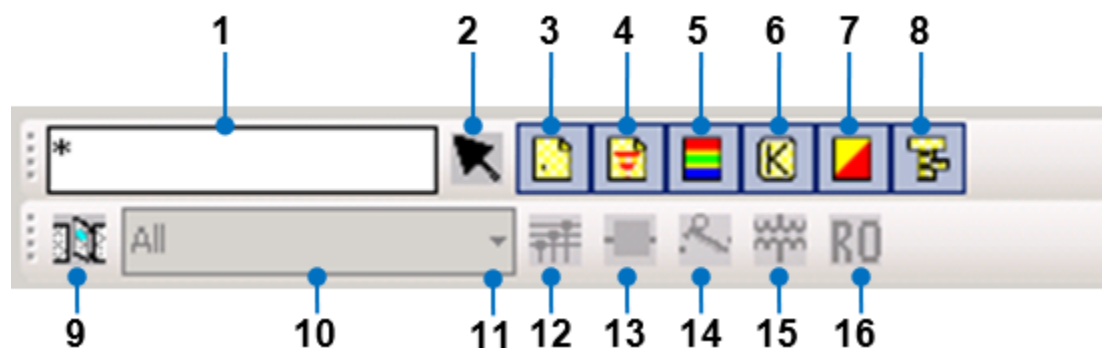


Figure 12. Explorer Elements Toolbar

| Explorer Elements Toolbar Key |                   |    |                  |    |                           |
|-------------------------------|-------------------|----|------------------|----|---------------------------|
| 1.                            | Filter Text Entry | 2. | Clear Filter     | 3. | Show Pictures             |
| 4.                            | Show Symbols      | 5. | Show Shared GABS | 6. | Show Key Sets             |
| 7.                            | Show CAMS         | 8. | Show Menus       | 9. | Show Only Network Objects |



| Explorer Elements Toolbar Key          |                              |                       |
|----------------------------------------|------------------------------|-----------------------|
| 10. Currently Displayed Equipment Type | 11. Equipment Type Selection | 12. Show Buses        |
| 13. Show Breakers                      | 14. Show Switches            | 15. Show Transformers |
| 16. Show Reference Only                |                              |                       |

The name of an element can be entered into the filter text entry field to display only that element. It is best to start entering the name of the desired element to the left of the asterisk (\*). This way, the list of displayed elements will be progressively refined as you type. Multiple wildcards (\*\*\*) can also be used in this field to provide a more effective filter.

The Show Only Network Elements button on the lower half of the toolbar filters the display to show only network objects. These are the display elements that have been associated with network devices (DNOM objects).

Clicking the RO (Reference Only) button filters the element display to show Reference-Only objects. For more information about associations, refer to the "Associations" section in the *Substation Editor Installation Guide*. In this document, also refer to [Network Object Associations References](#).

The objects in the Explorer Pane can be dragged onto the Main Pane to create a display.

To simplify finding the required network objects, the list can be filtered by a number of methods available from the toolbar. The type of object (breaker, bus, composite switch, etc.) can be displayed by selecting it from the pull-down menu, or a specific object type can be shown or hidden by toggling the buttons on the toolbar. The colored background on the button indicates that the object type is currently being displayed.

Each object on the Elements tab has a tooltip that indicates whether the object is a reference object (for example, xx\_RO\_XXX).

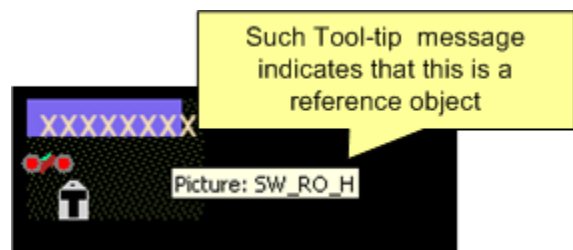


Figure 13. Reference-Only Object Indicated by Tooltip Message

### 3.3.4. Properties

The Properties tab displays the properties of the object currently selected on the display (refer to the image below). Selecting the background of the display shows the properties of the display itself. The Properties tab is object-specific and contains the set of properties with respect to the selected object. As each device on the display is selected, the properties table is automatically updated.

| Template Name | Template Value |
|---------------|----------------|
| %DEVICE%      | <b>5606</b>    |
| %DIVISION%    | <b>WST</b>     |
| %SUBSTN%      | <b>3LIONS</b>  |

| Name                                               | Value                                                                     |
|----------------------------------------------------|---------------------------------------------------------------------------|
| Associated Switch Model                            | <b>SW_GI</b>                                                              |
| Association Type                                   | Switch                                                                    |
| Attachment - Cam                                   | <None>                                                                    |
| Attachment - Key Set                               | <None>                                                                    |
| Attachment - Menu                                  | <None>                                                                    |
| Attachment - Shared Gab                            | <None>                                                                    |
| Boundary Station                                   |                                                                           |
| Branch: FromNode Id                                | b293fb91-c532-493b-a9e                                                    |
| Branch: ToNode Id                                  | 97fb1790-3ff2-44cf-a9ae                                                   |
| Composite Key                                      | <b>SUBSTN = 3LIONS<br/>DEVTYP = SW<br/>DEVICE = 5606<br/>POINT = STTS</b> |
| Composite Switch - Link ID                         |                                                                           |
| Compute Flows (false = collapse switch internally) | <b>True</b>                                                               |
| Division (display grouping)                        | <b>WST</b>                                                                |

Views Elements Properties Link

Figure 14. Explorer Properties Tab

The attributes that are displayed in bold characters are editable from the properties list. Attributes that have undefined values (blank filed) are also editable. Some properties values can only be selected from the drop-down list. Properties attributes can be sorted in descending or ascending order by clicking the Name header. Templates attributes can also be sorted in a similar way.

The default property values are assigned using the Associations Configuration tab, which is part of the administrative configuration. Refer to the "Associations" section in the *Substation Editor Installation Guide*.

### 3.3.5. Links

The Link tab presents the user with an optional tool that can be used to query a Habitat database for available composite keys. By selecting the appropriate composite key from the query results, the user can drag and drop the composite key onto (and thereby link) the corresponding picture

The relevant Habitat host is defined in the Substation Editor configuration (refer to the "Linker" section in the *Substation Editor Installation Guide*). Connect the Substation Editor to the Habitat host by clicking the Connect button. The connection status will be updated, and the list of composite keys will be displayed.



### 3.3.6. Measurements

### 3. Substation Editor User Interface Overview

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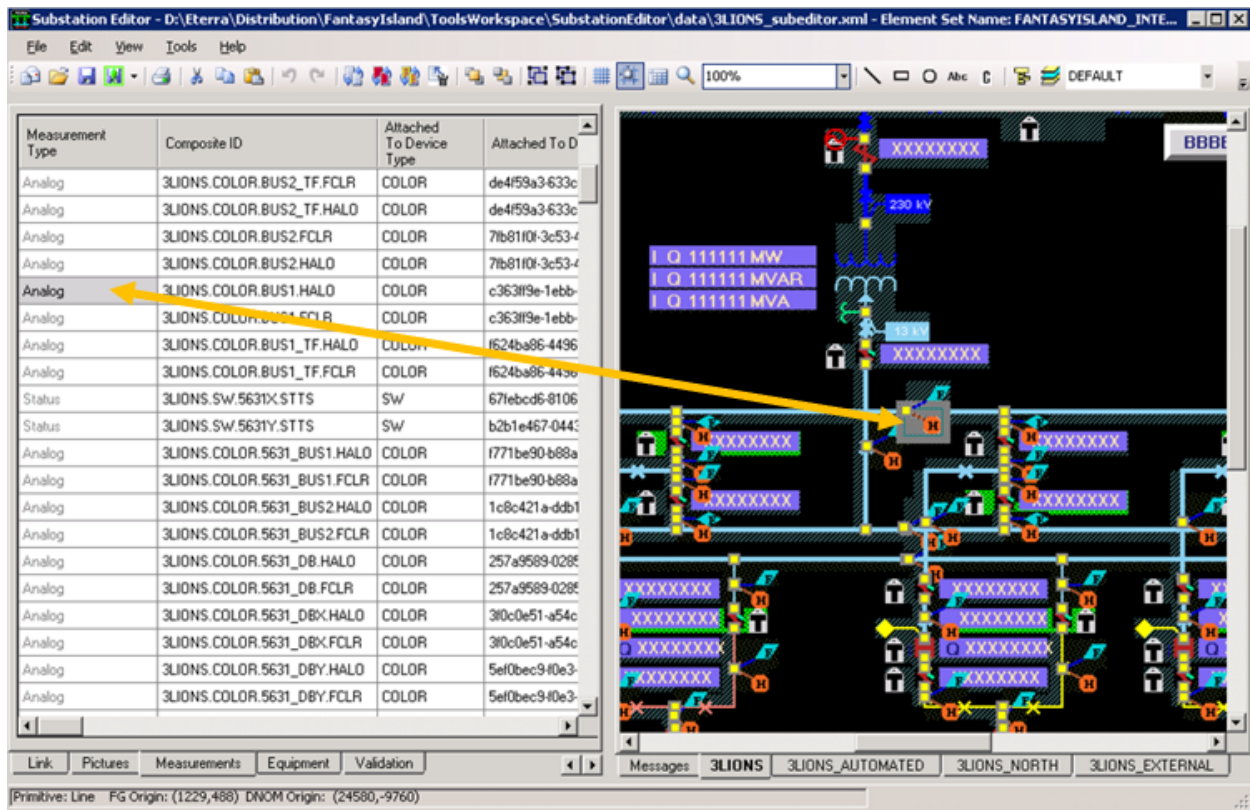


Figure 16. Explorer Measurements Display

Some other fields of DNOM objects are also editable. The columns in the Measurements tab can be sorted. Clicking the header cell of a column will change the sorting order of the column.

### 3.3.7. Pictures

The Pictures tab provides a tool for reviewing and editing the linkages between the display elements and the Habitat composite keys. When an item on the list (element name or composite key) is selected, the relevant display object is highlighted with a gray rectangle. Similarly, if a display object is selected, the relevant entry in the list is highlighted.

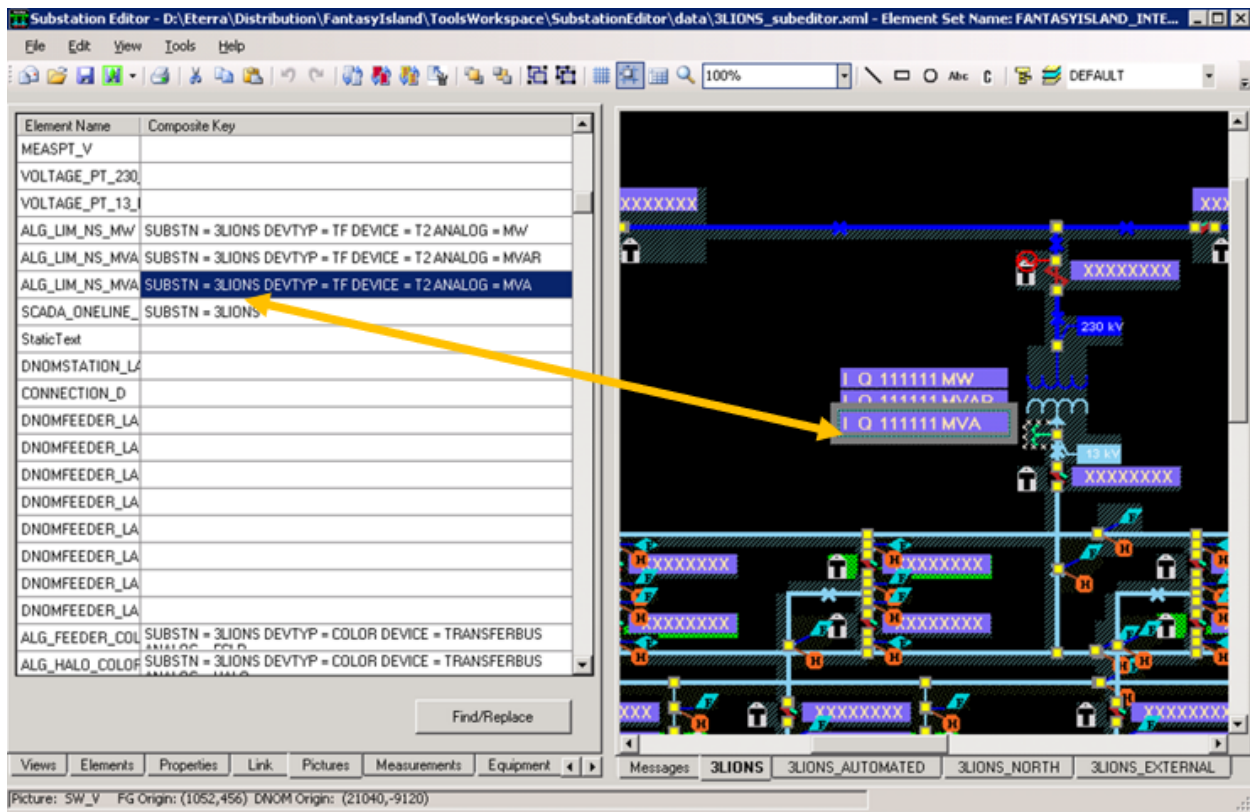


Figure 17. Explorer Pictures Display

If required, the composite key can be edited from the Pictures list. To simplify the task of making multiple edits to this list, a Find/Replace function is provided. Click the button at the bottom of the list to display the pop-up. Enter the Find and Replace text, and then use the function buttons to repeat the edits as required.

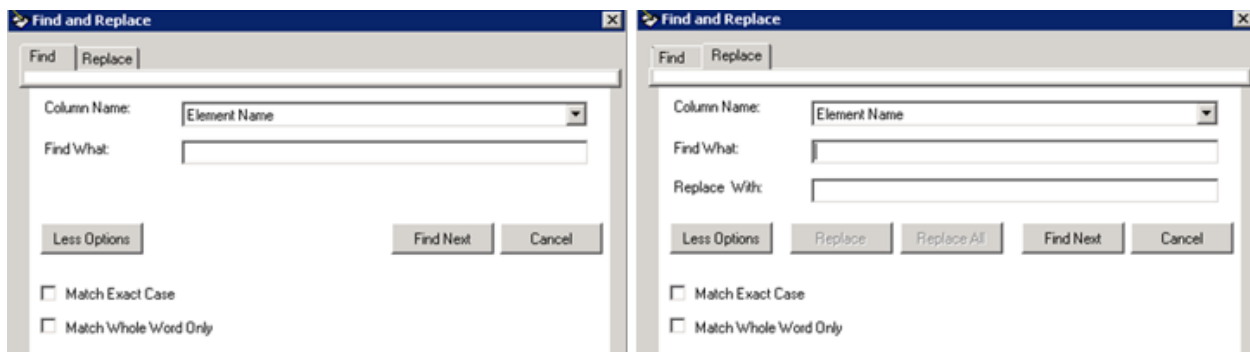


Figure 18. Explorer Pictures - Find and Replace Pop-Up

The Find and Replace dialog box allows specifying a text string to search for, along with the name of the column in which the search will take place. The Replace tab allows specifying a text string to search for, along with a column name and a replacement text string. The column name is a drop-down list of available columns.

If you need to replace all of the text string to another text string, click the Replace All button. If you



need to replace only the current text string, click the Replace button. To continue the search after you have found the specified text string, click the Find Next button.

You can access advanced search options by clicking the More button. Advanced search options are defined by two check boxes: Match Exact Case and Match Whole Word Only. Match Exact Case toggles a case-sensitive search. Match Whole Word Only toggles the ability to search for text substrings.

### 3.3.8. Equipment

The Equipment tab provides a tool for the review of DNOM IDs with the display objects. Selecting an ID on the list positions and highlights the relevant object on the display. Similarly, selecting an object on the display highlights the relevant ID on the list.

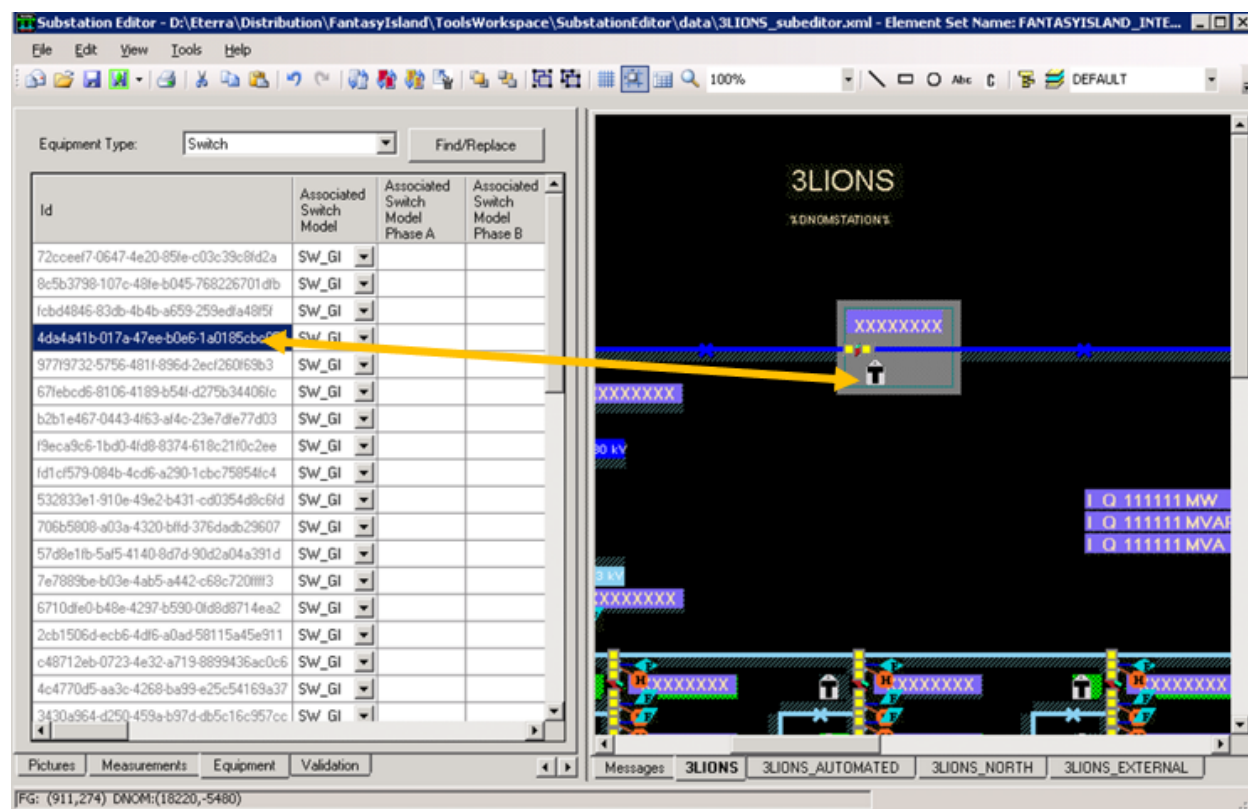


Figure 19. Equipment Tab

To simplify the location of specific IDs, the list can be filtered by selecting the required equipment type from the pull-down menu or by using the Find/Replace pop-up. By default, the list of objects is not filtered and the list contains only ID columns. When applying a filter to the list, the set of columns on the list also changes. A column set is device-specific. After filtering, the first record on the list is selected by default, and the relevant element is highlighted on the display. The list of elements can be sorted by clicking header cells.

### 3.3.9. Validation

The validation function allows a station model to be validated before it is exported. The model is tested against a set of rules, and violations are identified. The devices that do not conform to the validation rules can be easily identified. To validate a station model, click the Validation tab, and then click the Validate/Revalidate button. The result is a list of errors, warnings, and informational messages. Selecting a line on the validation error list highlights the relevant device on the corresponding display view. For more details about the use of the validation function, see [Validation](#).

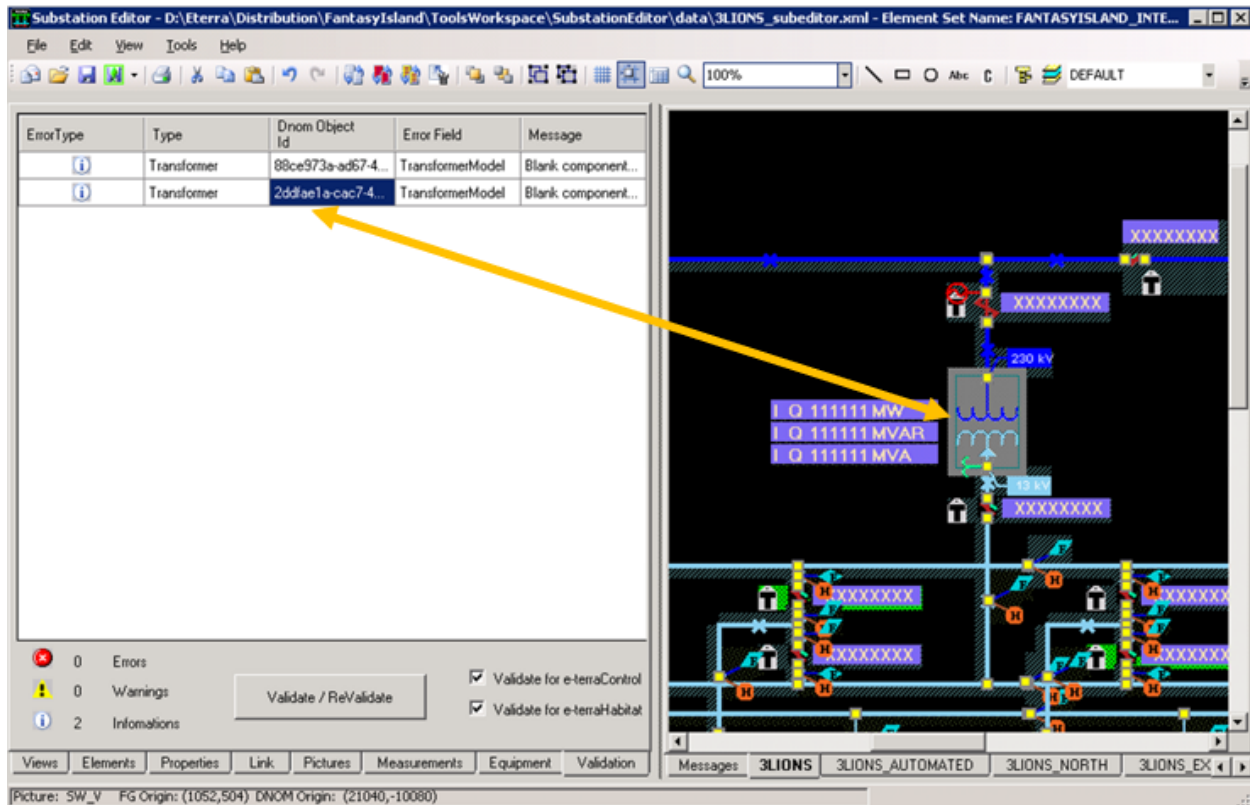


Figure 20. Validation Tab

## 4. Using the Substation Editor

### 4.1. Starting the Substation Editor

The Substation Editor is one of the modeling tools deployed when installing the ADMS configuration tools.

The Substation Editor is started from the Microsoft Windows Start menu, under **Programs > Eterra > Distribution > Tools > Modeling Tools > Substations Editor**.

#### 4.1.1. Main Pane View

In the Main Pane, the normal view is the display that is currently being created or edited. It is selected by clicking the tab with the name of the station file currently loaded.

The alternate view is the system Message log, which is displayed via the Messages tab at the bottom of the screen. The Message log keeps a complete time-tagged record of some of the activities during the current Substation Editor session, such as creating/deleting views, changing the configuration of the Substation Editor, etc.



Figure 21. Typical Main Pane Tabs

The arrangement of the views in both the Main Pane and the Explorer Pane is quite flexible. The splitter bar can be easily moved to adjust the sizes of the two panes. Views can be moved from one pane to the other by dragging the tabs (see [Arranging the Tab Views](#)). In this way, it is possible to arrange the two views needed for any operation to be side by side.

For example, if a Station Externals display has been imported, it can be placed in one pane while the display being built is placed in the other. This allows objects to be mapped from one display to the other simply by dragging.

Tabs from the Explorer Pane can also be moved to the Main Pane. Note that at least one tab must remain on the pane.

#### ! Important

Multiple instances of the Substation Editor can be run simultaneously. Display elements and pictures can be copied and pasted between these instances. In this way, an editing environment made up of multiple views can be created. Note, however, that if the same subeditor file is open in more than one instance of the Substation Editor, changes must only be made to one of the instances, or changes might be lost.



## 4.1.2. Arranging the Tab Views

To make more effective use of the display panes, you can move the tabs and their associated displays from one pane to the other. In this way, the displays in each pane can be organized as required. Click the tab, drag it across to the tabs on the other side, and release the mouse button.

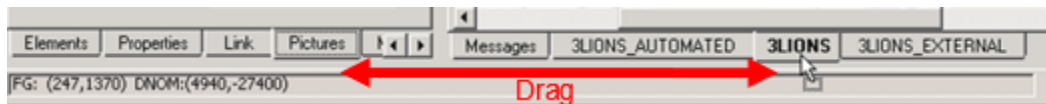


Figure 22. Dragging Tabs between Views

At least one tab must always remain in each pane. Tabs cannot be removed; only views can be removed from display panes. In order to remove a view, open the Views tab, select the required view, and click the Delete button.

It is also possible to create a new view using the Views tab. A new view will always be displayed on the Main Pane. However, it is possible to move views from one pane to the other.

## 4.2. Substation Editor Tasks

The Substation Editor provides the tools for a display to be constructed, and for the display objects to be linked to the relevant data models. The bounds of the display are automatically calculated to encompass all of the objects placed in the display.

The origin of the display is in the upper left corner. When laying out a display, it should grow out from the origin. Placements can be made outside of the current viewport, which will cause the bounds to be re-calculated. This will be seen as a re-alignment of the scroll bars for the display viewport.

Displays can be either created from scratch or loaded from an existing file. To load an existing Substation Editor file, use the Open File button on the Menu bar.

All Substation Editor files are of the form :

station\_subeditor.xml

Where `station` is the name of the relevant substation.

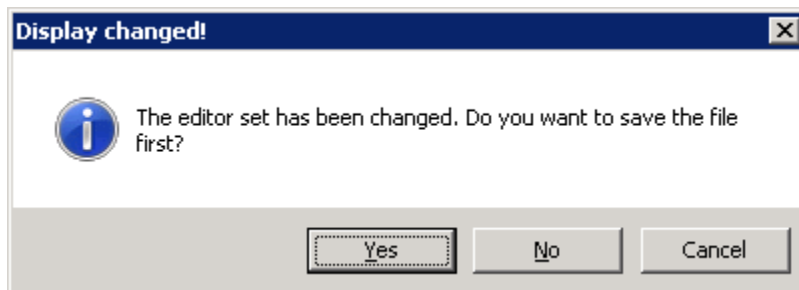
### **i** Note

The Converter tool uses the names of files to determine which ones belong to the same station. The "station" in the SUBEDITOR file name must exactly match the "station" in the station externals file name.

### 4.2.1. Display Drawing

A new display is created by selecting the File menu > New. This closes the current display and

presents a blank screen in the Main Pane. If there are any modifications to the current display, there will be a warning pop-up message. Clicking Yes saves all the modifications, closes the current display, and creates a new blank display. Clicking No closes the current display without saving the modifications. Clicking Cancel leaves the current display in the Main Pane; a new display will not be created.



*Figure 23. Warning Message to Save Changes on the Display*

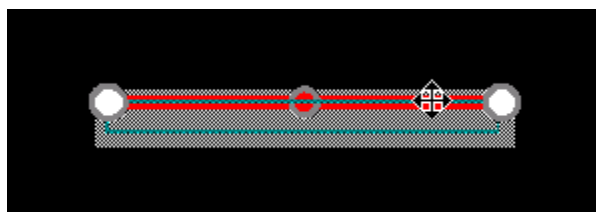
The most frequently used objects on the display will be the devices and pictures that are mapped to the network model. To display these in the Explorer Pane, select the Elements tab, and then click the Show Only Network Objects button.

By default, all available objects are displayed. Use the buttons on the Explorer Elements toolbar to filter the objects. The filtering buttons allow displaying only buses, breakers, switches, transformers, or reference-only elements. For example, de-select the buttons so that only the buses are displayed.

## Drag and Drop

Drag and Drop is the main method of display construction. Select an object in the Elements tab, hold down the mouse button, drag the object into the Main Pane, and then release the mouse button. For example, drag the bus object onto the Main Pane to form the basis of the display. Drag the ends of the bus out to the required length. The cursor shows a double-ended arrow to indicate that the object will be stretched rather than moved (refer to the images below).

Tabs can be moved from one pane to another. Therefore, it is possible that the Elements tab will be on the Main Pane, and a view to which the element needs to be dropped will be on the Explorer Pane.



*Figure 24. Moving an Object*

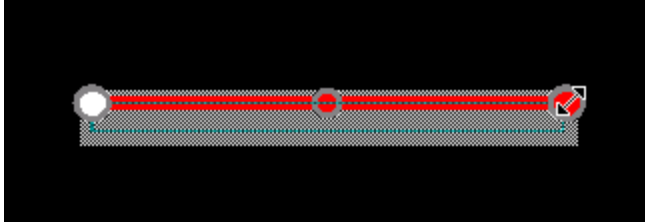


Figure 25. Stretching an Object

## Connections

The selection points at the ends of devices are the points at which they can be connected to other devices.

The one exception to this rule is the bus device, which will allow other devices to be connected to it at any point along its length. When two end points are moved over each other, or when an end point is moved over a bus, the topology process will run to determine the connection. A successful connection will be shown as a yellow square (refer to the image below). Note that the topology process does not run until the object is released, and it may take a couple of seconds on complex displays.

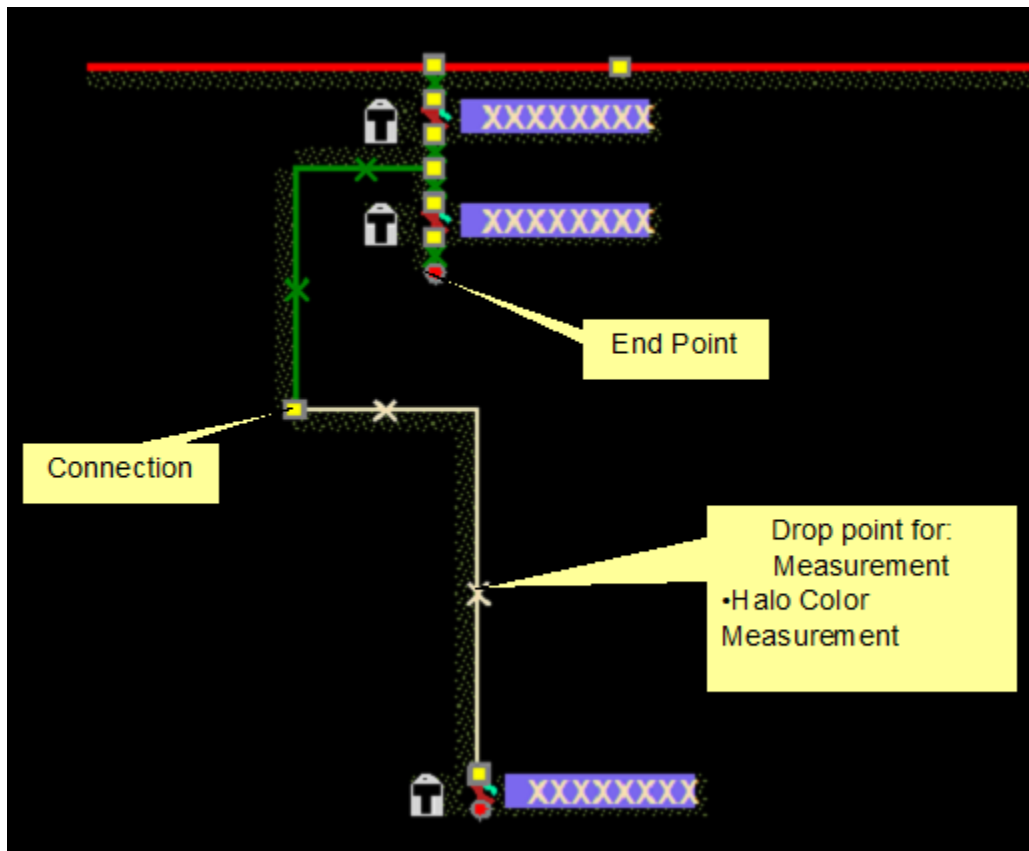


Figure 26. Connection Indications

## Lines

Most line sections are built using the jumper object. Once the jumper object has been placed on the display, it can be re-arranged by dragging either of the end points. Additional segments can be created by positioning the cursor over the end point of the line (the cursor shows a double-ended arrow) and pressing the "+" key on the numeric keypad. (Segments cannot be added to the starting point of a line.) The new end point can then be selected and the line stretched in the desired direction (refer to the image below). In a similar fashion, segments can be removed by pressing the "-" key on the numeric keypad.

On keyboards that do not have a numeric keypad, such as laptops, it may be necessary to press the Shift key in addition to the +/- keys.

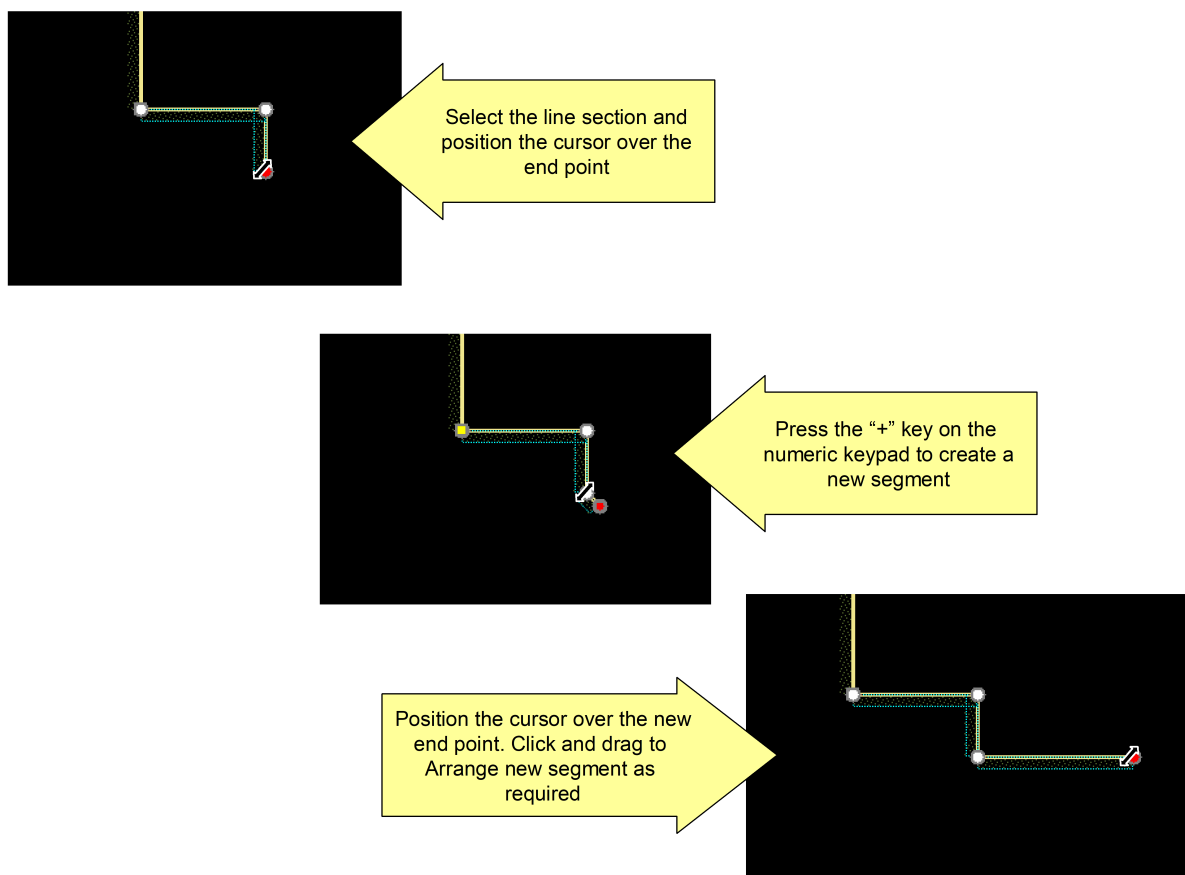


Figure 27. Adding a Line Segment

## Auto Straightening

In order to help the user draw straight vertical and horizontal lines, the Substation Editor automatically adjusts line segments where the starting point and end point are almost straight. If the difference in X or Y direction is less than 10 units, the assumption will be made that the user intends to make a straight vertical or horizontal line and the end point will be adjusted.

The auto-straighten feature processes each line segment in the line, starting at the beginning and progressing toward the end of the line. Thus, it is useful to anchor the beginning of the line in the

intended start position and progressively add additional line segments as needed at the end of the line to prevent the line from moving around. It is notable that additional points cannot be added to the beginning of the line because, if that were allowed, all other line segments would be moved and the line would appear to shift around.

## Deleting

You can delete objects from the display by selecting and then pressing the Delete key, or by right-clicking the mouse button and selecting Delete in the Placement menu. To revoke the delete operation, click the Undo button on the main toolbar; this will put the deleted object back in its previous position.

## Copy, Cut, and Paste

The standard copy, cut, and paste functions can be used on any of the display objects. The selected device (or group) can be copied or pasted using the normal Windows control functions on the Edit menu. In addition, the Copy, Delete, and Paste functions are available from the right-click Placement menu, as shown in the following image.

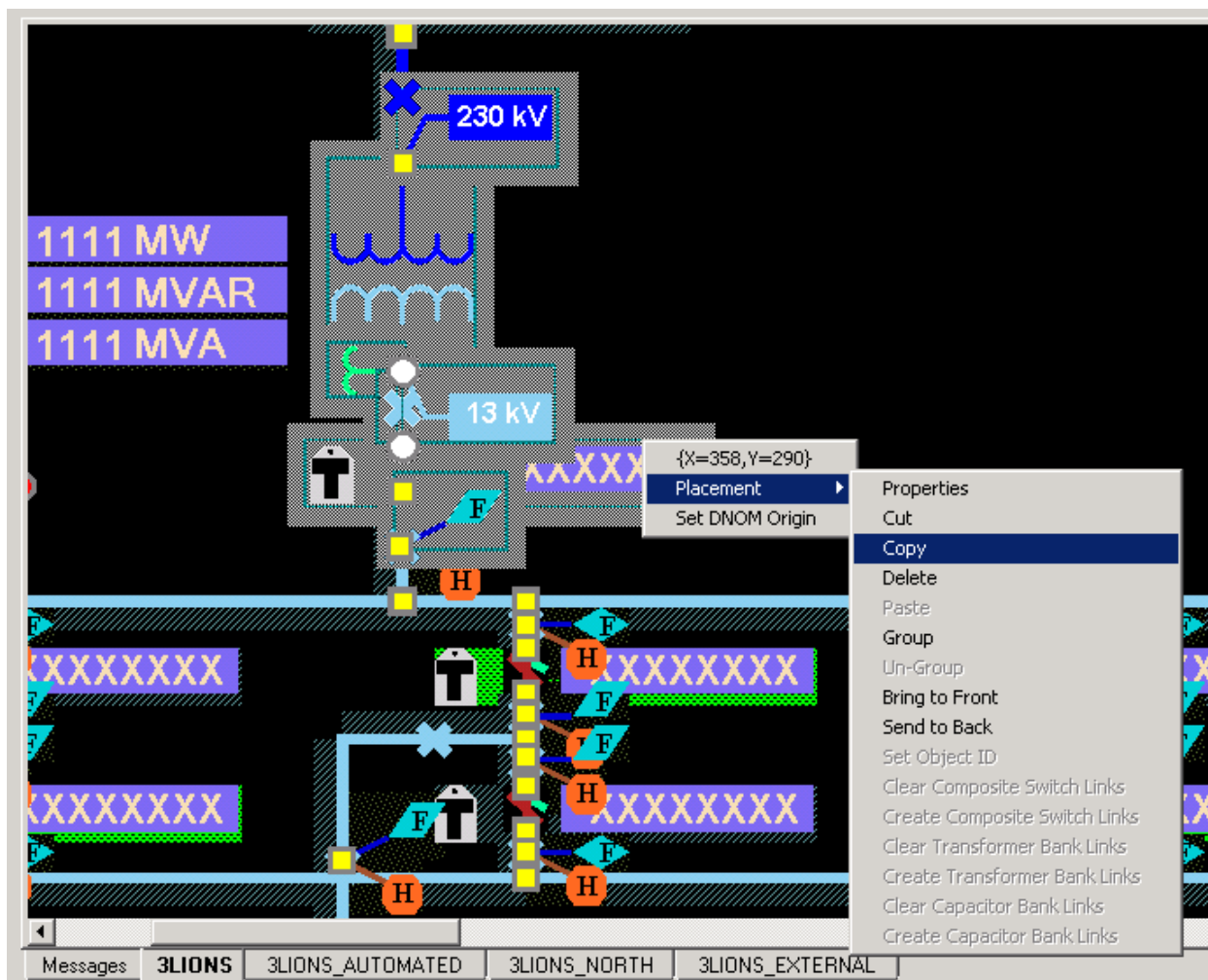


Figure 28. Right-Click Placement Menu

The standard Windows keyboard commands of Ctrl+C (Copy), Ctrl+X (Cut), and Ctrl+V (Paste) are also available.

A selected device can also be copied by selecting the device, holding down the Ctrl key, and then dragging the device to the desired position on the display. This will create a copy of the device when the mouse button is released.

## Selecting Multiple Objects

Multiple objects can be selected by holding down the Shift key and clicking the required devices. It is also possible to select multiple objects by clicking in a blank portion of the screen and dragging a selection box around the desired objects. An object can be unselected from the multiple selections by holding down the Shift key and clicking the required selected device.

## Grouping Objects

A number of objects can be selected and then grouped together so that they can be treated as a single object. Select a number of objects on the display, and then click the Group Items button on the toolbar. When the group is selected, a red border appears around the group. The objects can be ungrouped by selecting the group and clicking the Ungroup Items button on the toolbar. The right-click menu also allows grouping/ungrouping of objects. Select a number of objects on the display, and then right-click and select the Placement submenu. In the open Placement submenu, select Group or Ungroup Item.

## Right-Click Menu

To make the placement and editing process easier, some of the standard placement functions are available as a right-click menu.

The right-click menu contains the following items:

- **Coordinates:** Coordinates of the point on the display
- **Placement submenu:** This menu includes basic operations that are available from the main toolbar:
  - **Properties:** Opens the Properties tab
  - **Cut:** Removes the selected object from the display and copies it to the clipboard
  - **Copy:** Copies the selected object to the clipboard
  - **Delete:** Deletes the selected object
  - **Paste:** Pastes an object from the clipboard
  - **Group:** Creates a group of selected objects
  - **Un-Group:** Ungroups the selected group of objects
  - **Bring to Front:** Moves the selected objects in front of all other objects in the same display
  - **Send to Back:** Moves the selected objects behind all other objects in the same display
  - **Set Object ID:** Sets the unique identifier of the selected object
  - **Clear Composite Switch Links:** Clears the composite switch links

- **Create Composite Switch Links:** Creates the composite switch links
- **Clear Transformer Bank Links:** Clears the transformer bank links
- **Create Transformer Bank Links:** Creates the transformer bank links
- **Clear Capacitor Bank Links:** Clears the capacitor bank links
- **Create Capacitor Bank Links:** Creates the capacitor bank links
- **Set DNOM origin:** Shifts the default coordinates of DNOM origin

The right-click menu also gives the coordinates of the point on the display that was clicked, and it provides a means of setting the DNOM origin. This allows the origin for DNOM displays to be shifted from the default of 0,0 in the upper left corner of the window.

## Placeholders

A placeholder is a graphic device that provides a target for navigation within a simple layer. The user provides access to the placeholder by supplying its name as the object of a PLACEHOLDER qualifier in a DISPLAY command. This command is captured as the outcome of a menu option in a menu definition. Selection of the option causes the display to open over the placeholder.

A typical example is the use of a placeholder to link the feeder lines on a Substation Internals display to the filtered geographic display of the relevant feeder.

To insert a placeholder, click the Add Placeholder button on the toolbar, and then click the required location on the display. A text entry field is displayed to allow the object name to be entered. Press the Escape key to terminate the text entry and add the placeholder. The placeholder is then displayed, as shown in the image below.

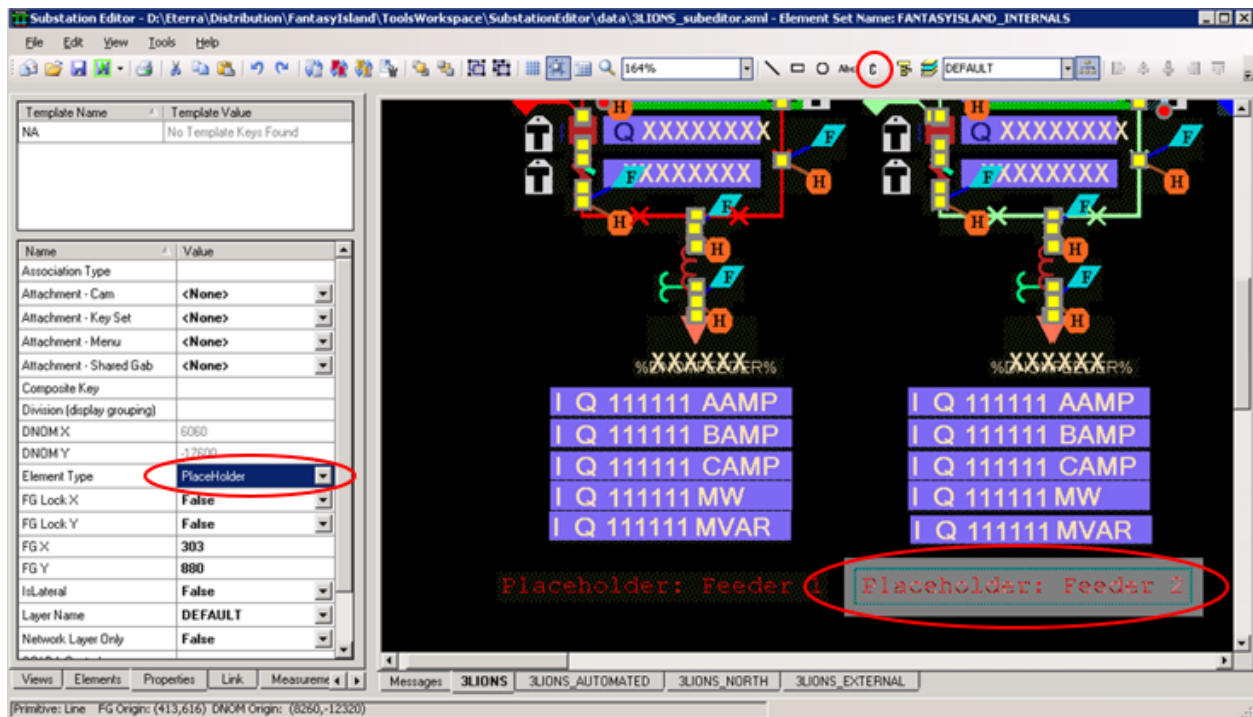


Figure 29. Adding a Display Placeholder

## Undo and Redo

Any action in the development of the display can be undone by clicking the Undo button. The undo action can be reversed, if needed, by clicking the Redo button.

### Note

Substation Editor configuration changes cannot be revoked with the Undo button.

## Voltage and Feeder Color Tracing

To simplify the display drawing tools, the Substation Editor allows automatic assignment of feeder colors and voltage levels to the relevant parts of the display by tracing out all connected devices. A feeder color can be automatically applied to all parts of the display downstream of a designated feeder head, and a voltage level can be applied to the section of the network bounded by a transformer and/or a feeder head.

To set the feeder color on a SCADA display, select the Show Shared GABS option on the Elements toolbar, select the desired Feeder Color shared GAB from the element list and drag it over the feeder head object on the display. All line sections downstream of the feeder head will be updated to the selected color, as shown in the image below.



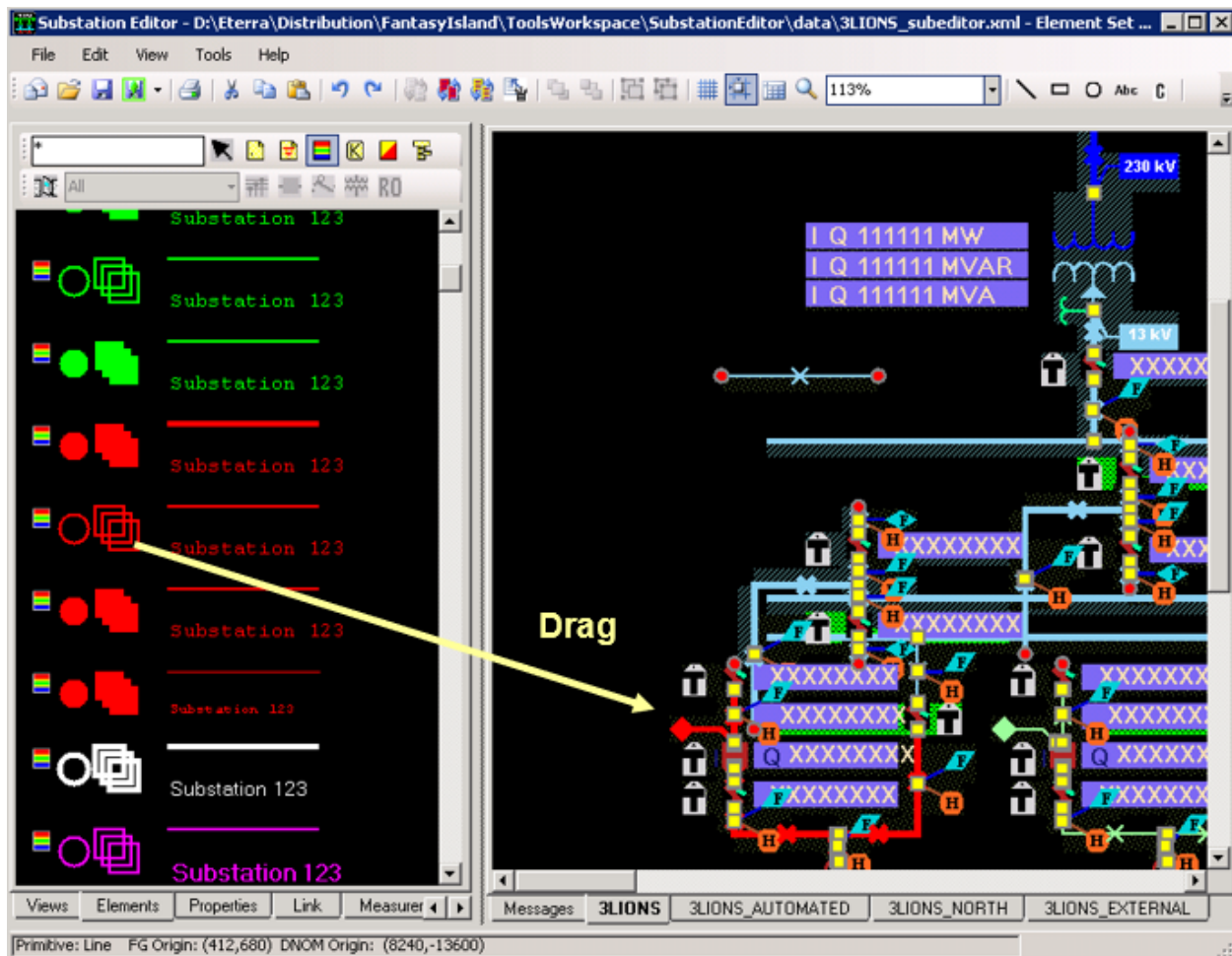


Figure 30. Setting the Feeder Color

To set the voltage level, select the appropriate voltage point picture and drag it onto a connection node in the relevant section of the network, as shown in the image below.

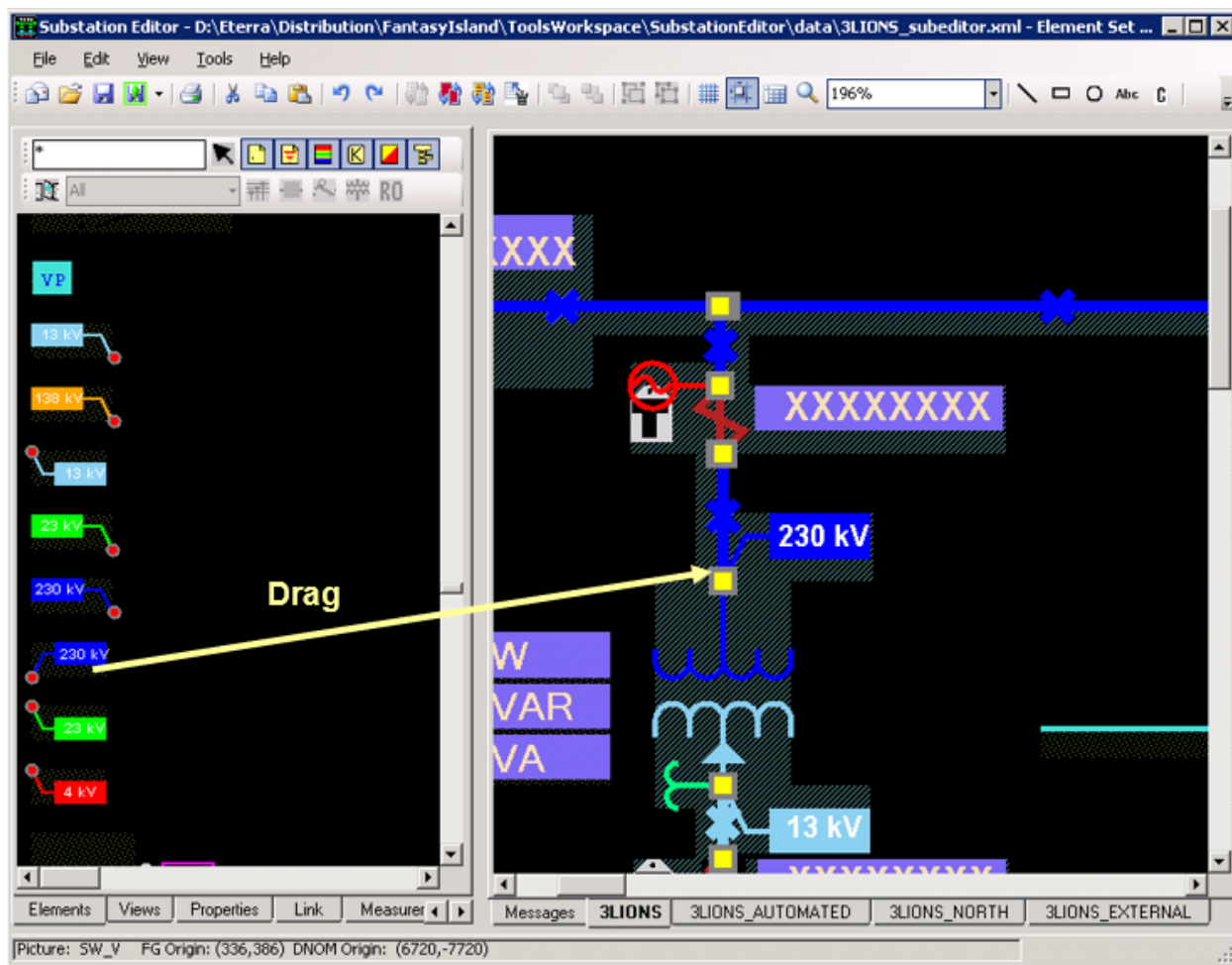


Figure 31. Setting the Voltage Level

## 4.3. Feeder Color and Halo Color Measurements

Feeder color and halo color measurements on lines and buses can be associated with a SCADA analog.

An example of this is shown in the image below. Feeder and halo color indicators have been added to two jumpers in the feeder bank. The screenshot on the left shows how they appear within the Substation Editor. The screenshot on the right shows how the same jumpers appear on the resulting SCADA display with the feeder color and halo colors enabled.

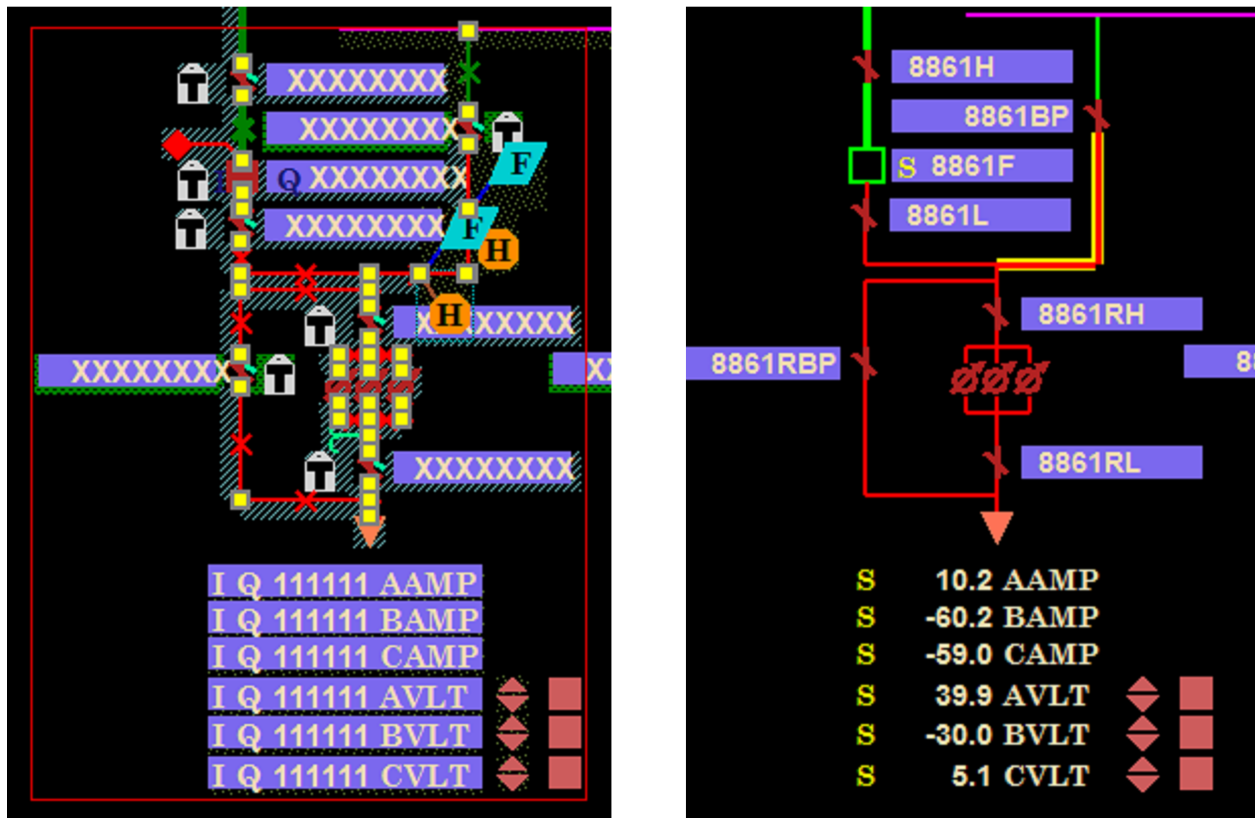


Figure 32. Examples of Feeder and Halo Color Measurement

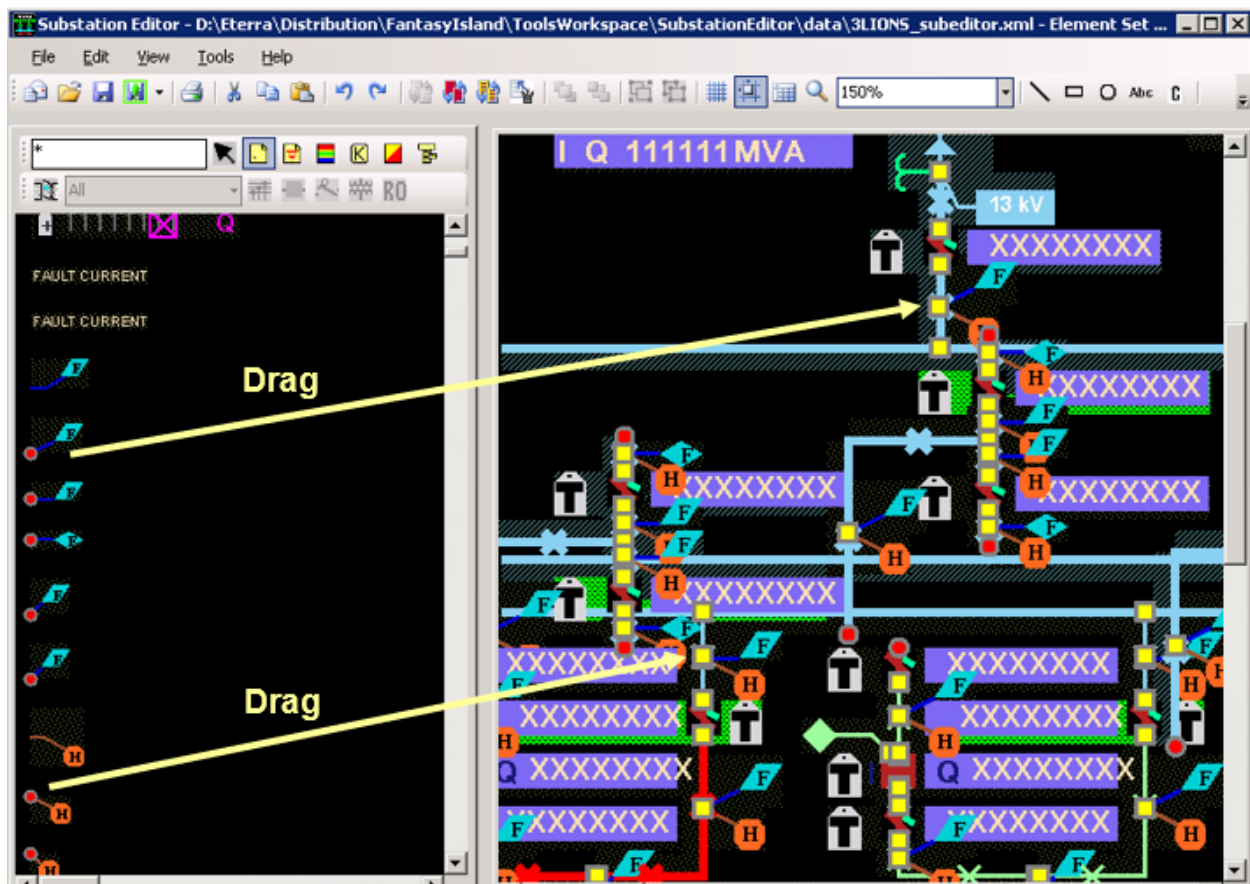


Figure 33. Applying a Feeder and Halo Color Measurement

### 4.3.1. Feeder Color Measurement

A feeder color measurement can be associated with a SCADA analog value. The measurement is associated by adding a picture, as shown in the image above. Feeder color indicators are connected to the mid-point (X) on jumpers and lines. On buses, they are placed close to the connection points.

The composite key for the SCADA analog value must be specified in the association. The MEAS type is set to FeederColor (refer to the following image).

**Substation Editor Configuration**

**Association:**

- Associated Element: **ALG\_FEEDER\_COLOR**
- Add Association:
- Mismatch: ☐
- DNOM Type:
- Assoc. Reference Picture: **ALG\_FEEDER\_COLOR**
- Connection Pt 1:  Connection Pt 2:
- Connection Pt 3:  Network Only: ☒ Delete Mismatch Assoc.
- Reference Only: ☐ FG Layer:
- Three Winding Transformer: ☐
- Create DNOM Symbol: ☒ Create Label: ☐ Clear Assoc.
- DNOM Symbol:  Offset:  Rotation:
- DNOM Label:  Offset:  Rotation:

**Measurement Info:**

- MEAS Type: **FeederColor**
- MEAS Group: **AnalogMeasurementGroup**
- Default Assoc:
- Default CAM: **ALG\_FEEDER\_COLOR**
- Create Meas: ☒

**Composite Key:** **SUBSTN = %SUBSTN%; DEVTYP = COLOR DEVICE = %DEVICE%; ANALOG = FCLR**

SCADA DataSource: **NR** SCADA Model Name:   Controls:

Division: **%DIVISION%**

**Measurement Properties Table:**

| Name                                                                      | Value                         |
|---------------------------------------------------------------------------|-------------------------------|
| Measurement: (HDB: CATALOG_ANALOG) Exception Category                     | <b>AL</b>                     |
| Measurement: (HDB: DELAY_ANALOG) Delay for Persistence Alarming (seconds) | <b>1</b>                      |
| Measurement: (HDB: HIREAS_ANALOG) High Reasonability Limit                | <b>0</b>                      |
| Measurement: (HDB: ID_MEAS) Measurement ID                                |                               |
| Measurement: (HDB: LOREAS_ANALOG) Low Reasonability Limit                 | <b>0</b>                      |
| Measurement: (HDB: NAME_DEVICE) Device Record Name                        |                               |
| Measurement: (HDB: NFLAGS_ANALOG)                                         | <b>NotSet</b>                 |
| Measurement: (HDB: TRACKID_DEVICE) Unique Tracking Id                     |                               |
| Measurement: Attached To Device                                           |                               |
| Measurement: Attached To Device Type                                      |                               |
| Measurement: Attached To Effective Device                                 |                               |
| Measurement: Base Value                                                   | <b>0</b>                      |
| Measurement: Composite ID                                                 |                               |
| Measurement: Default Quality                                              | <b>0</b>                      |
| Measurement: Default Value                                                | <b>0</b>                      |
| Measurement: FLIP Analog Value                                            | <b>False</b>                  |
| Measurement: Group                                                        | <b>AnalogMeasurementGroup</b> |
| Measurement: Measurement Utilized in Network Applications                 | <b>True</b>                   |
| Measurement: Network Device Id                                            |                               |
| Measurement: Network Device Name                                          |                               |
| Measurement: Permission Area Id (8 character max.)                        |                               |
| Measurement: Phase                                                        | <b>ABC</b>                    |
| Measurement: Simulate Only                                                | <b>False</b>                  |

Associations: General External Station Configurations Topology Linker Global Internals Template Names Poly Placements SCADA Configurations Colors & Halos

Figure 34. Feeder Color Measurement Definition

## 4.3.2. Halo Color Measurement

A halo color measurement can be associated with a SCADA analog value. The measurement is associated by adding a picture. Halo color indicators are connected to the mid-point (X) on jumpers and lines. They can be connected anywhere on buses.

The composite key for the SCADA analog value must be specified in the association. The MEAS type is set to HaloColor (refer to the image below).

**Substation Editor Configuration**

Associated Element: **ALG\_HALO\_COLOR**

Add Associations:

Mismatch: ☐ Copy:

DNOM Type:

Assoc. Reference Picture: **ALG\_HALO\_COLOR**

Connection Pt 1:  Connection Pt 2:

Connection Pt 3:  Network Only: ☒ Delete Mismatch Assoc.

Reference Only: ☐ FG Layer:

Three Winding Transformer: ☐

Create DNOM Symbol: ☒ Create Label: ☐ Clear Assoc.

DNOM Symbol:

Offset:

Rotation:

DNOM Label:

Offset:

Rotation:

**Measurement Info**

MEAS Type: **HaloColor**

MEAS Group: **AnalogMeasurementGroup**

Default Assoc:

Default CAM: **ALG\_HALO\_COLOR**

Create Meas: ☒

**Composite Key:** **SUBSTN = %SUBSTN% DEVTYPE = COLOR DEVICE = %DEVICE% ANALOG = HALO**

SCADA DataSource: **NR** SCADA Model Name:  Refresh

Division: **%DIVISION%**

Controls:

| Name                                                                      | Value                         |
|---------------------------------------------------------------------------|-------------------------------|
| Measurement: (HDB: CATALOG_ANALOG) Exception Category                     | <b>AL</b>                     |
| Measurement: (HDB: DELAY_ANALOG) Delay for Persistence Alarming (seconds) | <b>1</b>                      |
| Measurement: (HDB: HIREAS_ANALOG) High Reasonability Limit                | <b>0</b>                      |
| Measurement: (HDB: ID_MEAS) Measurement ID                                | <b>0</b>                      |
| Measurement: (HDB: LOREAS_ANALOG) Low Reasonability Limit                 | <b>0</b>                      |
| Measurement: (HDB: NAME_DEVICE) Device Record Name                        | <b>NotSet</b>                 |
| Measurement: (HDB: NFLAGS_ANALOG)                                         | <b>NotSet</b>                 |
| Measurement: (HDB: TRACKID_DEVICE) Unique Tracking Id                     |                               |
| Measurement: Attached To Device                                           |                               |
| Measurement: Attached To Device Type                                      |                               |
| Measurement: Attached To Effective Device                                 |                               |
| Measurement: Base Value                                                   | <b>0</b>                      |
| Measurement: Composite ID                                                 |                               |
| Measurement: Default Quality                                              | <b>0</b>                      |
| Measurement: Default Value                                                | <b>0</b>                      |
| Measurement: FLIP Analog Value                                            | <b>False</b>                  |
| Measurement: Group                                                        | <b>AnalogMeasurementGroup</b> |
| Measurement: Measurement Utilized in Network Applications                 | <b>True</b>                   |
| Measurement: Network Device Id                                            |                               |
| Measurement: Network Device Name                                          |                               |
| Measurement: Permission Area Id (8 character max.)                        |                               |
| Measurement: Phase                                                        | <b>ABC</b>                    |
| Measurement: Simulate Only                                                | <b>False</b>                  |

Associations: General External Station Configurations Topology Linker Global Internals Template Names Poly Placements SCADA Configurations Colors & Halos

OK Apply Cancel

Figure 35. Halo Color Measurement Definition

### 4.3.3. Grid Management

By default, objects are positioned on the display aligned to a grid of 20x20 pixels; this is the Snap to Grid mode. The grid can be displayed by toggling the Show Grid button on the toolbar, or by checking the Show Grid check box on the Grid Configuration pop-up, which is displayed by clicking the Configure Grid button. The Snap to Grid mode can be disabled by toggling the button on the toolbar. The grid spacing can be customized on the Grid Configuration pop-up.



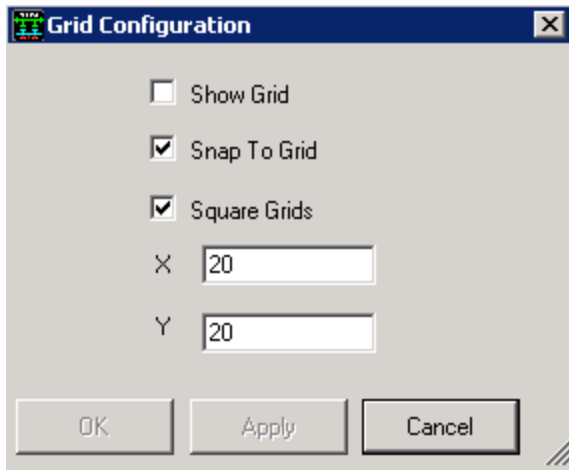


Figure 36. Grid Configuration Pop-Up

### 4.3.4. Alignment Tools

To assist in producing uniform layouts, a series of alignment tools is available on the toolbar. This toolbar is only enabled when more than one object is selected. To use these tools, select a number of objects on the display, and then click the button for the relevant tool; the objects will be aligned vertically or horizontally as required. For a full list of the alignment tools, refer to [Toolbar Guide](#).

### 4.3.5. Additional Drawing and Formatting

While most of the display will be created by dragging objects from the network list or the element set, it is possible to draw directly on the display. The shapes and lines created by using the standard drawing tools are not associated with network objects and, therefore, will not appear on a distribution network display. However, the objects can be part of a Browser display.

The standard drawing tools - Line, Rectangle, Circle, and Text - are available on the toolbar. Select the required tool, and then click the display area to give a starting point. Drag the cursor to create the required shape. To end the drawing process, press the Escape key.

The properties of drawn objects, such as the color and thickness of lines or the text font, can be changed by selecting the appropriate Shared GAB from the Explorer Elements view and dragging it over the object. The object will then inherit the properties of the Shared GAB.

## 4.4. View Templates

A display can be saved as a template by setting the True value in the Template combo box in the Display properties, as shown in the following image. Note that, if a display is marked as a template, only the Station.issg file is output when applying the Export function.

| Name                     | Value                                    |
|--------------------------|------------------------------------------|
| FG Origin (X)            | 0                                        |
| FG Origin (Y)            | 0                                        |
| Permission Test          | <ClassName [AND/OR<br>ClassName]>        |
| Refresh                  | 2                                        |
| Safety Doc Application   | SAFETY                                   |
| Safety Doc Database      | SAFETY                                   |
| Safety Doc Element Set   | FANTASYISLAND_INTE                       |
| Safety Doc Link Field    | D\$DOC_DOC                               |
| Safety Doc Movable Layer | False                                    |
| Export                   |                                          |
| Template                 | True                                     |
| Title                    | %DIS%,%DISAFF%<br>%HOST% (%VP%)<br>%REF% |
| wC Offset (X)            | 0                                        |
| wC Offset (Y)            | 10000                                    |
| X Scale                  | 0.05                                     |
| Y Scale                  | -0.05                                    |
| Zoom Scale               | 1                                        |

Elements Views Properties Link Measurer

Figure 37. Setting a Template Property for a Display

To build a display from a template, retrieve the template using the normal Open command. Go to the Display properties, set the template value to FALSE, and then click Save New Substation to create the new display. The display can then be updated with the linkages to the SCADA objects or DNOM objects as required.

### 4.4.1. Object Templates

Templates can be used within the object property definitions to simplify the data entry and to ensure consistency between device names and composite keys.

To use the object template, a template name of the form %XXXX% is entered into the Network Device Name field on the Properties page. When the template name is entered, an additional property is inserted into the object Properties page - Network Device Name (Template) - and a corresponding template entry appears at the top of the Properties page. The Network Device Name field is now grayed out because data can no longer be directly entered.

Similarly, the template name can be entered in other property fields for the object that needs the



network device name, such as the composite key in conjunction with template names for other properties.

In this way, key data, such as device name and substation name that appear in multiple property fields, can be linked to a set of template values at the top of the property page. The actual values are then entered in the Template Value fields, and these are automatically transferred to the respective fields throughout the Properties page.

In the example shown in the following image, the network device name has been defined to use the template name "%FEEDER%F", and the composite key is defined to use:

SUBSTN = %SUBSTN% DEVTYP = CB DEVICE = %FEEDER%F POINT = STTS

| Template Name | Template Value |
|---------------|----------------|
| %DIVISION%    | WST            |
| %FEEDER%      | 5634           |
| %SUBSTN%      | 3LIONS         |

| Name                                               | Value                                                         |
|----------------------------------------------------|---------------------------------------------------------------|
| Associated Switch Model                            | CB_GI                                                         |
| Association Type                                   | Breaker                                                       |
| Attachment - Cam                                   | <None>                                                        |
| Attachment - Key Set                               | <None>                                                        |
| Attachment - Menu                                  | <None>                                                        |
| Attachment - Shared Gab                            | <None>                                                        |
| Boundary Station                                   |                                                               |
| Branch: FromNode Id                                | 6688b64c-32e6-46d3-aa0e-65c383eff780                          |
| Branch: ToNode Id                                  | a0b2281d-5c58-4dc9-bd4a-42ba3037f15c                          |
| Composite Key                                      | SUBSTN = 3LIONS DEVTYP = CB DEVICE = 5634F POINT = STTS       |
| Composite Key (Template)                           | SUBSTN = %SUBSTN% DEVTYP = CB DEVICE = %FEEDER%F POINT = STTS |
| Composite Switch - Link ID                         |                                                               |
| Compute Flows (false = collapse switch internally) | True                                                          |
| Division (display grouping)                        | WST                                                           |
| DNOM X                                             | 14400                                                         |
| DNOM Y                                             | -12480                                                        |

|                                |                                      |
|--------------------------------|--------------------------------------|
| Network Device Id              | d0416f4c-28d9-412d-8fc8-7b6800d7eee0 |
| Network Device Name            | 5634F                                |
| Network Device Name (Template) | %FEEDER%F                            |
| Network Layer Only             | False                                |

Figure 38. Object Templates

The actual values for the templates are entered in the Template fields at the top of the page. The

values are then substituted into the appropriate property fields, shown in gray. The template values can now be changed, and all the respective fields in the property page are updated accordingly.

The EMS and DMS station names can be entered once as template name values, and they are then automatically entered into the properties of all devices placed in the display. This greatly reduces the effort required when defining a display; it also reduces data errors. The current naming convention is:

%DNOMSTATION%\_ means the DMS name.

%SUBSTN% means the EMS name.

This is configurable in the Substation Configuration Template Names tab (refer to the "Template Names" section in the *Substation Editor Installation Guide*).

When a new station file is being created, the property pane will be as shown in the image below.

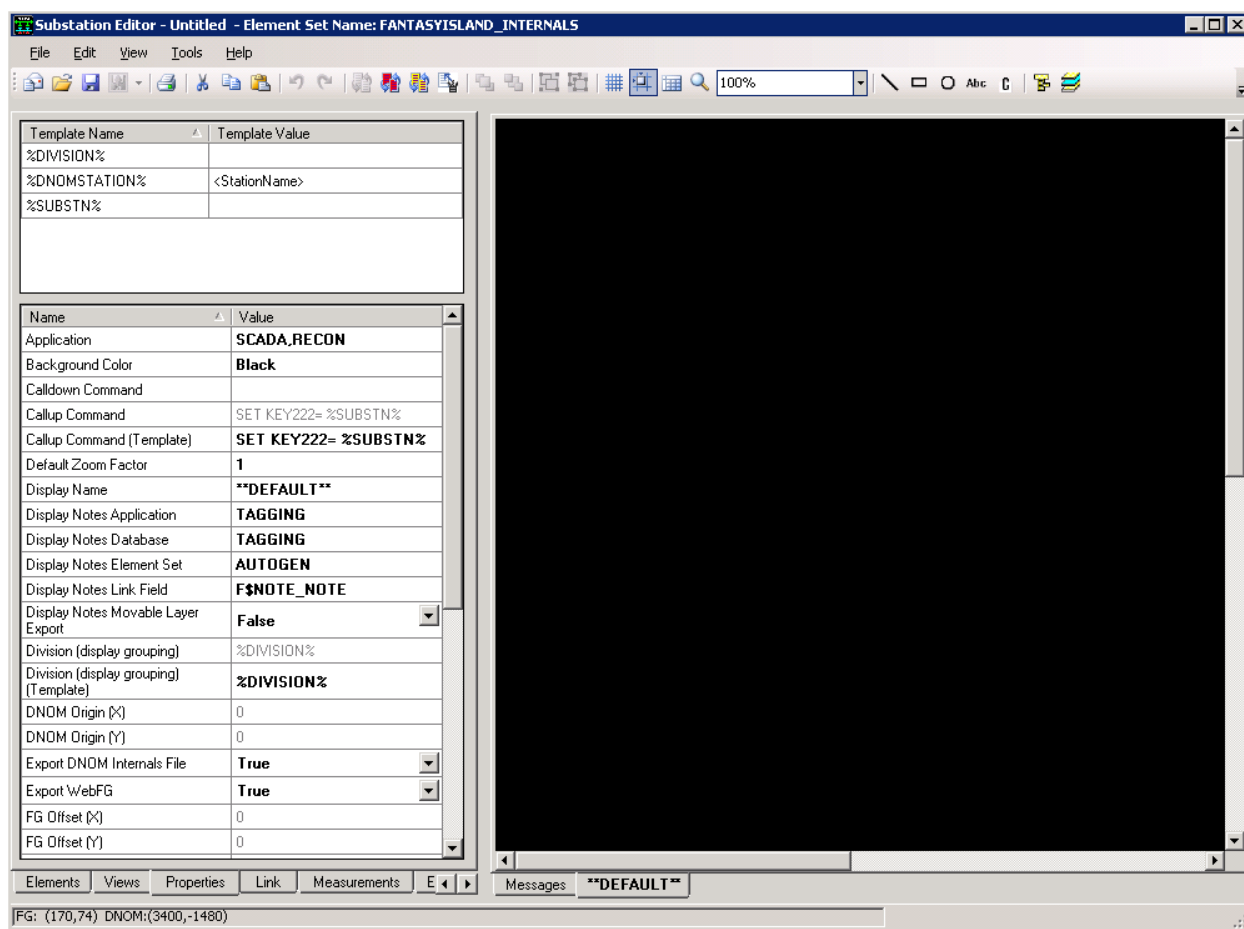


Figure 39. Template Name/Value Fields on the Properties Tab

When creating a new view, it is recommended that these values be entered first before creating any placements that may use the template values. When a new subeditor file is created, or when a new view within a subeditor file is created, the values of all the created templates should be entered.

By setting the values for %DNOMSTATION% and %SUBSTN% in the Template Value field, this

information will not have to be entered for each new placement.

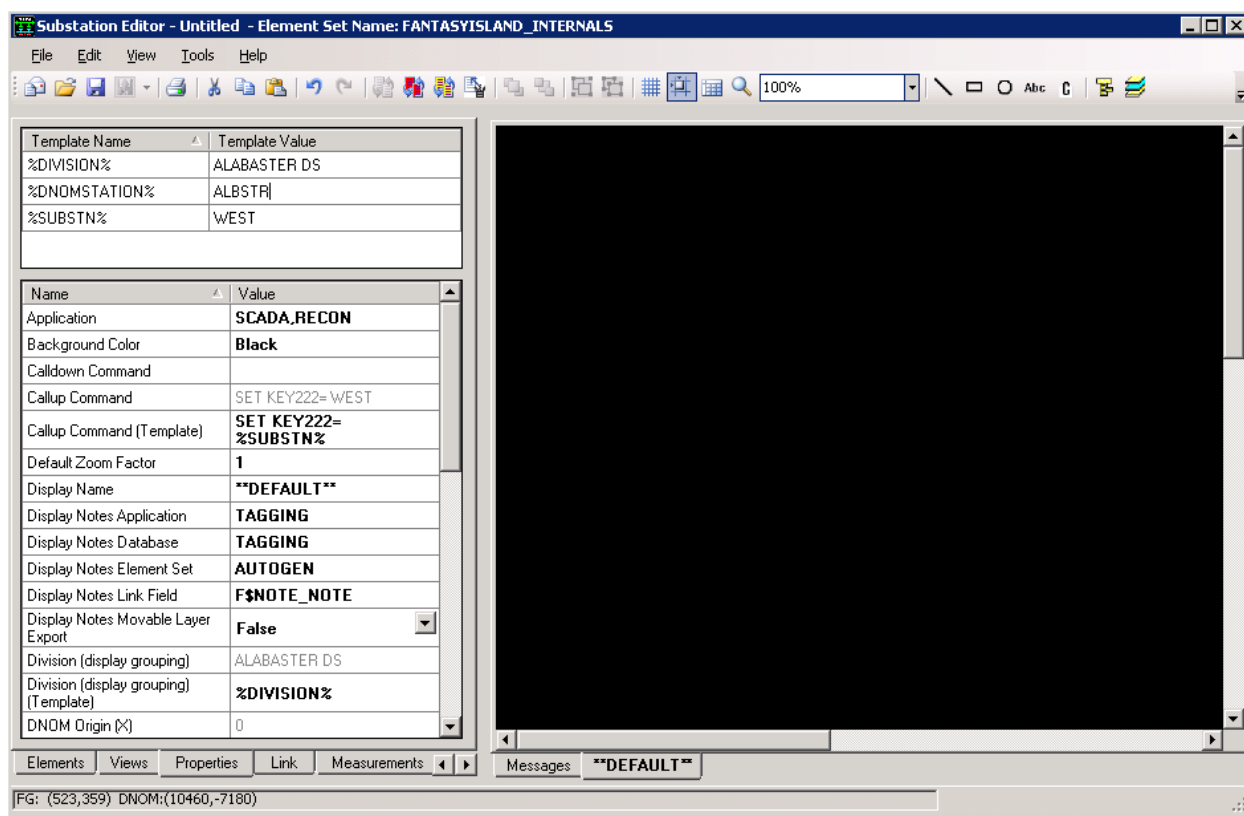


Figure 40. Station Names Entered into Template Name Values

### **i** Note

If the display has been auto-generated, it is necessary to ensure that the values for %DNOMSTATION% and %SUBSTN% are correct. If the station name is more than eight characters or contains spaces, the %SUBSTN% name must be changed to be an acceptable EMS name.

## 4.4.2. Reapply Template Values

To resolve the mismatch between template and resolved values during the display building process, click the Reapply Template Values button. This button is a way to reapply all the template values with one click. A mismatch can occur when a user manually modifies properties - for example, with the Find/Replace tool.

Managing resolved values manually can lead to a mismatch between a template value and the resolved value where the template is used.

When the user clicks the Reapply Template Values for View button, the correct value for a template attribute will be re-substituted into all occurrences where it is used. As a result, the composite key values will be fixed. The purpose of the Reapply Template Values for View button is to provide the

user with a way to fix any mismatches between template values and resolved instances where the template values are used.



Figure 41. Reapply Template Values Button

### 4.4.3. Reapply Template Values for Multi-File Processing

The ability to reapply template values for all the substations is available from the Tools menu.

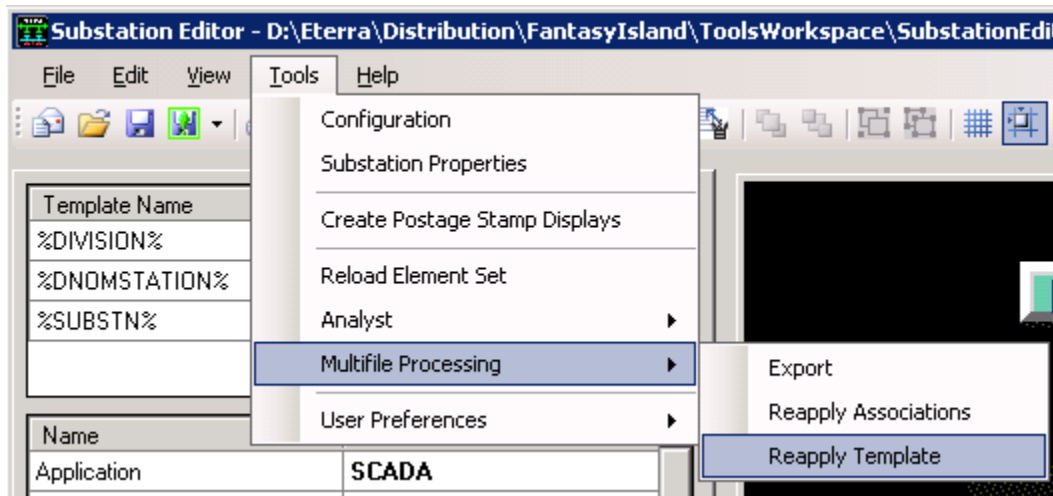


Figure 42. Reapply Template Values for Multi-File Processing

When you select Reapply Template, the Substation Editor requests the name of the directory containing the \*\_subeditor.xml files to be processed.

### 4.4.4. Substation Internals

A Substation Internals display is created by dragging the required objects onto the display, linking the objects to the appropriate composite keys for the devices and measurements in the SCADA model that will drive the display, and then making the necessary connections between objects.

Examples of Substation Internals displays are included with the sample models and the sample subeditor files.

### 4.4.5. Auto-Generate Station

To streamline the process of creating Substation Internals displays, the Auto-Generate Station function can be used to build the initial framework of the display by using the information contained in the related station external model file.

The Auto-Generate Station function is selected from the File menu. A dialog box is opened to allow

the user to select the appropriate station externals .XML file (the Converter tool input file). The external model file is then processed, and the user is asked to select the appropriate geometry (usually DETAIL or similar main geometry of the model).

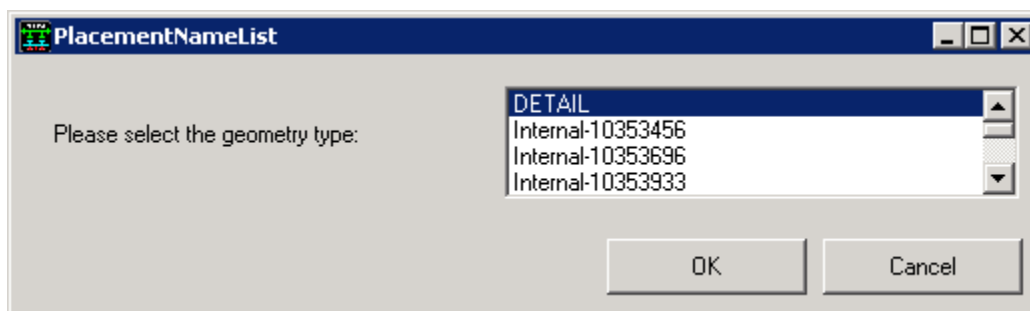


Figure 43. External Model Geometry Selection

After the geometry type is selected, a Save As dialog box is displayed to allow the user to specify the name of the substation internals model file. By default, the model file will be given the same name as the external model file.

Once the new model file is saved, the View properties for the display name and view-level template values should be entered prior to placing any more objects in the model. The save process generates a new minimum station model. Each feeder found in the external model is represented with a connection node, a feeder color, and an interface, as shown in the image below. The view is automatically created as "DEFAULT" (as shown in the 2nd image below).

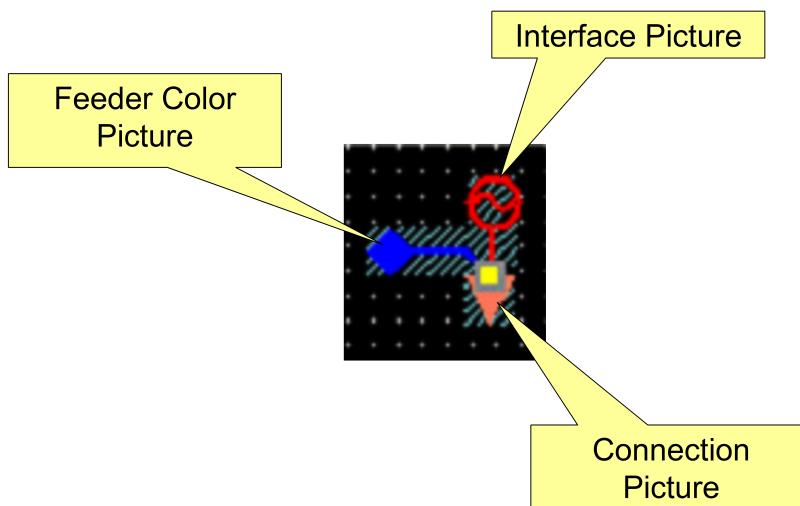


Figure 44. Auto-Generated Feeder Head

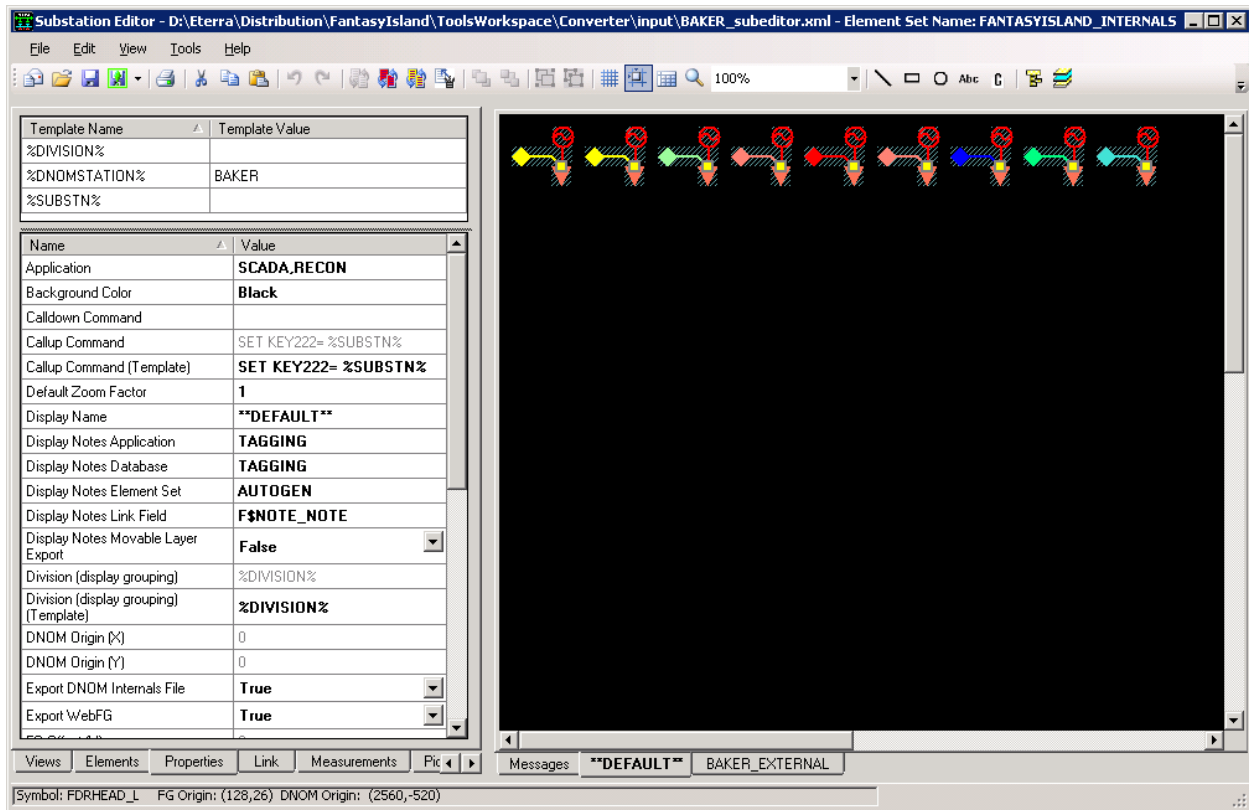


Figure 45. Auto-Generated Station Model Display

This provides the foundation for the creation of the remainder of the Station Internals model and display.

The picture symbols that are used to display these automatically created objects are defined as part of the Substation Editor configuration (General tab). Refer to the "General" section in the *Substation Editor Installation Guide*.

#### 4.4.6. Reapply Association for Placement

During the course of building a display, if the associations or default properties for an object need to be changed, it is then necessary to re-visit all the objects of this type that have been placed on the display and update the properties. To reduce the effort required to do this, the Reapply Association for Placement button is used.



Figure 46. Reapply Association for Placement Button

The properties are first updated on the Substation Editor Configuration page (usually an administrator task). Each of the affected devices on the display can then be selected, and the Reapply Association for Placement button clicked. This will update all the properties for that object, if the properties have not already been initialized.

To minimize inadvertent changes, each object can be updated separately.

#### **Note**

Choosing to reapply associations causes templates to be reapplied as well.

### 4.4.7. Reapply Associations for View

The Reapply Associations for View button can be used to iterate through all placements in a view, and reapply each placement association. This button is indented to be used only when it has been determined that it is safe to apply a certain configuration change to all the placements in the view.



Figure 47. Reapply Associations for View Button

In some instances, it can be very time-consuming to manually select each placement in order to reapply a given placement association. This button can help to speed up the process when needed.

### 4.4.8. Reapply Associations for Multi-File Processing

The ability to reapply associations for all the views in all the substations is available from the Tools menu.

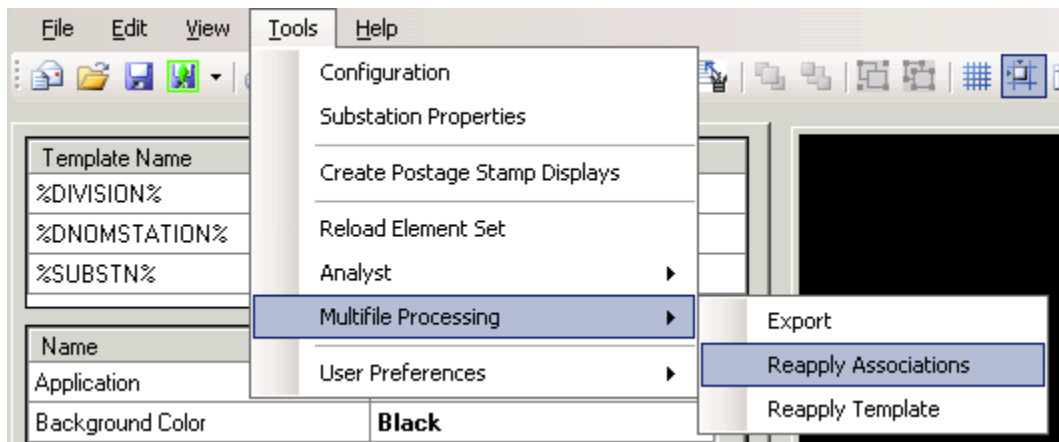


Figure 48. Reapply Associations for Multi-File Processing

When you select Reapply Associations, the Substation Editor requests the name of the directory containing the \*\_subeditor.xml files to be processed.

### 4.4.9. SCADA Model Linkage

The primary linkage to the SCADA model host is set in the Substation Editor configuration linker. The actual connection to the database is made from the Explorer Link page by clicking the Connect

button. The status indicator will show that the connection has been made. The list displays will be updated with the model contents.

Select the required station, equipment, and measurement type to show a list of relevant composite keys. Select the required composite key, and drag over the relevant object on the display to make the linkage.

## 4.4.10. Editing the Object Parameters and Linkages

Once the display has been constructed and the objects linked to the Scada database, there may be additional changes required to objects and/or parameters. This is done using the Properties, Measurements, Pictures, and Equipment tabs of the Explorer Pane.

From the Properties tab, selecting an object on the display shows the relevant table of properties. Properties that are shown in bold or unspecified (blank) can be edited.

## 4.4.11. Inserting a Station Name Label

To place a station name label on a display that will be automatically linked to the station name as defined in the template values, find the DNOMSTATION\_LABEL picture in the elements list. Drag the picture onto the display. Selecting the properties for the label will show that it has automatically been assigned the value %DNOMSTATION%. Note that for this to work, the DNOMSTATION\_LABEL object must be configured as a reference-only object associated with the DNOM substation object.

## 4.4.12. Permission Test

Permission area tests are used to control operator interaction with applications and their databases, by limiting whether a display can be called up at a workstation. In a permission area test, the user defines the access permission area ID(s) used in the display-based comparison check with the PERMIT database, or specifies where the area ID will be found.

Permission tests can be defined for the display as a whole, or for individual layers (shown in the following two images) by entering the required class names as a Boolean combination in the relevant properties field. For example:

- Class Name
- ClassName OR ClassName OR...
- ClassName AND ClassName AND ...

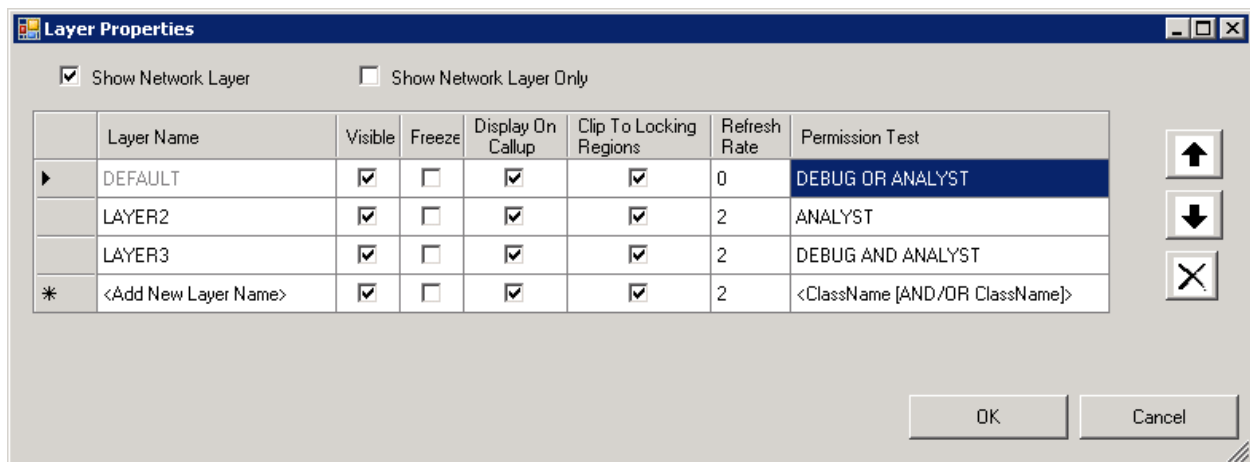
The available permission class names are defined within the application environment prior to building the display.



| Name ▲                          | Value ▲                                       |
|---------------------------------|-----------------------------------------------|
| DNOM Origin (Y)                 | 0                                             |
| Export DNOM Internals File      | <b>True</b> ▼                                 |
| Export WebFG                    | <b>True</b> ▼                                 |
| FG Offset (X)                   | 0                                             |
| FG Offset (Y)                   | 0                                             |
| FG Origin (X)                   | 0                                             |
| FG Origin (Y)                   | 0                                             |
| Permission Test                 | <b>DEBUG OR ANALYST</b>                       |
| Refresh                         | 2                                             |
| Safety Doc Application          | <b>SAFETY</b>                                 |
| Safety Doc Database             | <b>SAFETY</b>                                 |
| Safety Doc Element Set          | <b>FANTASYISLAND_INTERNAL</b>                 |
| Safety Doc Link Field           | <b>D\$DOC_DOC</b>                             |
| Safety Doc Movable Layer Export | <b>False</b> ▼                                |
| Template                        | <b>False</b> ▼                                |
| Title                           | <b>%DIS%,%DISAPP%<br/>%HOST% (%VP%) %REF%</b> |
| wC Offset (X)                   | 0                                             |
| wC Offset (Y)                   | <b>10000</b>                                  |
| X Scale                         | <b>0.05</b>                                   |

Figure 49. Defining a Permission Test as a Display Property

The Layer Properties dialog box can be displayed by clicking the Layers  button on the SSE toolbar.



The Layer Properties dialog box has a title bar with standard window controls. It contains two checkboxes: ☒ Show Network Layer and ☐ Show Network Layer Only. Below these is a table with the following columns: Layer Name, Visible, Freeze, Display On Callup, Clip To Locking Regions, Refresh Rate, and Permission Test. The table contains four rows: DEFAULT, LAYER2, LAYER3, and a template row starting with an asterisk. The DEFAULT row is selected, and its Permission Test is 'DEBUG OR ANALYST'. To the right of the table are three buttons: an up arrow, a down arrow, and a close button (X). At the bottom right are OK and Cancel buttons.

| Layer Name             | Visible                             | Freeze                   | Display On Callup                   | Clip To Locking Regions             | Refresh Rate | Permission Test                |
|------------------------|-------------------------------------|--------------------------|-------------------------------------|-------------------------------------|--------------|--------------------------------|
| ▶ DEFAULT              | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | 0            | DEBUG OR ANALYST               |
| LAYER2                 | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | 2            | ANALYST                        |
| LAYER3                 | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | 2            | DEBUG AND ANALYST              |
| * <Add New Layer Name> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | 2            | <ClassName [AND/OR ClassName]> |

Figure 50. Defining a Permission Test as a Layer Property

## 4.4.13. Defining Display Menus

The displays created within the Substation Editor can be linked to the appropriate Browser menus by defining the Display Menu properties.

The Display Menu properties are selected by clicking the button on the toolbar, which will display the dialog box pop-up.

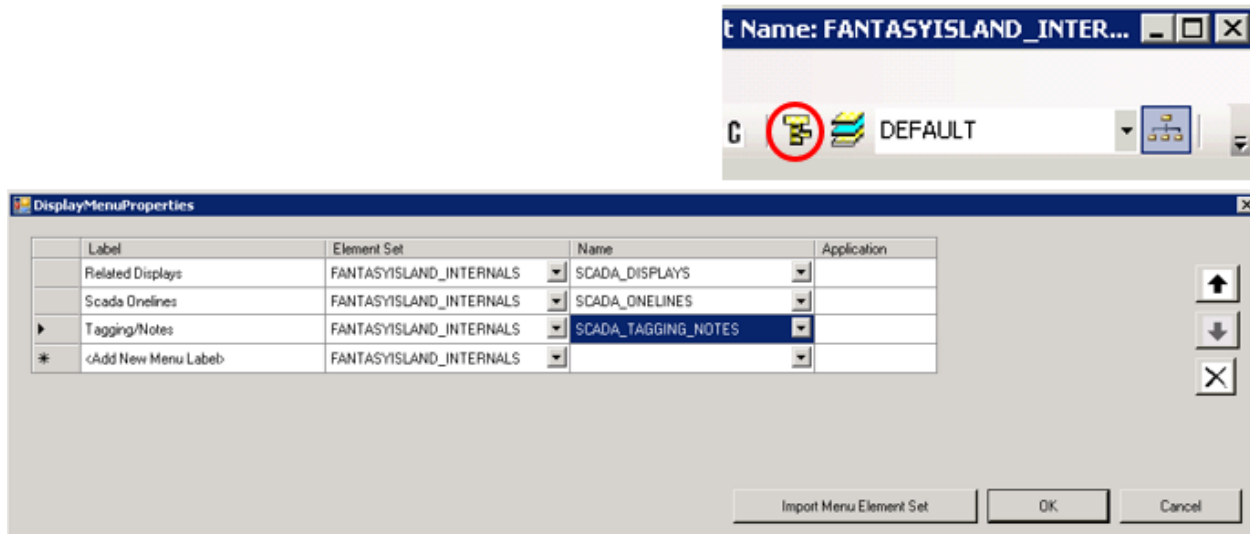


Figure 51. Display Menu Properties

The Label entry must be directly specified by entering the text into the field. The element set and menu name are selected by clicking the combo box menu.

A Display Menu entry can be deleted from the properties by selecting a field on the desired line and clicking the **X** button at the right of the data panel.

Menus can be referenced from multiple element sets. The primary element set is specified on the General Configuration tab; refer to the "General" section in the *Substation Editor Installation Guide*. An optional element set for menus can also be specified from this tab. Additional menu element sets can be imported directly by clicking the relevant button in the Display Menu Properties dialog box (shown in the image below). All element sets are assumed to be located in the same directory as the location of the primary element set. The imported element set will be automatically added to the pull-down combo box for selection.

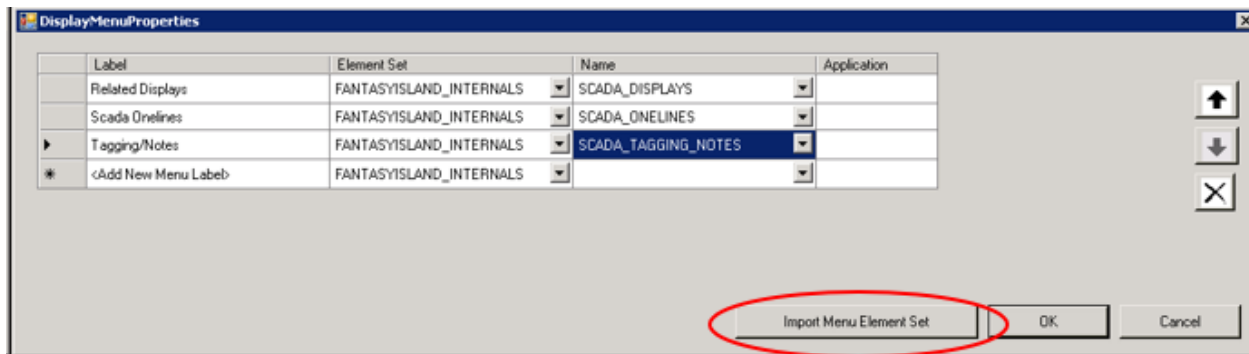


Figure 52. Importing an Additional Menu Element Set

#### 4.4.14. Substation Externals

Substation Externals displays show a combination of substation information as well as devices that are on the feeders (i.e., equipment that is both inside and outside the substation fence). An example of this is shown in the following image.

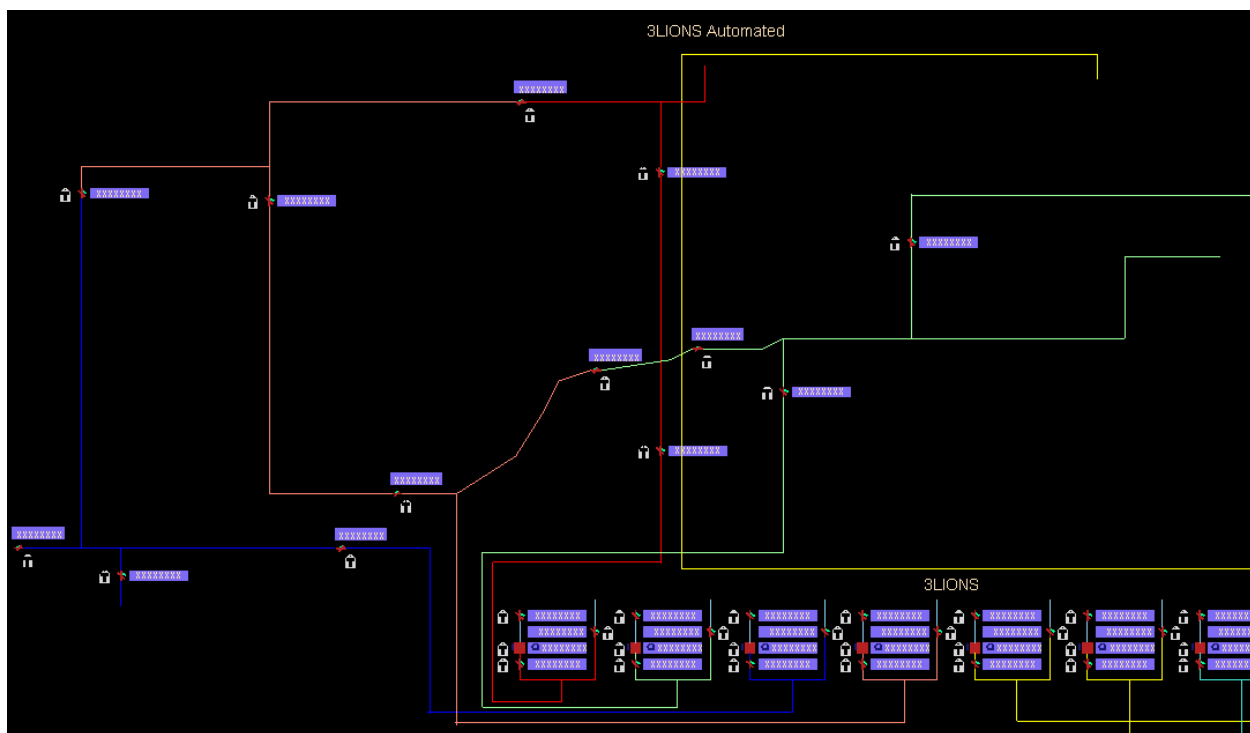


Figure 53. Example of a Substation Externals Display

The display is drawn using the same methods and tools as the substation internals, but the devices are mapped in different ways depending on their modeling source. Equipment that is outside the substation fence - such as feeder line sections, manual switches, etc. - is expected to be modeled in DNOM and is, therefore, mapped to DNOM device IDs. This information is entered on the Properties page. Equipment that is inside the substation fence is expected to be modeled in the Substation Editor (main Substation view).

Substation Externals displays are also used to add SCADA measurements to equipment that is modeled in DNOM. An example is the addition of SCADA measurements to automated feeder switches. Objects such as automated fault indications can be directly modeled on the Substation Externals display without any reference to the DNOM model, removing the need to include them in the GIS modeling process.

Substation Externals views can have a mix of reference-only placements for equipment modeled elsewhere, and regular placements for objects that are modeled by the fact that they have been placed on the Substation Externals view.

## 4.4.15. Reference Objects

Two types of objects can be used to construct displays:

- **Regular Objects:** These are display objects that are linked to network device objects that are created as part of the display building process. In the case of Substation Internals displays, these objects are then usually linked to SCADA model measurements.
- **Reference Objects:** These are display objects that are linked to existing DNOM objects. These are used to produce alternate displays based on the existing DNOM station models. The properties for these objects contain fields that link to the reference object ID (DNOM device ID) and the reference object station name (DNOM station).

Therefore, when constructing a display, it is important to select the correct types of objects. When selecting an object from the element list, the tooltip indicates that the object is a reference object (for example, xx\_RO\_xxx).

## 4.4.16. Postage Stamp Displays

"Postage stamp" displays show detailed information about equipment defined in the station externals file. These displays can be linked to both SCADA (measurement) and DNOM objects. They are initially created using the Create Postage Stamp Displays function, and thereafter may be edited like any other display.

## 4.4.17. Import Station Externals

To streamline the creation of Postage Stamp and reference-only displays, the Import Station Externals function can be used to allow the easy identification and selection of external reference objects and to directly map them to the new display.

The Import Station Externals option is selected from the File menu, allowing the user to load the required substation externals model files.

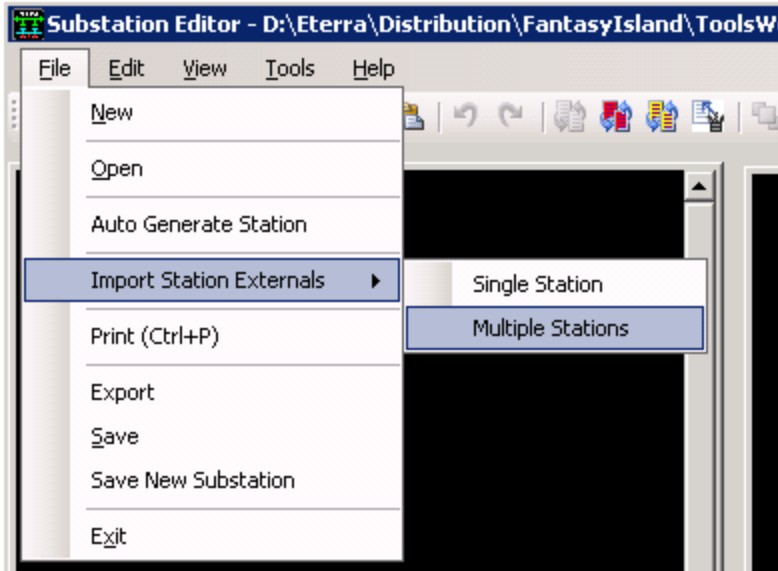


Figure 54. Import Station Externals

Once files are selected, the user is asked to select which geometry is to be displayed - usually DETAIL or a similar main geometry.

The station external model is then presented as a normal geographic Network view in the Main Pane, with a tab showing the station name + External, as shown in the following image.

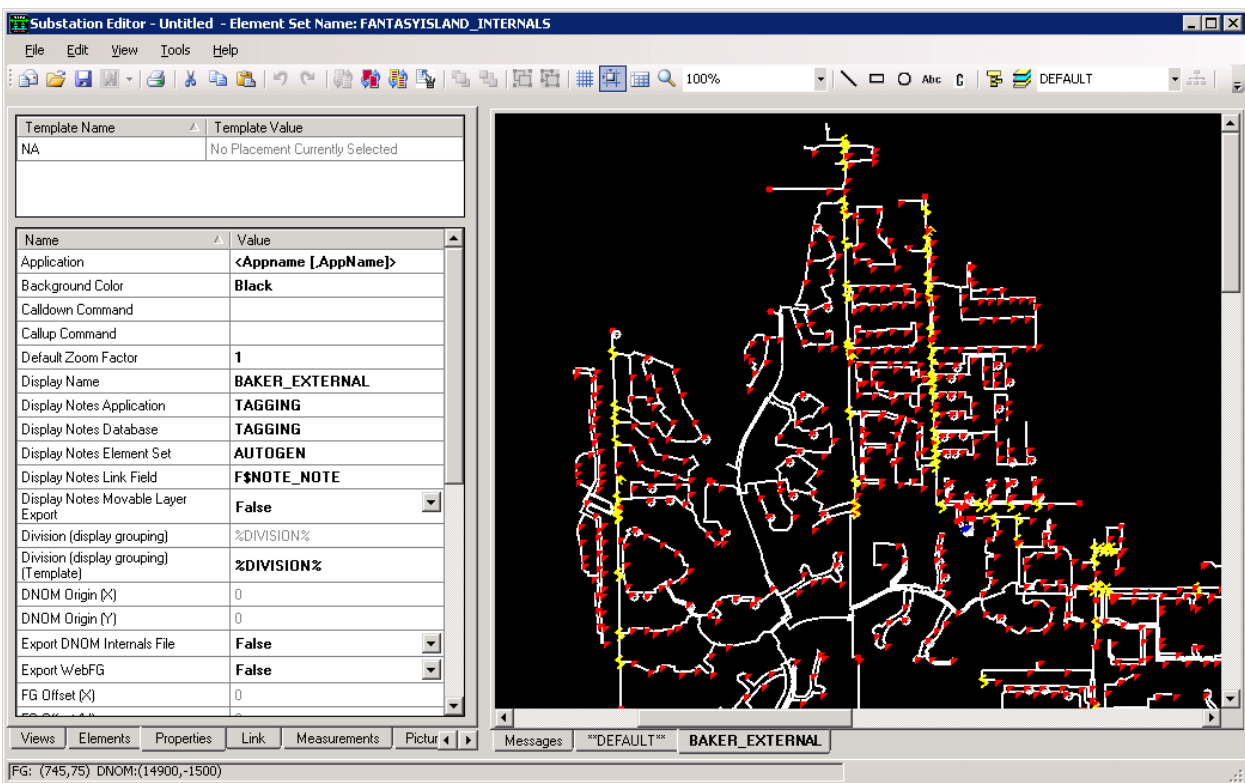


Figure 55. Imported Station Externals Model

The Station Externals display can be manipulated by using the zoom (Ctrl + mouse wheel) and scroll functions. The display can also be de-cluttered by toggling the laterals and labels on and off using the toolbar buttons or the Edit menu.

When a device on the Externals display is selected, its details are displayed in the Properties tab of the Explorer Pane.

To find a specific device on the Substation Externals display, use the Find DNOM Object function, as shown in the following image. This is invoked by Ctrl+F, or by selecting it from the Edit menu.

**Note**

The Substation Externals display must have focus before pressing Ctrl+F.



*Figure 56. DNOM Object Find*

Enter the device ID or name in the text box, and click the Find button. If the device exists in the currently loaded model, the display will be zoomed in and centered on the device. The located device will automatically be selected.

The usual method of using the imported Substation Externals display is to arrange it in one pane of the Substation Editor, with the new Reference-Only display being built in the other pane, as shown in the following image. This allows reference objects to be created by dragging the required device from the Externals display to the new Reference-Only display.

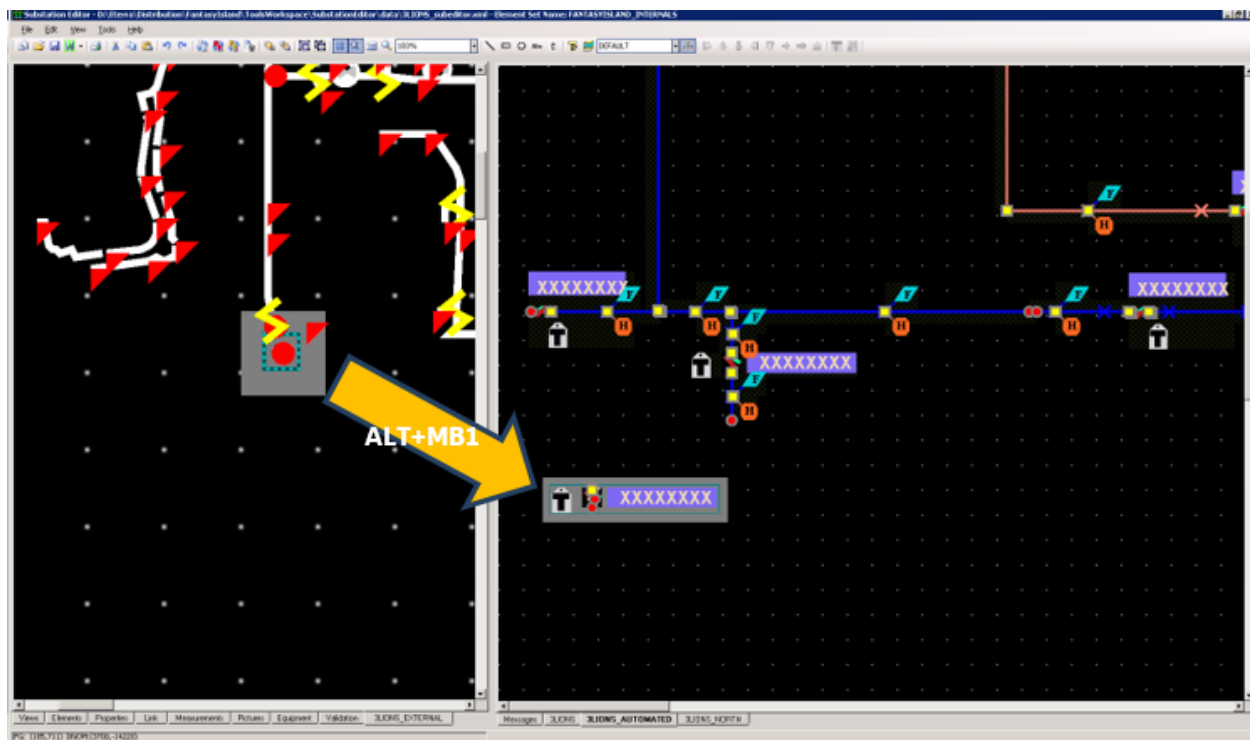


Figure 57. Creating a Reference Object Using Drag and Drop

The selected device must be dragged while holding down the Alt key and the left mouse button. When the device placement is dropped on the Reference-Only view, a reference-only object is created and all required DNOM properties (including DNOM ID and substation name) are automatically entered.

Note that this requires that an associated reference picture has been configured for the device that is dragged onto the Reference-Only view.

This mapping is defined by setting the "Assoc. Reference Picture" value on the Substation Editor Configuration Associations tab (for more details, refer to the "Associations" section in the *Substation Editor Installation Guide*).

#### **Note**

SCADA device information will be automatically entered if the newly created reference placement contains %DEVICE% as a template value. This assumes that the correct device name has been entered in the GIS model.

It is important to note that, when an external switch is dragged onto the display, the connecting line segments are also copied over, if present. This is done with the assumption that the line segments on each side of the switch will be used on the Reference-Only display. If this is not the case, then the line segments should be deleted.

The mapping of symbols to external objects used to draw the Substation Externals display is defined in the Substation Editor Configuration External Associations tab (refer to the "External Station Configuration" section in the *Substation Editor Installation Guide*).

## 4.4.18. Creating Reference Objects from Internal Devices

In a similar manner to that described in the previous section, it is possible to use the drag-and-drop method to create reference-only objects from substation internals devices.

As shown in the image below, the Substation Internals view and the Reference-Only view are arranged side by side in the two panes. The required object is selected on the Internals view, dragged while holding down the Alt key and the left mouse button, and then dropped on the Reference-Only view. The Reference-Only object is automatically created, and all the necessary properties are automatically entered.

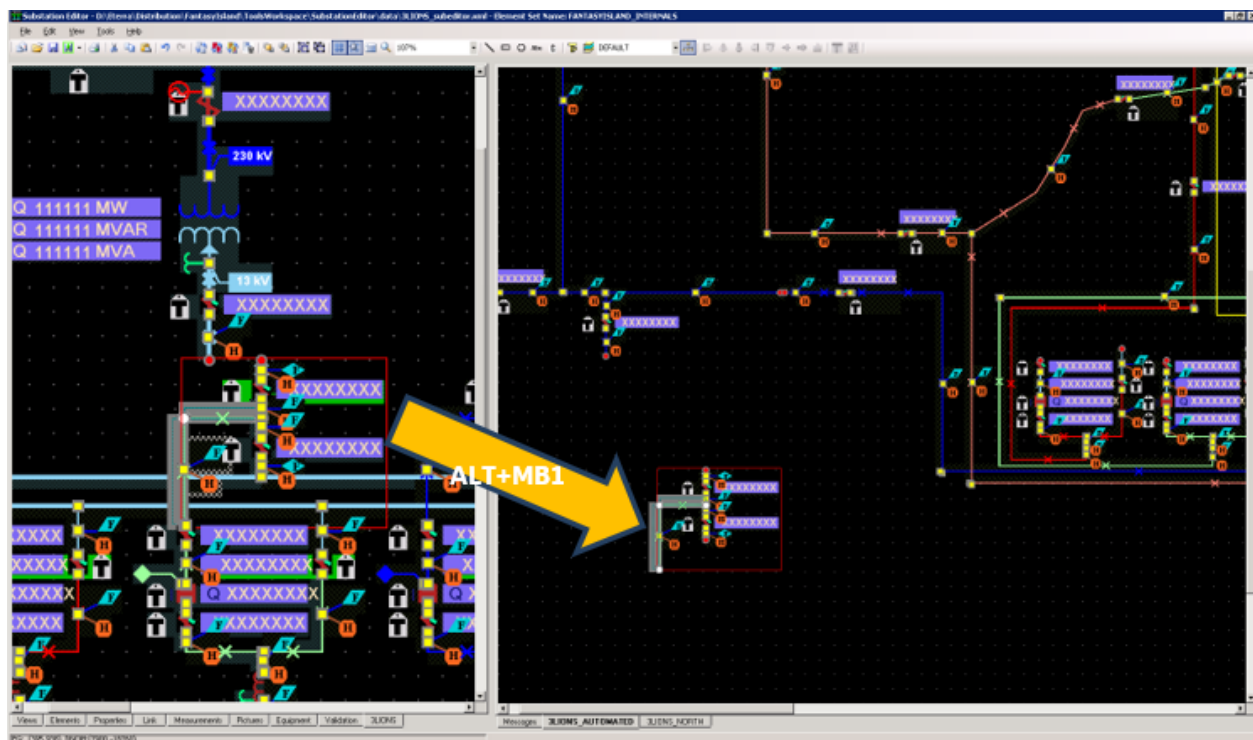


Figure 58. Creating a Reference Object from an Internal Device

The mapping between regular network objects and reference objects is defined by setting the "Assoc. Reference Picture" value in the Substation Editor Configuration Associations tab (refer to the "Associations" section in the *Substation Editor Installation Guide*).

## 4.4.19. Creating SCADA Measurement Mapping to Measurement Point

Before creating a mapping between the Analog measurement and the Measurement point, ensure that all the elements are properly loaded and views are created, and that a SCADA one-line diagram or the display layout in Display Builder (.dset file) is available.

1. In the Substation Editor, navigate to Tools > Configuration. In the Add Association drop-down list, select the analog element that needs association.



2. Check the Create DNOM symbol. If the label needs to be displayed on the client, check the Create Label box.
3. In the Measurement Info panel:

- In the MEAS Type drop-down list, select a proper type (i.e., MeasAmpA), which measures the "A" Phase current.

Select the corresponding MEAS Type depending on the analog element chosen. In this case, ALG\_LIM\_NS\_AAMP is the analog phase "A" current measurement.

- Since the measurement is analog, it belongs to the Analog measurement group. Therefore, the MEAS Group field should be AnalogMeasurementGroup.
- In the Default Assoc drop-down list, select the measurement point element. The measurement point measures voltage, current, power factor, etc. at the node where the element is connected.

**i Note**

To successfully create the measurements, ensure that the Create Meas box is always checked. Depending on the Analog element, select the proper MeasType and MEAS Group.

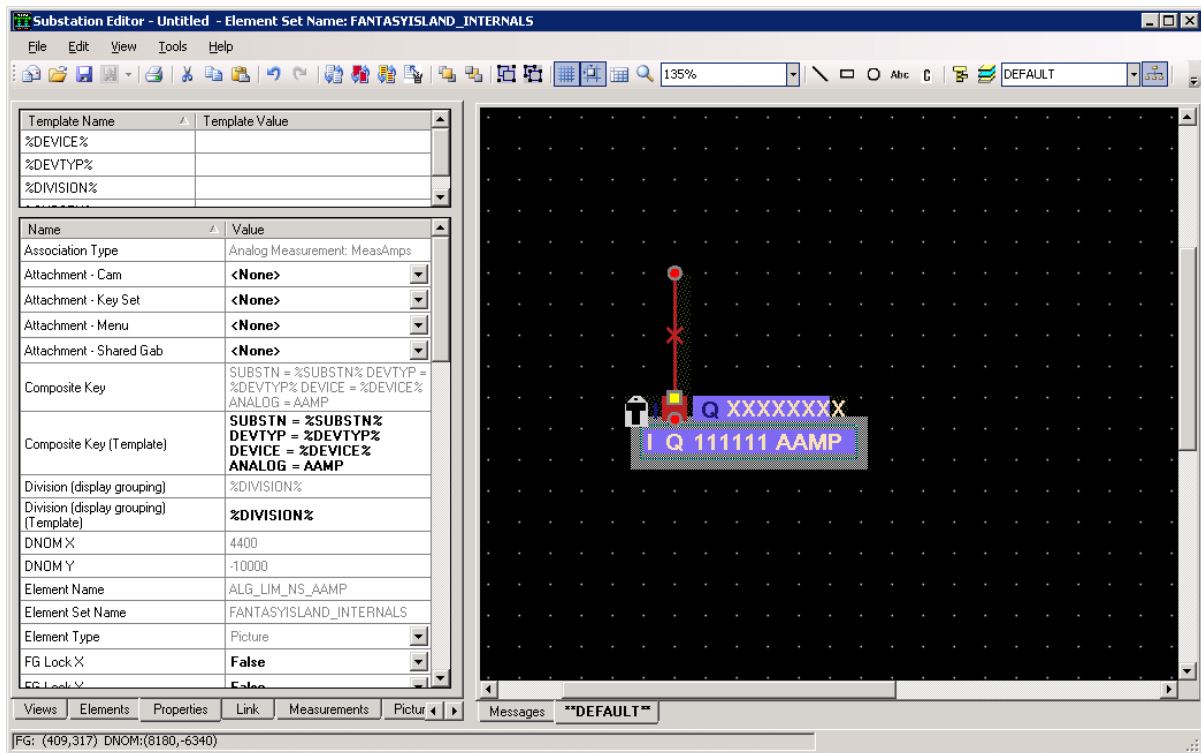
4. Verify that all the above-mentioned fields are properly entered.
5. Enter the Composite key in the following standard format for analog measurements:  
  
SUBSTN=%SUBSTN% [Space] DEVTYP=%DEVTYP% [Space] DEVICE=%DEVICE% [space] ANALOG =  
AAmp
6. Enter the SCADA site name in the SCADA Data Source field.
7. Select the appropriate SCADA Model Name from the drop-down list.
8. Enter the area name in the DIVISION field or create a template %DIVISION%.
9. In the Name/Value grid on the right side of the display, enter the Measurement Composite ID as shown in the example below. The Measurement Composite ID connects with SCADA Measurement information.

%SUBSTN%.%DEVTYP%.%DEVICE%.AAmp

| Name                                                                  | Value                               |
|-----------------------------------------------------------------------|-------------------------------------|
| Measurement: Attached To Device                                       |                                     |
| Measurement: Attached To Device Type                                  |                                     |
| Measurement: Attached To Effective Device                             |                                     |
| Measurement: Base Value                                               | 1                                   |
| Measurement: Composite ID                                             | %SUBSTN%. %DEVTYPE%. %DEVICE%. AAMP |
| Measurement: Country                                                  |                                     |
| Measurement: Default Quality                                          | 0                                   |
| Measurement: Default Value                                            | 0                                   |
| Measurement: Division Level Measurement                               | False                               |
| Measurement: Enable Measurement Recency Check                         | False                               |
| Measurement: FLIP Analog Value                                        | False                               |
| Measurement: Group                                                    | AnalogMeasurementGroup              |
| Measurement: Location Description                                     |                                     |
| Measurement: Measurement Utilized for Triggering Network Applications | True                                |
| Measurement: Measurement Utilized in Network Applications             | True                                |
| Measurement: Municipality                                             |                                     |
| Measurement: Network Device Id                                        |                                     |
| Measurement: Network Device Name                                      | %DEVICE%                            |
| Measurement: Permission Area Id (8 character max.)                    |                                     |
| Measurement: Phase                                                    | A                                   |
| Measurement: Simulate Only                                            | False                               |
| Measurement: State                                                    |                                     |
| Measurement: System Level Measurement                                 | False                               |
| Measurement: Type                                                     | MeasAmps                            |
| Measurement: Units                                                    | Base                                |

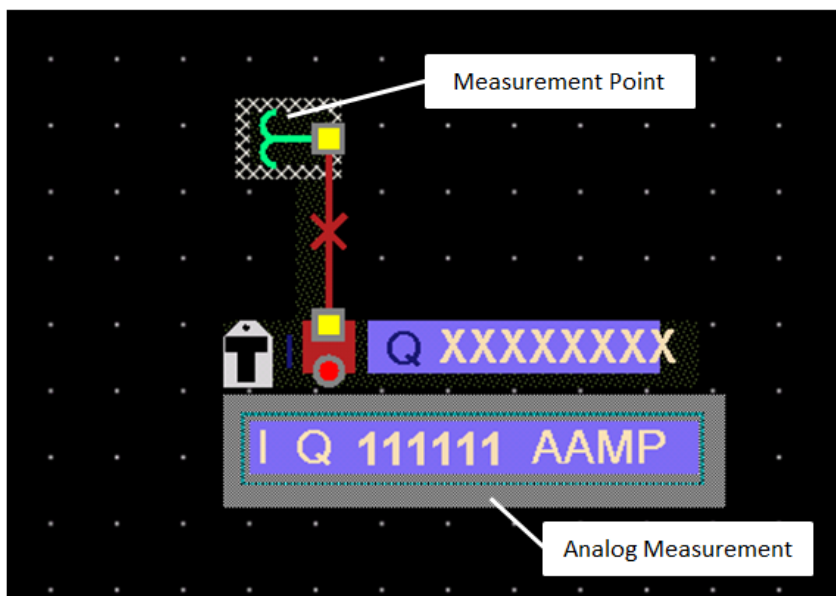
Depending on the measurement, change the measurement unit. Base=0, Kilo=1, Mega=2, PerUnit=3.

10. Once all the fields are properly filled, click Apply and close the configuration.
11. In the Substation Editor's Elements tab, select on the left panel the element to add measurements to. Refer to the example below that demonstrates a selected breaker with the current measurements.
12. Drag and drop the element on to the right side scratch pad.
13. Drag and drop the Analog current measurement element that was selected in step 1.
14. Fill in all fields, including DEVTYPE, DEVICE, SUBSTN, and DIVISION. This information is obtained from either the SCADA one-line display or the Display Layout from the Display Builder.



15. To map the measurement point and the analog measurement element, do the following:

- Click on the Analog measurement.
- Press and hold Ctrl+A.
- Click on the measurement point (for example, MEASPT\_V).
- White cross-hatching around the measurement point indicates that the analog measurement has been mapped with the measurement point, as shown below.



- Release Ctrl+A.
- The "Measurement: Attached To Device" property of the analog measurement is populated with the ID of the associated measurement point, as shown below.

|                                           |                                    |
|-------------------------------------------|------------------------------------|
| Measurement: Attached To Device           | 70a1589c-1fa1-4fa2-a75a-0fe367f19f |
| Measurement: Attached To Device Type      |                                    |
| Measurement: Attached To Effective Device |                                    |

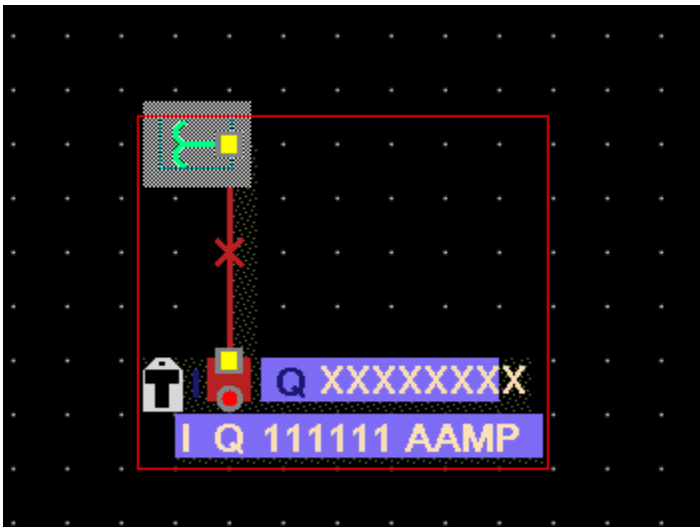
Properties Views Elements Link Measurements Pictures Equipme

### **Note**

The analog measurement mapping works only in this order.

To clear the mapping, select the measurement and press Ctrl+S. This works for all measurement types.

- Select both elements and click Group. The elements are rubber banded by a red box as shown below. When a measurement point is selected, a small white bubble appears across the element indicating that the devices are mapped.



## 4.4.20. Validation

The Validation function is used to provide the user with an indication of potential errors or omissions in a station model file before it is exported. This greatly reduces the amount of effort required to produce working displays and models.

To perform a validation check, select the Validation tab and click the Validate/Revalidate button. The results will be shown in a list, with a summary of the errors, warnings, and informational messages

at the bottom. Two check boxes are shown at the bottom of the page; check these boxes as required to indicate whether the validation should be performed for a Habitat or a Control export.

It may be necessary to shift the pane splitter bar to show the whole of the error message.

Selecting a line on the validation error list will highlight the relevant device on the main display view, as shown in the image below. The properties for this device can now be displayed by selecting the Properties tab, and entering or correcting the relevant data.

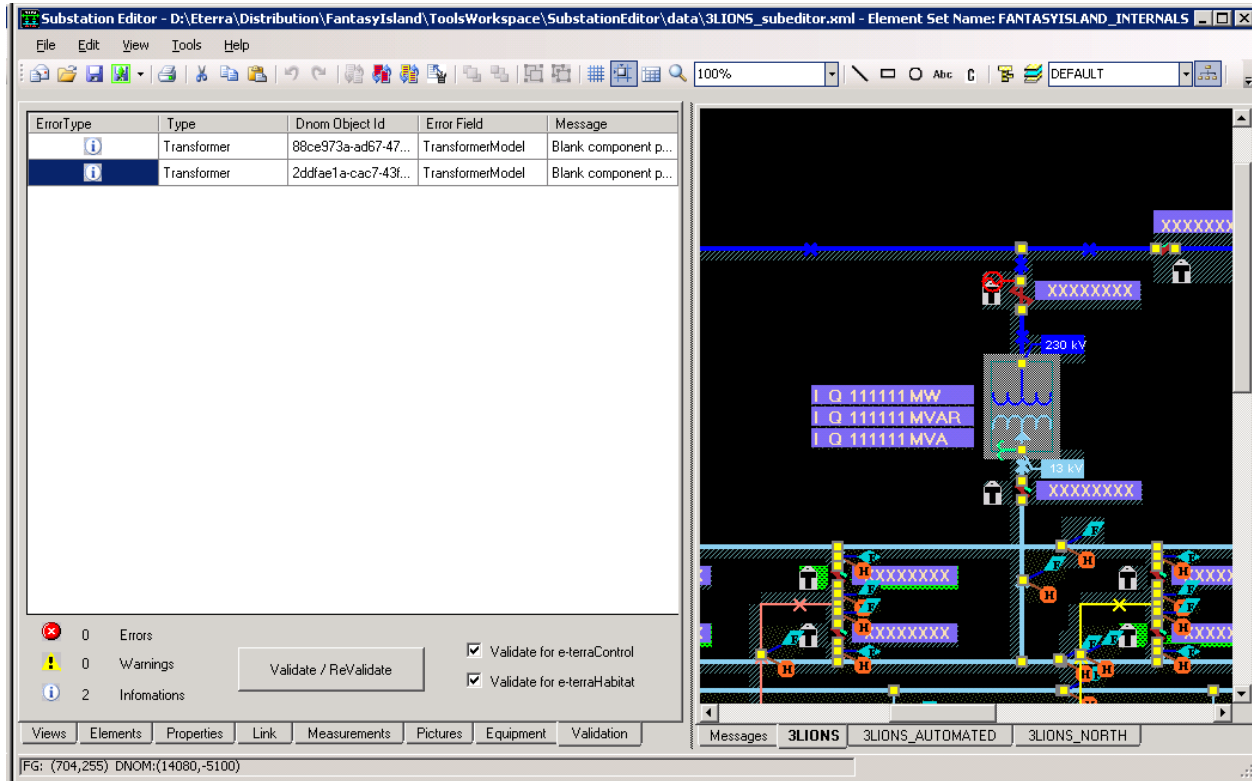


Figure 59. Validation Display

The validation process checks for the following errors in the model file:

- Blank DNOM object ID, station name, or object type in reference-only objects
- Blank model names (switch model for a switch device, transformer model for a transformer, etc.)
- Blank names for any DNOM object, except for lines and jumpers
- Well-formatted key pairs (for example, A=B C=D)
  - Have fewer than two "%" in any composite key, name
  - For editor placement, verify the composite key
  - Verify CAM, feeder color measurement, and halo color measurement
- For measurement objects, a blank composite ID, or one that is not well-formed with four parts
- Blank view names

- SCADA model name for status points (POINT records)
- SCADA area is required when exporting HDB files. The area is the "Area of Responsibility" the placement is related to. This is not required for Control exports.

## 4.4.21. Exporting a Display

Once a display is completed (and saved), it is exported from the Substation Editor to be processed into the main system models and display files.

Clicking the "Export" button on the toolbar, selecting the "Browser Display" export type, and then clicking the "Export" button on the toolbar generates a file named "editorfilename\_EDITOR\_DSET.DDL". The display set contains a display for each of the views contained in the subeditor file.

The file will be exported to the folder defined in the export path. If this is not defined on the command line or in the shortcut, it can also be defined by using the Tools > User Preferences menu and selecting "Edit/View Export Directory". Once this is set, it will be stored in the preferences file and re-used thereafter.

This DDL file will then need to be loaded into the Display Builder in order to be compiled into a Browser display.

### 4.4.21.1. Safety Documents Layer Notes

If the property "Safety Doc Movable Layer Export" is set to True, the Safety Documents layer is included in the generated DDL file.

To compile the generated DDL file, ensure the correctness of the following properties:

- Safety Doc Application
- Safety Doc Database
- Safety Doc Link Field

For more information regarding the values of these properties, refer to the "General" section in the *Substation Editor Installation Guide*.

The Safety Layer contains references to the element set DDL file (for example, `fantasyisland_internals_eset.ddl`, where *fantasyisland\_internals* is the element set name).

## 4.4.22. Exporting Station Internals

If the "Station Internals" export type is selected from Tools > User Preferences > Export Types, then clicking File > Export will write two or more XML files to the same folder where the Substation Editor file is saved.

For example, the export files for the Stationxx subeditor file are:

Stationxx\_INTERNALS.XML

and one or more

stationxx\_PLACEMENTS\_geoname.XML

(if a reference-only display has been defined).

The export of files can also be performed by a command-line operation to allow batch mode operation.

### 4.4.23. Control Export

Clicking the File > Export menu will write out a Control import file containing all SCADA information directly pertaining to the station being modeled if the "Control" export type is selected from Tools > User Preferences > Export Types. For example, the export file for the BAKER subeditor file is BAKER\_SMP.XML. This file can be imported into Control from the File > Import menu item.

### 4.4.24. Habitat Export

Clicking the File > Export menu will write out a Habitat import file containing all SCADA information directly pertaining to the station being modeled if the "Habitat" export type is selected from Tools > User Preferences > Export Types. For example, the export file for the BAKER subeditor file is BAKER\_HDB.TXT. This file can be imported into Habitat using the hdbimport command.

## 4.5. Set User Preferences

Different user preferences can be set through Tools > User Preferences. These user preferences are saved into local application data files when the Substation Editor exits and are then reloaded again when the Substation Editor is re-started.

[image101] | *image101.png*

*Figure 60. User Preferences - Export Types*

The available user preferences are:

- Edit/View Import Directory: Sets the import directory for import operation
- Edit/View Export Directory: Sets the export directory for the export operation
- Export Types: Sets different export types for the export operation

## 4.6. Converting and Viewing a Model

The exported internals and placement XML files are included in the conversion process to generate the .MOD file that is loaded into the ADMS Network view. The Global\_Internals.xml file is also required (refer to the "Global Internals" section in the *Substation Editor Installation Guide*).

### To convert and view the model:

1. Move the files to the same folder that contains all the other standard inputs for the Converter - for example, the station external file, stationxx\_sub.xml (or .csv).
2. Start the Converter by selecting it from the **Start > Eterra > Distribution > Tools > Modeling Tools menu**.
3. Set the Converter options if necessary.

Confirm that the input and output folders are correctly set.

4. Select the required stationxx.xml file, and click Convert.
5. Copy the resultant .mod file to the models folder (for example, Eterra\Distribution\FantasyIsland\models\RT).
6. Start the client.
7. Manually navigate to the Substation Internals display by using the command:

```
netview://rt/view/geometryxx/stationxx
```

Where geometryxx is the geometry name (for example, DETAIL).

8. The Reference-Only display can be viewed by using the command:

```
netview://rt/view/geoname/stationxx
```

Where geoname is the name of a reference-only display.

## 4.7. Printing Displays

The Substation Editor provides the standard Windows printing methods. The image sent to the printer is the display visible in the Main Pane. The background color of the display is inverted for the printed output. When printing, the current scale of the display is used. Only one page is printed. If the display needs to be printed with large scale on many pages, this needs to be done manually.

The methods of invoking a print display are:

- Clicking the Print button on the toolbar
- Selecting the Print option via the File menu
- Pressing Ctrl+P

Each of these methods displays the standard Windows Print dialog box, allowing the printer and printing options to be selected.





Figure 61. Display Printing Methods

## 5. Creating Displays

### 5.1. Creating a Substation Internals Display

The following is an example of how to create a Substation Internals display, using the tools of the Substation Editor.

It is assumed that the Substation Editor has been properly installed and configured, and that the required element sets, etc., have been loaded. For configuration guidance, refer to the *Substation Editor Installation Guide*.

1. Information required prior to commencing: station externals model file (for example, RAVEN.xml. You can find the example file at the folder: %IDMS\_ROOT%\ToolsWorkspace\Converter\Input), substation one-line diagram, or display layout. Basic parameters for substation equipment (switches, transformers, etc.).
2. Start the Substation Editor, and select Auto-Generate Station from the File menu.  
  
If you have made any changes to the current display, you will be asked whether the changes need to be saved.
3. Select the Station Externals file.
4. Select the required geometry (usually DETAIL or a similar main geometry).
5. The system will save the Substation Editor file. Verify that the Substation Editor file name is correct, and remove "EXTERNAL" from the name as appropriate.
6. The foundation of the substation internals model (i.e., feeder heads) will be created. Select the Default tab to view it.
7. Go to the Properties page on the Explorer Pane.
  - Check that %DNOMSTATION% and DISPLAY NAME have been entered correctly.
  - Check that the APPLICATION is set to SCADA, RECON.
  - Enter a template value for %SUBSTN% (SCADA name). This is the eight-character name used in the SCADA system.
  - Enter a template value for %DIVISION%.
8. Select all the feeder head symbols by rubber-banding, and move them to the bottom of the display.
9. Insert a Station Name label by dragging the DNOMSTATION\_LABEL picture from the Elements tab.

Add a text placement as the station name/display name to be used as the header on the SCADA display. Normally, these two placements can be grouped and placed together.

#### Note

Place a DNOMSTATION\_LABEL (or DNOMCAPACITOR\_LABEL) picture on each view. Using

this element, the modeler defines the external "Station Name" by which the substation is identified in the external model files. The Station Name is also used to name the placement files exported by the Substation Editor. The naming convention for the placement files is:

```
<DNOMXXX_LABEL StationName>_PLACEMENTS_<View Name>.XML
```

10. If there is an existing station internals display that has a similar layout, use it as a starting point by copying it to the current display.

The best way to do this is to open a second instance of Substation Editor, and open the station\_subeditor.xml file for the station to be copied from. Use the rubber-banding function to draw a rectangle around the required area of the substation display and then click Edit, Copy. Move back to the new Substation Internals display and click Edit, Paste.

It is usual to only copy the buses and the upstream devices.

**Note**

Make sure that the appropriate template values are changed (i.e., the station name for the copied elements).

11. On the Elements tab of the Explorer Pane, select "Show Only Network Objects".

Drag the required pictures from the list onto the Substation Internals display, for the devices downstream of the buses for the first feeder.

Arrange the pictures so that the appropriate connections are made to the busses, shown by the small yellow boxes on the connection points.

12. Using the Properties tab, identify each of the feeder heads that have been created by the auto-generate process.

Select the first feeder to be placed on the internals diagram, usually the left-most feeder.

13. Place the feeder color picture for the first feeder (diamond shape) on the high side of the feeder breaker (refer to the image below).

Ensure that the connection indication symbol is displayed.

14. Place the feeder connection picture on the outlet line segment of the feeder.

Ensure that the connection symbol is displayed.

15. Delete the interface picture for the first feeder, as it is no longer required.

**Note**

The interface picture is needed to energize the feeder. If the upstream devices have been added in a previous step and an interface has been added, then the auto-generated interface should be deleted. Normally, an interface is placed on the high side of a transformer to simulate the transformer being energized. All connected equipment on the low side of the transformer will then be marked as energized.

16. Check the properties of each of the pictures that make up the feeder devices, and enter the feeder number as required.
17. Go to Validation and click the Validate/Revalidate button.
18. For any errors or warnings, select the message to highlight the affected device on the display, then select the Properties tab and enter or correct the relevant fields to address the problem.
19. Repeat the exercise for the remaining feeders.
20. Update the properties for the power transformers and other devices within the substation.
21. Perform a final validation check.
22. Click the Save/Export button on the toolbar.
23. Select the folder for the output files. The folder will now be populated with several files:
  - Station.issg: This file contains data about station internals and station placements files output by the Substation Editor.
  - Station\_EDITOR\_DSET.DDL: This file is used by the display builder to build a Browser display.
  - Station\_INTERNALS.XML: This file is used by the Converter tool to create the station model file (.mod).
  - Station\_PLACEMENTS\_geoname.XML: This file will be created for each reference-only display that has been defined.
  - Station\_SMP.XML: This file is imported into Control to build the Scada database.
  - Station\_HDB.TXT: This file can be imported into Scada to build the Scada database.
  - global\_internals.xml: This file is used by the Converter tool to create the station model file (.mod).

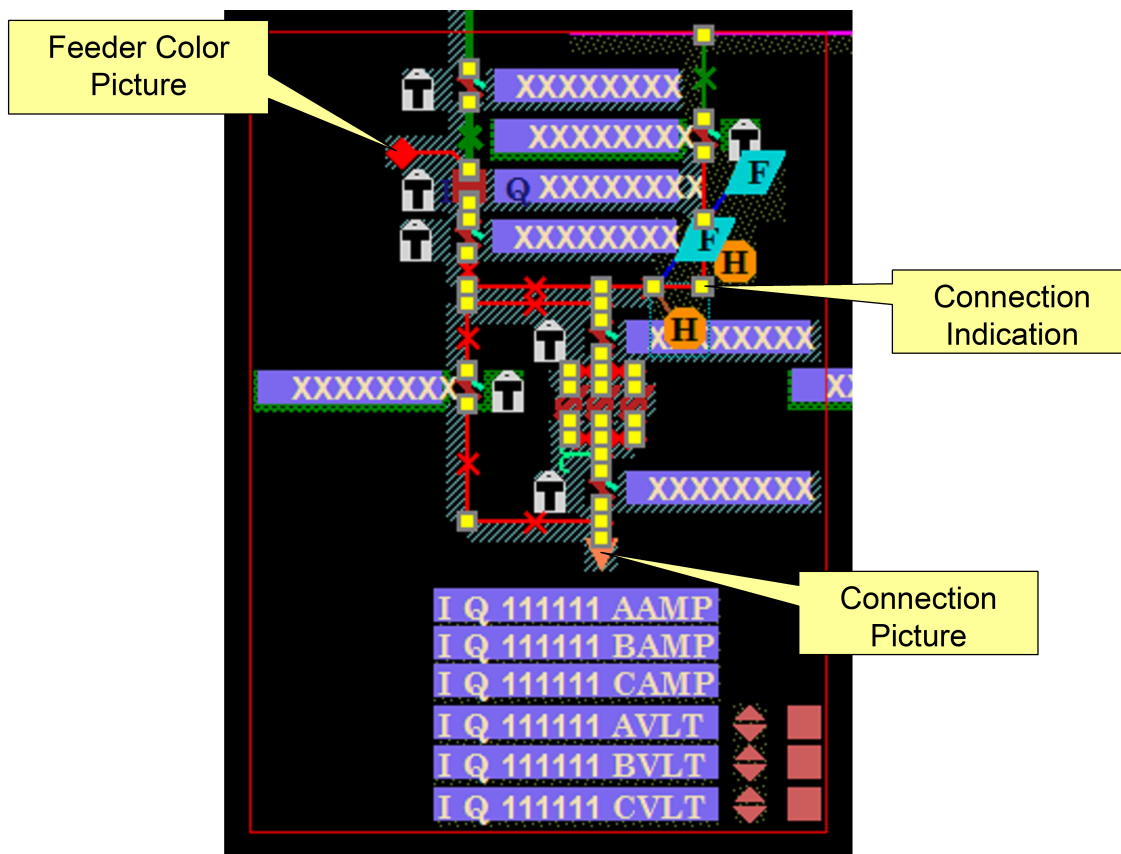


Figure 62. Typical Layout of Feeder Internals

Converting the Substation Editor output to generate the DMS "Mod" file:

1. Copy the output files (Station\_internals.xml, Station\_PLACEMENTS\_geoname.xml) to the nominated input file for the Converter. Also check that the folder contains the Global\_internals.xml.

If a Global\_internals.xml file is required, refer to the "Global Internals" section in the *Substation Editor Installation Guide*.

#### **Note**

Rather than copying the files to the Converter's input folder, the Substation Editor's export folder can be configured so that the exported files are written directly to the Converter's input folder (i.e., the two applications use the same work area).

2. Start the Converter by selecting it from the **Start > Eterra > Distribution > Tools > Modeling Tools menu**.
3. Set the Converter options if necessary. Confirm that the input and output folders are correctly set.

4. Select the required stationxx\_sub.xml file, and click Convert.
5. Move the resultant .mod file to the models folder (for example, Eterra\Distribution\FantasyIsland\models\RT).
6. Start the client.
7. Manually navigate to the Substation Internals display by using the command:

```
netview://rt/view/geometryxx/stationxx
```

Where geometryxx is the geometry name (for example, DETAIL)

8. The Reference-Only display can be viewed by using the command:

```
netview://rt/view/geoname/stationxx
```

Where geoname is the name of a reference-only display.

### 5.1.1. Modeling Equipment Groups

When modeling substation internals, large groups of related equipment will often need to be modeled.

An example of this is a feeder bay. A given substation can have a number of feeder bays: a large substation usually has up to 20 bays. Each feeder bay can contain many related pieces of equipment. A feeder bay with six switches, one circuit breaker, three regulators, and various measurements.

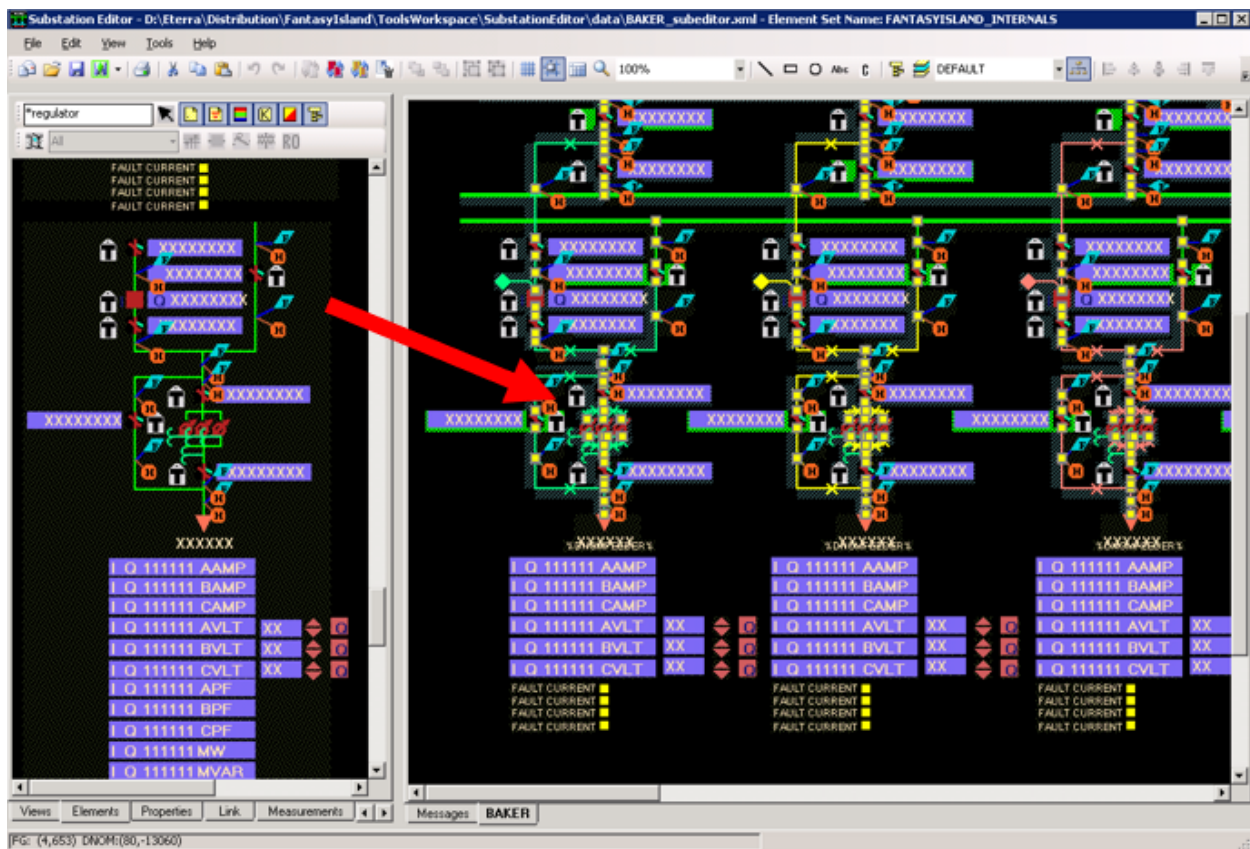


Figure 63. Feeder Bay - Example of Grouped Equipment

It would be very tedious and labor-intensive to model this one device at a time. In order to make this work manageable and to reduce the chance of manual data entry mistakes, common groups of equipment can be pre-configured so that when a substation is modeled, only a small number of groups need to be placed rather than repeatedly placing individual devices.

Note that groups can easily be changed after they have been placed. If the actual feeder bay being modeled does not have a particular measurement or control, the ungroup function can be used to make modifications, removals, or additions. The equipment can then be "re-grouped". This functionality removes the need to pre-configure all possible variations of equipment groups. A smaller number of groups can be created, which might include equipment that, in some cases, will be removed as part of the modeling.

Another significant advantage of modeling equipment using pre-configured groups is the fact that the groups can be designed so that only a minimum number of property values have to be entered. In some cases, only the feeder number (the ID of the feeder group) has to be specified.

The following image shows the property pane where the value for %FEEDER% has been set to 5467. All composite keys, device names, etc. are filled in from this one value, along with common property values such as the substation name and permission area.

| Template Name | Template Value |
|---------------|----------------|
| %DIVISION%    | CTL            |
| %FEEDER%      | 5467           |
| %SUBSTN%      | BAKER          |

| Name                                               | Value                                                              |
|----------------------------------------------------|--------------------------------------------------------------------|
| Associated Switch Model                            | SW_GI                                                              |
| Association Type                                   | Switch                                                             |
| Attachment - Cam                                   | <None>                                                             |
| Attachment - Key Set                               | <None>                                                             |
| Attachment - Menu                                  | <None>                                                             |
| Attachment - Shared Gab                            | <None>                                                             |
| Boundary Station                                   |                                                                    |
| Branch: FromNode Id                                | dfafec4b-ff1c-4fa8-afdd-96d44ec                                    |
| Branch: ToNode Id                                  | e9a3fbd1-7155-454e-85d1-5ba50                                      |
| Composite Key                                      | SUBSTN = BAKER DEVTYP = SW DEVICE = 5467H POINT = STTS             |
| Composite Key (Template)                           | SUBSTN = %SUBSTN%<br>DEVTYP = SW DEVICE = %FEEDER%<br>POINT = STTS |
| Composite Switch - Link ID                         |                                                                    |
| Compute Flows (false = collapse switch internally) | True                                                               |
| Division (display grouping)                        | CTL                                                                |
| Division (display grouping) (Template)             | %DIVISION%                                                         |
| DNOM X                                             | 1720                                                               |
| DNOM Y                                             | -9760                                                              |
| DSO Override                                       | NotSet                                                             |
| Element Name                                       | SW V                                                               |

Views Elements Properties Link Measurements

Figure 64. Property Pane for Feeder Group

The Substation Editor relies on graphical elements defined in the Browser Display Builder tool. In addition to the commonly used picture and symbol definitions, the display builder also supports the creation of "nested" or "grouped pictures". The Substation Editor takes advantage of these nested pictures by reading them from the element set and breaking them up, or exploding them out, when the user places the group in the Substation Editor. The Substation Editor then groups the individual placements back into a loosely coupled group that can easily be modified.

The following image shows the group used in the example above in the display builder.



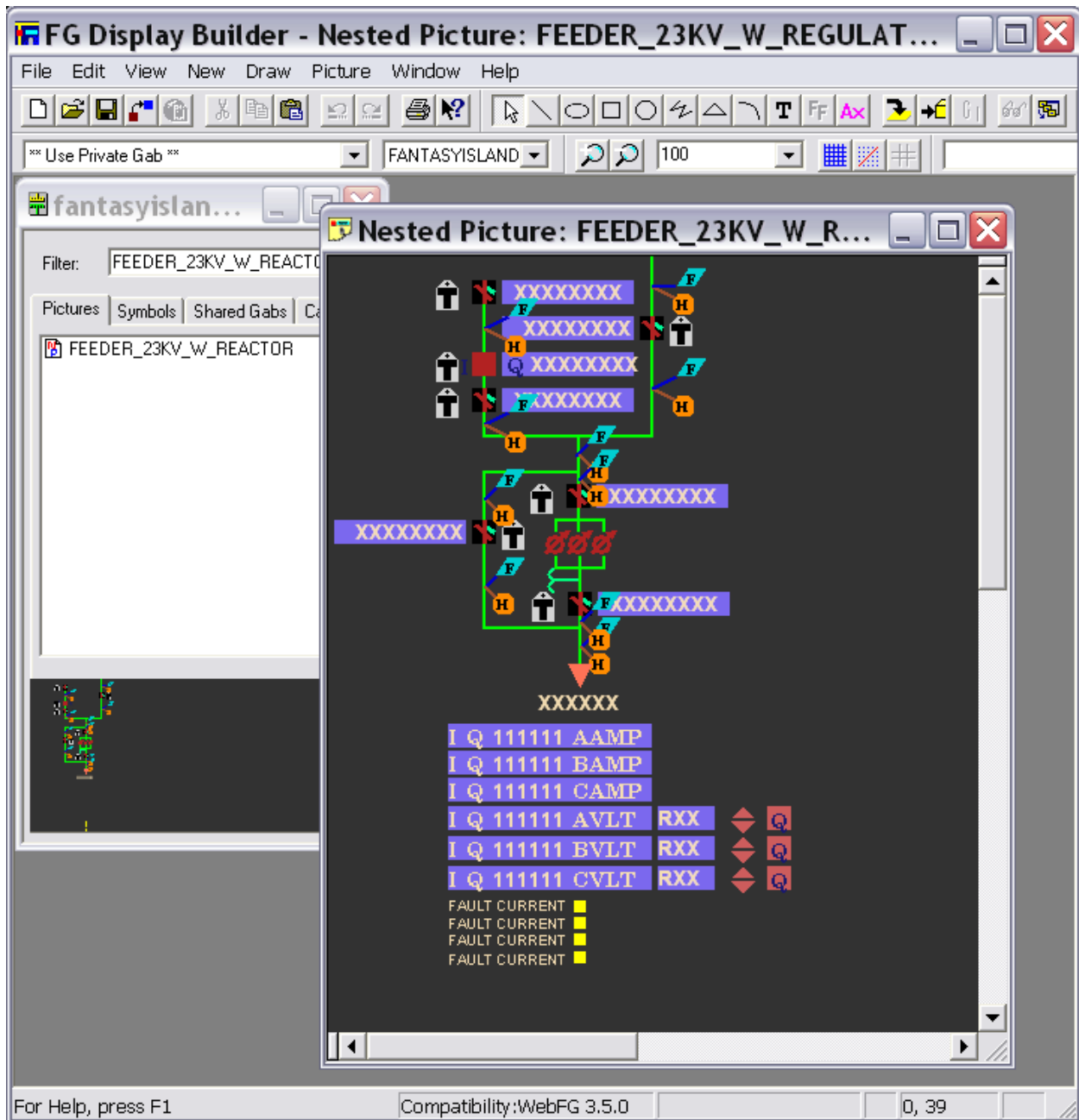


Figure 65. Display Builder - Grouped Pictures

A nested picture can be added to the element set from the New menu by clicking the Nested Picture menu item. It is recommended to give the group a meaningful name that can easily be filtered on to make it easy for the end user to find and use the group.

A nested picture can contain any number of picture placements, symbol placements, and primitives such as poly lines. The group should be laid out exactly as it should appear in the substation model and eventually in the SCADA one-line.

For information about how to use the display builder, refer to the *Display Builder User Guide*.

When a picture is placed on display, it needs to be linked to the appropriate database record instances by specifying a composite key. For example, a picture representing a switch could have a composite key such as:

```
SUBSTN = BAKER  
DEVTYP = SW  
DEVICE = 5467H  
POINT = STTS
```

In order to create useful, reusable equipment groups, the composite keys specified in a nested picture should be generic. This can be done by taking advantage of the Substation Editor's template value feature. For example, a picture representing a high side switch in a feeder bank would have a composite key such as:

```
SUBSTN = %SUBSTN%  
DEVTYP = SW  
DEVICE = %FEEDER%H  
POINT = STTS
```

As the equipment group is placed in the Substation Editor, some of the commonly used template values, such as %SUBSTN%, are automatically filled in without the user having to type in the value. Specific template values, such as %FEEDER% used in this example, have to be manually specified by the user to properly link the group.

By using the same consistent template values (for example, %SUBSTN% and %FEEDER%) in all composite key definitions for the entire equipment group, all picture placements can be placed and linked properly by specifying only one value. In the example above, the feeder number is specified by entering the value 5467 for the template %FEEDER%. When designing equipment groups, the goal should be to require the end user to have to specify as few values as possible when placing the equipment groups.

When modeling equipment in the Substation Editor, various properties related to the electrical devices have to be specified, in addition to the composite keys. In order to define complete equipment groups, which require the end user to manually set the minimum number of property values, the Substation Editor has been designed to look for additional information in the nested picture's composite key definition. By being able to specify individual property values in each picture's composite key definition, complete equipment groups can be designed where the end user only needs to specify one single value, such as the feeder number. This is accomplished by specifying additional information in the composite key definition, as shown in the example below:

```
DNOM[NETWORKDEVICENAME = %FEEDER%H]  
MEAS[MEASUREMENT:COMPOSITEID = SUBSTN%.SW.%FEEDER%H.STTS]  
MEAS[MEASUREMENT:NETWORKDEVICENAME = %FEEDER%H]  
SMP[SCADADATASOURCE = NS]
```

In the example above, four property values are specified. Notice that the template values %SUBSTN% and %FEEDER% are used to make the definitions reusable. In this example, one DNOM, two measurements, and one database (Control)-related property are specified.

The syntax used to specify a DNOM property value is:

DNOM[<propertyName> = <propertyValue>]

The syntax used to specify a measurements property value is:

MEAS[<propertyName> = <propertyValue>]

The syntax used to specify a database-related property value is:

SMP[<propertyName> = <propertyValue>]

The syntax used to specify the property names is simply the condensed form of the property name where all blank spaces are removed and the entire property name is capitalized. The available property values are specific to each DNOM type.

The image below shows the Substation Editor Configuration dialog box's Associations tab. The DNOM and measurement properties are located on the right side. The various database-related properties are located on the bottom of the dialog box.

The screenshot shows the 'Substation Editor Configuration' dialog box, specifically the 'Associations' tab. The 'Associated Element' is set to 'SW\_H'. The 'DNOM Type' is 'Switch'. The 'Measurement Info' section shows 'MEAS Type' as 'SwitchStatusMeasurementType' and 'MEAS Group' as 'StatusMeasurementGroup'. The 'Composite Key' is defined as 'SUBSTN = %SUBSTN% DEVTYP = SW DEVICE = %DEVICE% POINT = STTS'. The 'SCADA DataSource' is 'NS' and the 'SCADA Model Name' is 'Sw, Normally closed, Event'. The 'Division' is '%DIVISION%'. The right side of the dialog lists various properties and their values, such as 'Measurement: (HDB: ID\_MEAS) Measurement ID', 'Measurement: (HDB: NAME\_DEVICE) Device Record Name', 'Measurement: (HDB: NFLAGS\_POINT)', 'Measurement: (HDB: PNTTYP\_POINT) Point Type Identifier', 'Measurement: (HDB: REMOTE\_POINT) True when control is allowed from a Remote Site', 'Measurement: (HDB: SITPRM\_POINT) SITPRM ID - for Intersite Permission Checking', 'Measurement: (HDB: THREE\_POINT) Status Flag', 'Measurement: (HDB: TRACKID\_DEVICE) Unique Tracking Id', 'Measurement: Attached To Device', 'Measurement: Attached To Device Type', 'Measurement: Attached To Effective Device', 'Measurement: Composite ID', 'Measurement: Default Quality', 'Measurement: Default Value', 'Measurement: Group', 'Measurement: Network Device Id', 'Measurement: Network Device Name', 'Measurement: Permission Area Id (8 character max.)', 'Measurement: Phase', 'Measurement: Simulate Only', 'Measurement: Type', 'Measurement: Units (0=Base, 1=Kilo, 2=Mega)', 'Network Device Id', and 'Network Device Name'. The bottom of the dialog has tabs for 'Associations', 'General', 'External Station Configurations', 'Topology', 'Linker', 'Global Internals', 'Template Names', 'Poly Placements', 'SCADA Configurations', and 'Colors & Halos'. The 'Associations' tab is currently selected. The 'OK', 'Apply', and 'Cancel' buttons are at the bottom right.

| Name                                                                             | Value                       |
|----------------------------------------------------------------------------------|-----------------------------|
| Measurement: (HDB: ID_MEAS) Measurement ID                                       |                             |
| Measurement: (HDB: NAME_DEVICE) Device Record Name                               |                             |
| Measurement: (HDB: NFLAGS_POINT)                                                 | NotSet                      |
| Measurement: (HDB: PNTTYP_POINT) Point Type Identifier                           | TNN                         |
| Measurement: (HDB: REMOTE_POINT) True when control is allowed from a Remote Site | False                       |
| Measurement: (HDB: SITPRM_POINT) SITPRM ID - for Intersite Permission Checking   | False                       |
| Measurement: (HDB: THREE_POINT) Status Flag                                      | False                       |
| Measurement: (HDB: TRACKID_DEVICE) Unique Tracking Id                            |                             |
| Measurement: Attached To Device                                                  |                             |
| Measurement: Attached To Device Type                                             |                             |
| Measurement: Attached To Effective Device                                        |                             |
| Measurement: Composite ID                                                        | %SUBSTN%.SW.%DEVICE%.STTS   |
| Measurement: Default Quality                                                     | 0                           |
| Measurement: Default Value                                                       | False                       |
| Measurement: Group                                                               | StatusMeasurementGroup      |
| Measurement: Network Device Id                                                   |                             |
| Measurement: Network Device Name                                                 | %DEVICE%                    |
| Measurement: Permission Area Id (8 character max.)                               |                             |
| Measurement: Phase                                                               | ABC                         |
| Measurement: Simulate Only                                                       | False                       |
| Measurement: Type                                                                | SwitchStatusMeasurementType |
| Measurement: Units (0=Base, 1=Kilo, 2=Mega)                                      | 0                           |
| Network Device Id                                                                |                             |
| Network Device Name                                                              | %DEVICE%                    |

Figure 66. Associating Measurement Properties for a Switch

In the example above, the translation between the actual property name and the condensed form

is as follows:

- Network Device Name = NETWORKDEVICENAME DNOM[NETWORKDEVICENAME = %DEVICE%]
- Measurement: Composite ID = MEASUREMENT:COMPOSITEID

MEAS[MEASUREMENT:COMPOSITEID = %SUBSTN%.SW.%DEVICE%.STTS]

- Measurement: Network Device Name = MEASUREMENT:NETWORKDEVICENAME

MEAS[MEASUREMENT:NETWORKDEVICENAME = %DEVICE%]

- SCADA DataSource = SCADADATASOURCE

SMP[SCADADATASOURCE = NS]

As another example, a given picture placement for a normally open switch could have the definition:

DNOM[NORMALLYOPEN = TRUE]

All the information needed by the Substation Editor is specified on the Define Link tab of the Picture Placement Properties dialog box of the Display Builder tool. The dialog box is activated by simply double-clicking on a given picture placement.

The image below shows the composite key definition for the example used above. In this case, the picture placement represents a switch on the high side of the circuit breaker in a feeder bank.

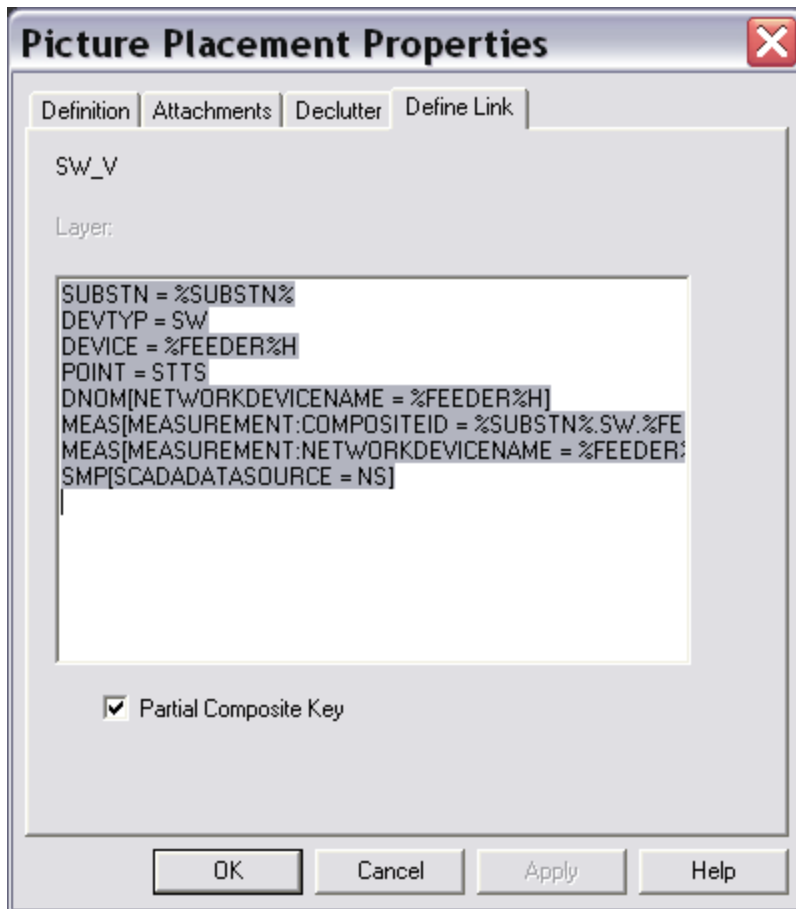


Figure 67. Composite Key Definition for a Switch

Note that, before an electrical device can be modeled in the Substation Editor, the graphical representation of the device (i.e., the picture or symbol defined in the element set) must have an association with its DNOM equipment type. This is done on the Substation Editor Configuration dialog box's Association tab. For each association, default property values can be configured. All picture and symbol placements used in a nested picture must have a valid association in order for the Substation Editor to know which equipment type the given placement should represent. If a picture or symbol does not have an associated DNOM equipment type, the picture will become a Browser-only picture and will only appear on the SCADA display.

As described above, property values can be specified both in the associations and in the nested group's composite key definitions. The goal should be to specify as much as possible in the association, and to only have to specify the minimum number of properties in the composite key definition. More specifically, the property values that need to be different for each instance of the picture placement should be specified in the composite key definition.

For example, a picture definition for a normally open switch picture is named NORMALLY\_OPEN\_VERTICAL\_SWITCH. In this case, the Normally Open property could be set to True in the association definition. If the picture is named VERTICAL\_SWITCH, the Normally Open property could be set to False in the association definition.

If the picture VERTICAL\_SWITCH is used multiple times in a nested picture, an instance representing a normally open switch could have the following property value specified in the composite key definition:

DNOM[NORMALLYOPEN = TRUE]

## 5.2. Creating a Reference-Only Display

The following is an example of how to create a Reference-Only display, using the tools of the Substation Editor.

It is assumed that the Substation Editor has been properly installed and configured, and that the required element sets, etc., have been loaded.

1. Information required prior to commencing: An overall design for the Reference-Only display, a list of the devices to be included, and a substation externals model (station.xml or .csv) file.

It is preferable to create the substations internal model first, so that devices from this display can also be included in the Reference-Only display.

2. Create a new view by selecting the Views tab on the Explorer Pane and clicking the Add button on the toolbar.

Enter a name for the view.

The new view will appear as a tab on the Main Pane. Note that the name given to the view will become the geometry name for the display within the DMS.

Ensure that the display name is set to the same name as the newly created view.

3. Import the substation externals file (station.xml (or csv)). Go to the File menu, click Import Station Externals, select the file, and select the geometry (usually DETAIL).

A new view will be created in the Main Pane.

4. Drag the tab of the Substation Externals view across to the Explorer Pane, and then adjust the splitter bar to make the two panes roughly equal in size.

5. De-clutter the Substation Externals view by turning off the laterals using the toolbar button. Optionally, the labels can be turned on or off.

It is not practical to display both the labels and the laterals together.

6. Identify the first device on the Substation Externals display to be mapped to the Reference-Only display. Use the DNOM Find (Ctrl+F), and enter the name or ID of the device.

The device will be centered on the screen and selected (outlined with a fine green rectangle).

7. Holding down the Alt key and the left mouse button, drag the device across to the Reference-Only display.

The reference-only object that has been associated with the device will be automatically created

and placed on the Reference-Only display.

8. Check the properties of the new reference-only object, and enter the values for the SCADA substation and COMPOSITE KEY.
9. Continue to select devices from the Substation Externals display.

By default, the Substation Externals display is zoomed in (to 300%) and centered on the placement found by the Find logic. The found placement is automatically selected, so it is ready to be copied over to the Reference-Only display by simply pressing the Alt key and the left mouse button, and then dragging the placements over.

Note that, by performing this selection multiple times in the same location, placements stacked on top of each other may be selected in a round-robin fashion. If necessary, it is also possible to move devices out of the way to ensure that the correct device is selected; this will not affect the external model file, as the display is a read-only copy.

10. Similarly, the Substation Internals display can be opened, and devices can be dragged and dropped onto the Reference-Only display.
11. If devices are needed from other stations (for example, when building overview displays where equipment from multiple stations is required), a second instance of the Substation Editor should be used. Simply select the needed reference-only placements, and copy and paste them into the overview.

#### Note

The Alt key and left mouse button drag-and-drop feature cannot be used between separate instances of the Substation Editor.

12. Use the Validation tab to identify and correct any problems with the display.
13. Click the Save / Export button to create the output files.

## 5.3. Creating a Postage Stamp Display

Postage stamp displays are SCADA displays for specific elements contained in the GIS. For more information, see [Postage Stamp Displays](#) and [<UIContainer> Postage Stamp Display](#).

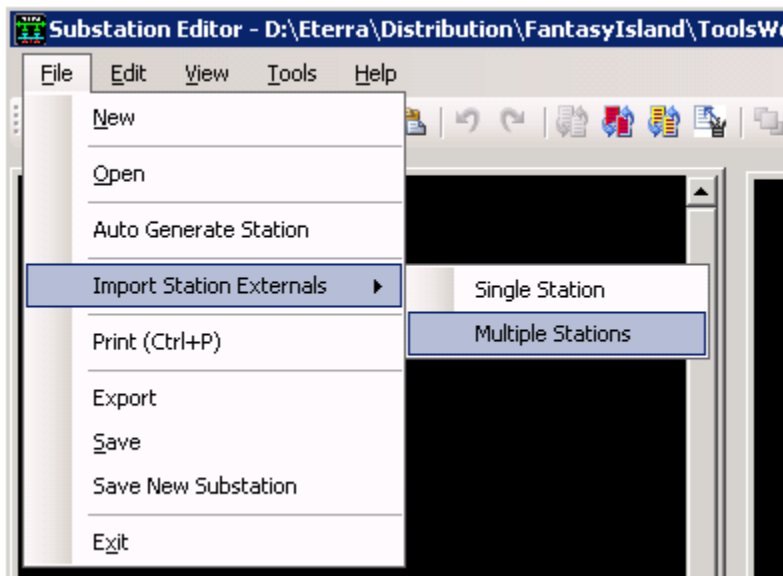
The Create Postage Stamp Display function, accessible from the Tools menu, supports modeling SCADA displays and measurements for equipment that is "outside the fence" of the substation. The characteristics of the equipment are provided in the station externals files, and the characteristics of the measurements are defined during creation using proxies and the Define Link properties of the ESET pictures, as described in [Modeling Equipment Groups](#).

### 5.3.1. Importing the Station Externals

After the necessary pictures have been created in the Element Set, and the Element Set has been

loaded, the first step is to open the SUBEDITOR file of the station that owns the target equipment. (If you are targeting a piece of boundary equipment, you must execute these steps for the SUBEDITOR files of both stations.)

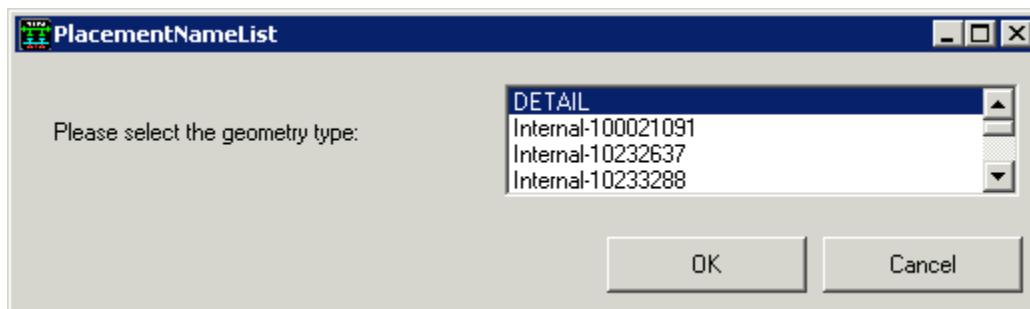
Now import the station externals file(s); select Single Station if you are not targeting boundary equipment, or Multiple Stations if you are.



If you select Single Station, Substation Editor closes the Open dialog box after you double-click on a single file. If you select Multiple Stations, Substation Editor allows you to continue to double-click on files until you click Cancel. (Instead of double-clicking the file name, you may also select the file name and click Open.)

You may select multiple stations in any order, and you may select the same station multiple times without any issues. Before importing stations, Substation Editor drops duplicates and sorts them in alphabetical order.

If you specify more than one placement <GeometryType> in your externals files, Substation Editor asks you which one you want to use in the externals view that you are loading the equipment into.

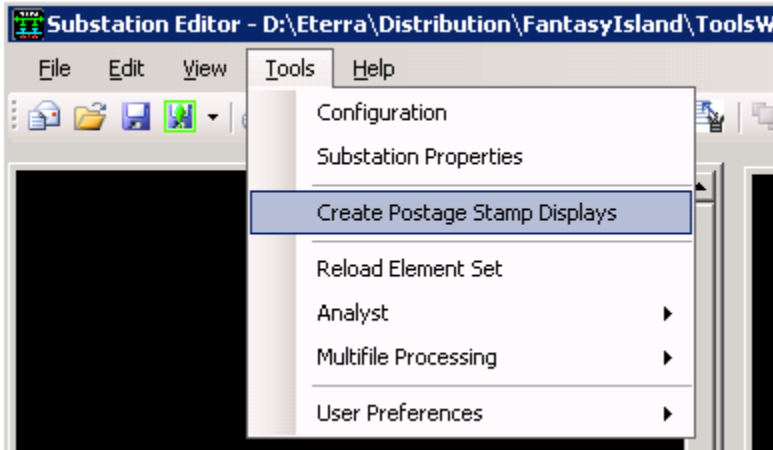


The equipment whose placements are in an unselected <GeometryType> will not be represented in the externals view, but they will be loaded into the DNOM model. They will be available as targets of Create Postage Stamp Displays.

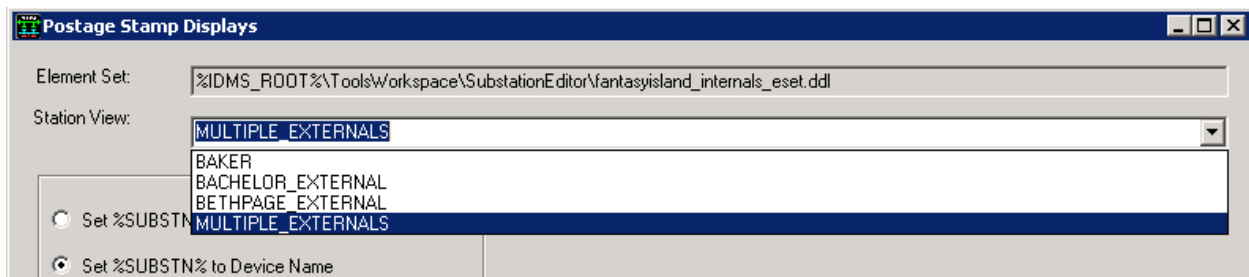


## 5.3.2. Selecting Equipment for View Creation

Once the externals view is created, you are ready to select the equipment for which you want to create views. From the Tools menu, select Create Postage Stamp Displays.

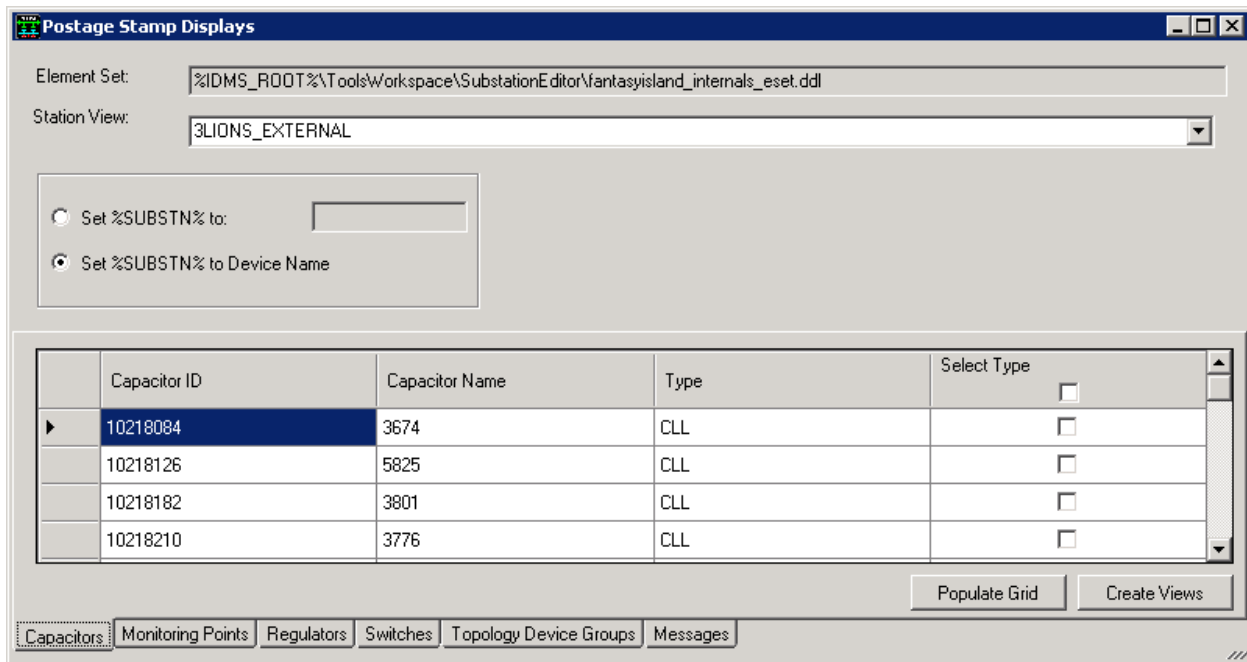


In the Postage Stamp Displays window, make sure the externals view you just created is the one selected in the Station View list. Use the drop-down menu to change to it, if necessary:



## 5.3.3. Populating Grids

Initially, the grid areas on each tab are blank. To load data, select the tab for an equipment type and click the Populate Grid button. A grid then appears with ID, Name, Type (Circuit Pattern), and Select Type columns. One row is filled for each piece of equipment found.



### **Note**

Any piece of boundary equipment for which only one station was loaded is still listed, but with an incomplete "Type" (Circuit Pattern).

### **Note**

It can take some time to load the Switches tab because there are so many of them.

## 5.3.4. Postage Stamp Equipment Tabs

The Postage Stamp Displays window has six tabs at the bottom. The Messages tab gives you access to a log, which shows high-level progress and error messages. The other tabs correspond to categories of CIMDNOM objects.

### 5.3.4.1. Capacitors

Every <Capacitor> that was found in the loaded externals is eligible to appear on this tab.

When a set of capacitors are in parallel, only the capacitor with the alphabetically-lowest <Id> is listed in the grid. The others are represented in the circuit pattern. For example, for a capacitor with the other two capacitors in parallel with it, the circuit pattern in the Type column will include <CCC> symbols.

Create Postage Stamp Displays does not support <CapacitorBank>, per se. A placement for the Capacitor Bank may certainly be included in the picture ("CAPACITOR\_<circuitpattern>"; for

example, "CAPACITOR\_<CCC>LL"). The picture placement for the Capacitor Bank may define associated measurements, and its properties can be filled using the Postage Stamp proxies, just like any other picture placement.

Note that only capacitors that are in parallel with the first capacitor are included in the created view, regardless of how many capacitors are defined in the Capacitor Bank.

### 5.3.4.2. Monitoring Points

Almost every <SensorLocation> and <MeasurementPoint> found in the loaded externals is listed on this tab, except for the objects that are located on the same node as a piece of equipment which is listed in a <UIContainer>.

Multiple monitoring points are not expected at the same node. Monitoring points may be located at a node shared by other devices that are in parallel.

### 5.3.4.3. Regulators

Regulators are defined as Transformers that have at least one tap changer (for example, <Transformer><TapChanger>2</TapChanger>). Every regulator found in the loaded externals is eligible to appear on this tab.

Parallel regulators are handled the same way parallel capacitors are handled: the first regulator is listed and the others are included in the circuit pattern.

Create Postage Stamp Displays does not support <TransformerBank>, per se. As with Capacitor Banks, a placement for the Transformer Bank may be included in the picture (for example, "REGULATOR\_<TTT>LL"), and the placement may define measurements and utilize the Postage Stamp proxies to fill properties. But no special handling or recognition is given to the Transformer Bank itself.

Again, note that only regulators that are in parallel with the first regulator are included in the Postage Stamp view. This means that proxies are not defined for them, and any placements provided for them may or may not have correct linkages created.

### 5.3.4.4. Switches

Most switches, and some reclosers, found in the loaded externals are listed on this tab. A switch or recloser is excluded from this tab when it is listed in a <UIContainer>, or it shares a node with a <SensorLocation> or <MeasurementPoint> listed on the Monitoring Points tab.

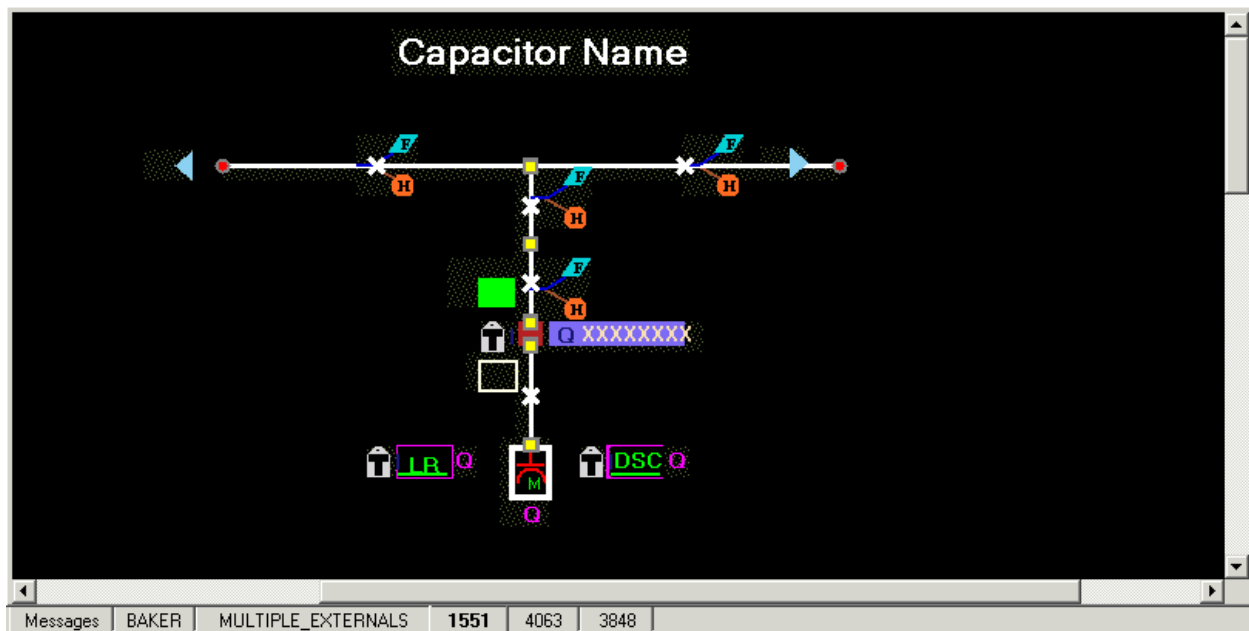
Parallel switches are handled the same way parallel capacitors are handled: the first switch is listed, and the others are included in the circuit pattern.

### 5.3.4.5. Topology Device Groups

The only Topology Device Group currently supported by Create Postage Stamp Displays is <UIContainer>. All validly modeled <UIContainer>s are listed on this tab; error messages are logged for the ones that cannot be traced.

### 5.3.5. Creating Views

Find your target equipment in the grid and check the circuit pattern. If the pattern is correct, and if your ESET contains a picture named according to the rules in [Naming the Picture](#), select the "Select Type" check box on its row and then click the Create Views button.



When view creation is successful, a view labeled with the name of your target equipment appears in the right-hand pane of the Substation Editor window. If a problem is encountered, one or more message boxes will appear. The problems are also saved in the log file.

## 6. Tasks and How to Accomplish Them

### 6.1. Correcting Measurement Attachments

Because a correct model depends upon several different sources of information, and because models change so frequently, it is fairly easy for individual properties of a model object to become incorrect. One of the most troublesome can be the owning station (property "dnomStation") of the measurements on external devices.

The Substation Editor provides the "Correct Measurement Attachments" functionality to help with this problem.

#### 6.1.1. Invocation

##### For a single substation

1. Click File > Open to open the SUBEDITOR file of the target station.
2. Click File > Import Station Externals > Multiple Stations to import the target station's externals file, plus the externals files of its neighbors.
3. Click Tools > Analyst > Correct Measurement Attachments.

##### For multiple substations, from the command line

- Invoke SubstationEditor at the command prompt, with the following parameters:
  - -CONFIGFILE: Location and name of the Substation Editor Configuration file
  - -INPUTDIR: Location of the SUBEDITOR files
  - -EXTDIR: Location of the station externals files.

For example:

```
>%DMS_TOOLS_BINDIR%\SubstationEditor.exe  
-CONFIGFILE %SSEDIR%\SubstationEditorConfiguration.xml -EXPORTDIR %SSEDIR%\data -EXPORTTYPE  
"INTERNALS"  
-OUT %SSEDIR%\output -EXTDIR %CNVDIR%\input
```

The environmental variables used in the example might be:

```
DMS_TOOLS_BINDIR= D:\eterra\Distribution\Tools\bin  
SSEDIR= D:\eterra\Distribution\FantasyIsland\ToolsWorkspace\SubstationEditor  
CNVDIR= D:\eterra\Distribution\FantasyIsland\ToolsWorkspace\Converter
```

Unfortunately, when invoked from the command line, control is returned immediately to the console window, even though SubstationEditor has barely begun to work. You can watch the Windows Task Manager, or look for the "Main exit" message in the SubstationEditorTrace log, to

determine when the work is finished.

## 6.1.2. Processing

For every measurement (Analog, Status, Ctrl, or Setpnt) in the SUBEDITOR file, SubstationEditor looks for the measurement's "attachee" (the DNOM object whose <Id> matches the <Device> property of the measurement). An attachee may be:

- A.** A piece of electrical equipment defined in the station internals (the SUBEDITOR file)
- B.** A piece of external electrical equipment (defined in one of the station externals files)
- C.** A MeasurementPoint
- D.** A ReferenceObject

If the attachee is A or B, then it is the "ultimate attachee", and the owning station of the measurement is the same as the owning station of its ultimate attachee. The measurement's <dnomStation> property is set to the name of the station which defines the electrical equipment.

If the attachee is C or D, then SubstationEditor recurses through the attachments until an ultimate attachee is found.

If no ultimate attachee can be found, then SubstationEditor logs an error and does not change the measurement's <dnomStation> property.

## 6.1.3. Results

If you are running from the GUI, then no <dnomStation> changes are preserved until you explicitly save the SUBEDITOR file. If you are running from the command line, then the <dnomStation> changes are saved automatically when the file is saved.

### Note

When run from the command line, only SUBEDITOR files which have <dnomStation> changes are re-saved.

Errors are logged in the SubstationEditorTrace log.

## 6.2. Problems and Solutions

### 6.2.1. Boolean Values in XML Files

**Symptom:** In XML modeling files (for example, station input files such as <Station>.xml and Substation Editor files such as <Station>\_subeditor.xml), values of "1" and "0" are resolved as "True"

and "False" in Boolean fields.

A Boolean is a simple type, which serializes using the XML standard, and assumes that "1" and "0" are correct Boolean values for "True" and "False". No exception is shown in this case.

**Solution:** It is recommended to use "True" or "False" values for Boolean fields, and avoid using "1" and "0".

## 7. Reference Information

### 7.1. Menu Guide

#### 7.1.1. File Menu

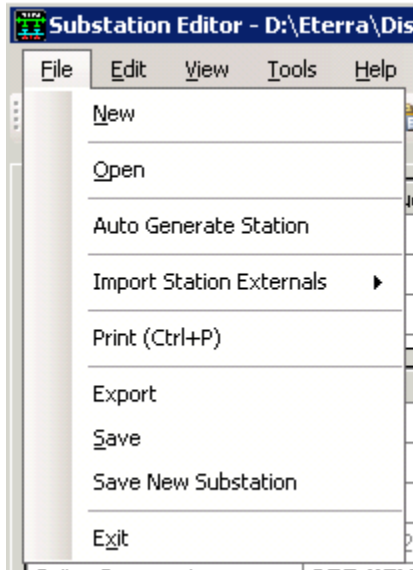


Figure 68. File Menu

**New:** Starts a blank display.

**Open:** Opens an existing Substation Editor file.

**Auto-Generate Station:** Automatically creates a Substation Internals display containing the feeder head devices for each feeder defined in the relevant substation externals file.

**Import Station Externals:** Imports and displays single or multiple substation externals files so that it can be used to generate reference-only objects and Postage Stamp displays.

**Print (Ctrl+P):** Displays the Print dialog box.

**Export:** Exports the Station Internals, Browser, Control SCADA, and Habitat display and database files depending on what is chosen for "Export Types".

**Save:** Saves the current Substation Editor file.

**Save New Substation:** Copies the current Substation Editor file, with newly generated GUIDs, into a new Substation Editor file.

#### **i** Note

If you give the new file the same name as the current one, the current file is replaced. This may



break the substation's connections to equipment and historical data, because the GUIDs no longer match.

**Exit:** Closes the Substation Editor. The system will check whether you want to save any unsaved changes before closing.

## 7.1.2. Edit Menu

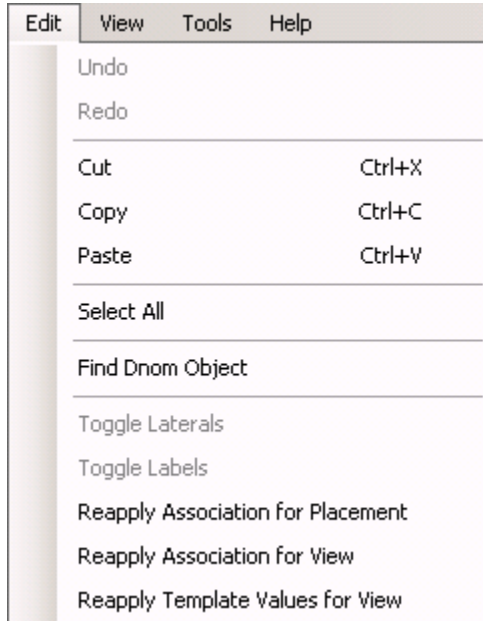


Figure 69. Edit Menu

**Undo:** Undoes the last operation.

**Redo:** Reapplies the last "undone" operation.

**Cut:** Copies the selected object to the clipboard and deletes it from the current display.

**Copy:** Copies the selected object to the clipboard.

**Paste:** Copies the current contents of the clipboard to the display.

**Select All:** Selects all objects on the display.

**Find Dnom Object:** Displays the Find DNOM Object pop-up. Used to find objects in an imported substation externals model file. Ctrl+F performs the same function.

**Toggle Laterals:** Toggles the display of laterals on and off for an imported station externals model file. Button on toolbar performs the same functions.

**Toggle Labels:** Toggles the display of labels on and off for an imported station external model file. Button on the toolbar performs the same functions. This button is only enabled for Station External display.

**Reapply Association for Placement:** Reapplies the associations to the selected device. Used when there has been a change to associations in the Substation Editor configuration.

**Reapply Association for View:** Reapplies the associations for the entire view. Used when there has been a change to associations and it needs to be applied to a large number of objects.

**Reapply Template Values:** Reapplies the default template values to the current display. Used when the default template values have been changed.

### 7.1.3. View Menu

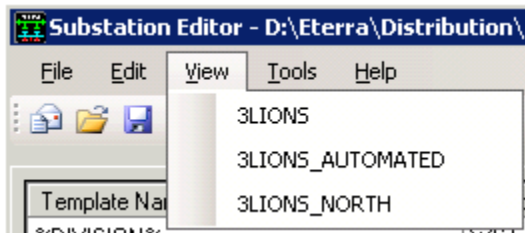


Figure 70. View Menu

Displays a list of available views in the currently loaded model file.

### 7.1.4. Tools Menu

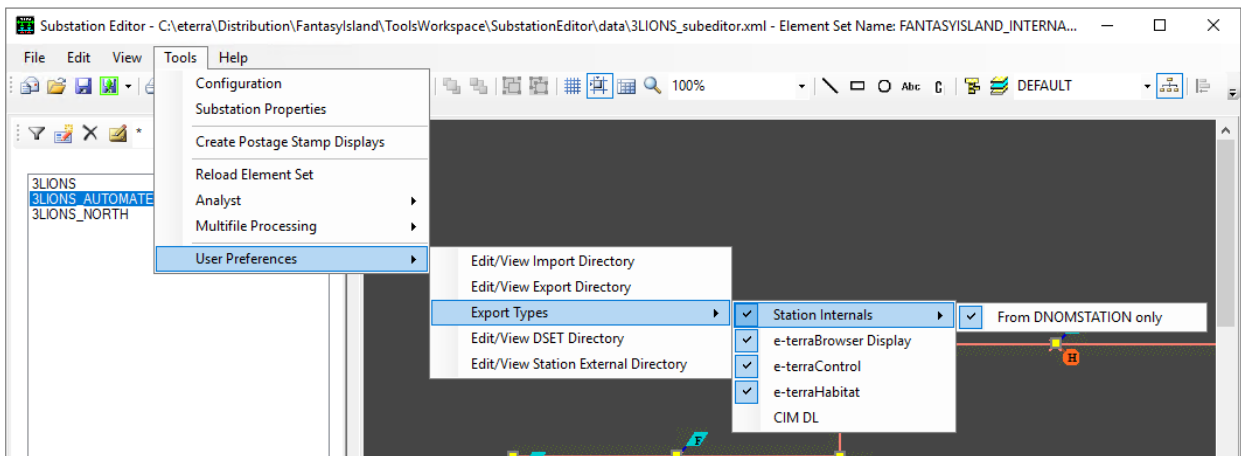


Figure 71. Tools Menu

**Configuration:** Opens the Substation Editor Configuration pop-up.

**Substation Properties:** Sets the default substation name to be used for the display and editing session.

**Create Postage Stamp Displays:** Opens the Postage Stamp Displays pop-up.

**Reload Element Set:** Reloads the element set to refresh the system after configuration changes.

## Analyst:

- **Dependent Object Search:** Invokes the Dependent Objects Search pop-up, where the object and its associated objects are displayed, as shown in the following image. Any device presented on the display can be either searched by ID, or selected via the devices tree list at the left. To export the search data to an .xml file, clicking the Export button.

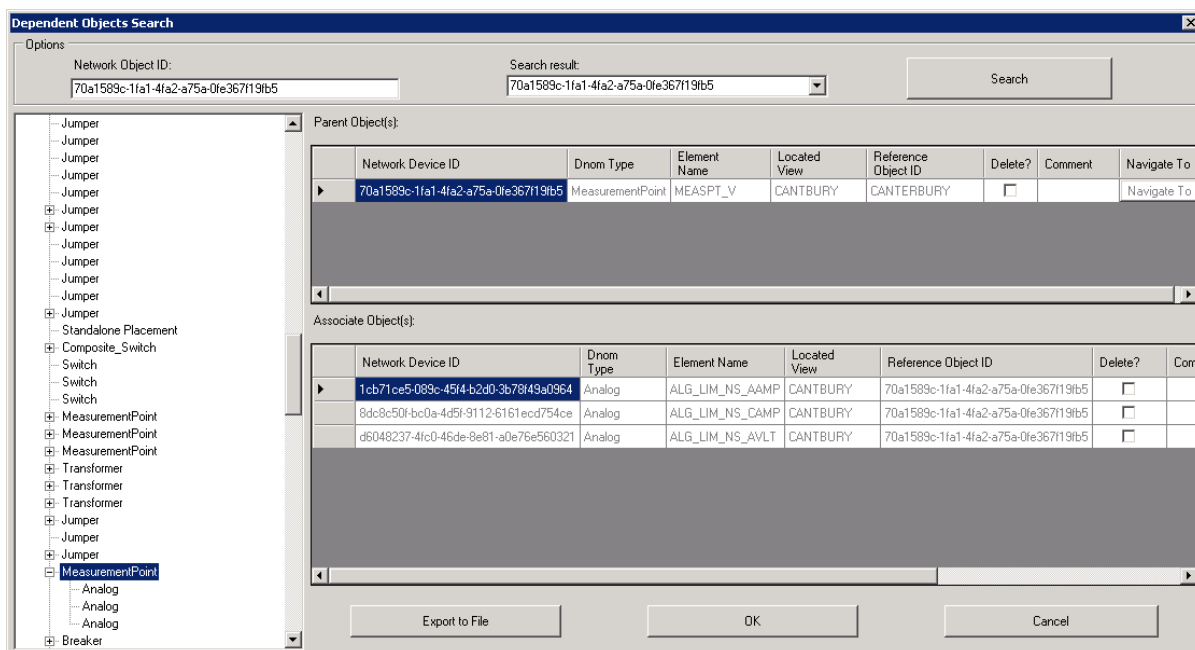


Figure 72. Dependent Object Search

- **Correct DNOM Station for Measurements:**

This option is to be executed if a SUBEDITOR file is known to contain measurements that are not included in the Station Internals file, or that are included in the wrong Internals file.

It executes the same function as is called by Create Postage Stamp Displays to fill the <dnomStation> field (where the name of a measurement's owning station is stored) of newly-created measurements. If the <dnomStation> field is filled correctly, and the Tools > User Preferences > Export Types > Station Internals > From DNOMSTATION Only flag is set, then no incorrect measurements will appear in the Station Internals file; therefore, the Converter tool will not encounter problems by being unable to find the device they belong to.

## Multifile Processing:

- **Export:** Performs the Export function for all the Subeditor files containing in the Import directory (by default, the location is D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\SubstationEditor\data).
- **Reapply Associations:** Reapplies associations for all the views in all the substations (see [Reapply Associations for Multi-File Processing](#)).
- **Reapply Template:** Reapplies templates for all the views in all the substations (see [Reapply Template Values for Multi-File Processing](#)).

## User Preferences:

- **Edit/View Import Directory:** Opens a pop-up displaying the current Import directory. The desired Import directory can be changed by directly editing the path field, or by selecting a folder tree view.
- **Edit/View Export Directory:** Opens a pop-up displaying the current Export directory. The desired Export directory can be changed by directly editing the path field or by selecting a folder tree view.
- **Export Types:** Allows selecting the file types to be exported:
  - Station Internals
    - **From DNOMSTATION Only:** Select this option to exclude objects and placements belonging to other station. For a boundary device (a device which appears in two station externals files), the station with the alphabetically lower name is the owning station
  - **Browser Display:** Select to export the internals in the XML format for viewing in Browser.
  - **Control:** Select to export the internals in the XML format for editing in Control.
  - **Habitat:** Select to export the internals in the TXT format for storing in Habitat.
  - **CIM DL:** Select to export the internals in the Common Information Model (CIM) diagram layout format. This is a future functionality.

## 7.1.5. Help Menu

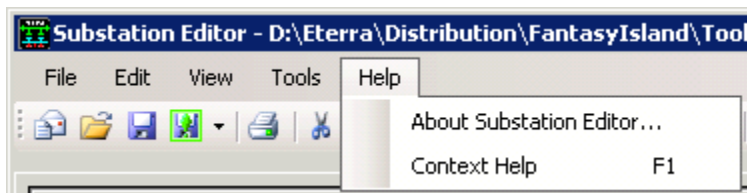


Figure 73. Help Menu

**About Substation Editor:** Opens the pop-up window with information about the current version of the Substation Editor being used.

**Context Help:** Runs the context help mode.

## 7.2. Toolbar Guide

| Menu Selection                                                                      | Description                                                | Section         |
|-------------------------------------------------------------------------------------|------------------------------------------------------------|-----------------|
|  | New: Starts a blank display.                               | Display Drawing |
|  | Open a New File: Opens an existing Substation Editor file. |                 |

| Menu Selection                                                                      | Description                                                                                                                                                                              | Section                           |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
|    | Save: Saves the current Substation Editor file.                                                                                                                                          |                                   |
|    | Export: Exports the station internals, Browser, Control SCADA, and Habitat display and database files depending on what is chosen for "Export Types".                                    |                                   |
|    | Print (Main Pane)                                                                                                                                                                        | Printing Displays                 |
|    | Cut                                                                                                                                                                                      | Display Drawing                   |
|    | Copy                                                                                                                                                                                     | Display Drawing                   |
|    | Paste                                                                                                                                                                                    | Display Drawing                   |
|    | Undo the last action                                                                                                                                                                     | Display Drawing                   |
|    | Redo the last action that was undone                                                                                                                                                     | Display Drawing                   |
|    | Reapply Association for Placement: Reapplies the associations to the selected device. Used when there has been a change to associations in the Substation Editor configuration.          | Reapply Association for Placement |
|   | Reapply Association for View: Reapplies the associations for the entire view. Used when there has been a change to associations and it needs to be applied to a large number of objects. | Reapply Associations for View     |
|  | Reapply Template Values: Reapplies the default template values to the current display. Used when the default template values have been changed.                                          | Reapply Associations for View     |
|  | Correct DNOM Station for Measurements: Used to fill the <dnomStation> field of newly-created measurements.                                                                               | Tools Menu.                       |
|  | Bring to Front: Brings the selected object before the overlapping objects.                                                                                                               |                                   |
|  | Send to Back: Puts the selected object behind the overlapping objects.                                                                                                                   |                                   |
|  | Group Items                                                                                                                                                                              | Display Drawing                   |
|  | Ungroup Items                                                                                                                                                                            | Display Drawing                   |
|  | Show/Hide Grid                                                                                                                                                                           | Grid Management                   |
|  | Snap to Grid                                                                                                                                                                             | Grid Management                   |
|  | Configure Grid                                                                                                                                                                           | Grid Management                   |
|  | Zoom                                                                                                                                                                                     |                                   |
|  | Current Zoom Level/ Select Zoom Level                                                                                                                                                    |                                   |

| Menu Selection                                                                      | Description                                                                                                    | Section                           |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------|
|    | Draw a Line                                                                                                    | Additional Drawing and Formatting |
|    | Draw a Rectangle                                                                                               | Additional Drawing and Formatting |
|    | Draw a Circle                                                                                                  | Additional Drawing and Formatting |
|    | Insert Text                                                                                                    | Additional Drawing and Formatting |
|    | Add a Placeholder to the display                                                                               | Display Drawing                   |
|    | Open the Display Menu Properties dialog box                                                                    | Defining Display Menus            |
|    | Layers                                                                                                         | Reapply Association for Placement |
|    | Active Layer/ Select the Active Layer                                                                          | Reapply Association for Placement |
|    | Show/Hide the Network Layer: Toggles the display of the label elements, and measurement and connection points. |                                   |
|    | Align Left                                                                                                     | Alignment Tools                   |
|    | Align Vertically                                                                                               | Alignment Tools                   |
|  | Align Connection Points Vertically                                                                             | Alignment Tools                   |
|  | Align Right                                                                                                    | Alignment Tools                   |
|  | Align Top                                                                                                      | Alignment Tools                   |
|  | Align Horizontally                                                                                             | Alignment Tools                   |
|  | Align Connection Points Horizontally                                                                           | Alignment Tools                   |
|  | Align Bottom                                                                                                   | Alignment Tools                   |
|  | Turn Laterals On/Off: Used when a station externals display has been imported.                                 |                                   |
|  | Turn Labels On/Off: Used when a station externals display has been imported.                                   |                                   |

## 7.3. Function Keys Reference

| Key           | Function                                                                        |
|---------------|---------------------------------------------------------------------------------|
| <b>Ctrl+A</b> | Associate a measurement with a measurement point or other device.               |
| <b>Ctrl+C</b> | Copy the selected object to the clipboard.                                      |
| <b>Ctrl+D</b> | Copy data from the top fields into the lower fields in the data entry tabulars. |

| Key                             | Function                                                                                                                                                                                                                                          |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ctrl+F</b>                   | Display the Find DNOM object pop-up. Used for imported substation externals model files.                                                                                                                                                          |
| <b>Ctrl+P</b>                   | Open the Print dialog box.                                                                                                                                                                                                                        |
| <b>Ctrl+S</b>                   | Disassociate the measurement from the measurement point or other device.                                                                                                                                                                          |
| <b>Ctrl+V</b>                   | Paste the object from the clipboard to the current cursor location.                                                                                                                                                                               |
| <b>Ctrl+X</b>                   | Cut the selected object to the clipboard.                                                                                                                                                                                                         |
| <b>Esc</b>                      | Complete the current drawing process.                                                                                                                                                                                                             |
| <b>Left Click</b>               | Select the object at the current cursor position.                                                                                                                                                                                                 |
| <b>Shift</b>                    | Holding down the Shift key while positioning the cursor over the edge of the display will pan the display in that direction.                                                                                                                      |
| <b>Shift + Left Click</b>       | Holding down the Shift key while clicking the left mouse button will select multiple objects.                                                                                                                                                     |
| <b>Ctrl + Left Mouse Button</b> | Select and copy object (when cursor is moved away).                                                                                                                                                                                               |
| <b>Alt + Left Mouse Button</b>  | <ol style="list-style-type: none"> <li>1. Drag an object from an imported Station Externals display onto a Substation Internals display.</li> <li>2. Drag an object from a Substation Internals display onto a Reference-Only display.</li> </ol> |
| <b>Right Mouse Button</b>       | Display X,Y coordinates, a menu of placement functions - properties, copy, paste, etc., and the Set DNOM origin function.                                                                                                                         |
| <b>+</b>                        | Create an additional segment to a line/jumper object (when positioning the cursor over the end point of the line).                                                                                                                                |
| <b>-</b>                        | Delete the segment of a line/jumper (when positioning the cursor over the end point of the line).                                                                                                                                                 |
| <b>Ctrl + Mouse Scroll Up</b>   | Zoom in.                                                                                                                                                                                                                                          |
| <b>Ctrl + Mouse Scroll Down</b> | Zoom out.                                                                                                                                                                                                                                         |

## 7.4. Network Object Associations References

This section contains a list of object associations that have been defined for the demonstration system network model, including information about how the parameters can be tuned. It also forms a reference guide for the definition of associations for other network models.

Note that only relevant associations between display elements and DNOM object types need to be defined. For more information, refer to the "Associations" section in the *Substation Editor Installation Guide*.

The tables below give information about the definitions of the parameters for the DNOM object types that are typically used in most display models.

## 7.4.1. Substation Editor Configuration Associations - Breaker

| Element Name                       |         |                                                                                                                                                                                       |
|------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PARAMETER                          | VALUE   | COMMENTS                                                                                                                                                                              |
| DNOM Type                          | Breaker |                                                                                                                                                                                       |
| Connection Pt 1:                   |         | This field is the coordinate for the first connection point (format is X, Y). A branch device has two connection points.                                                              |
| Connection Pt 2:                   |         | This field is the coordinate for the second connection point (format is X, Y). A branch device has two connection points.                                                             |
| Connection Pt 3:                   |         | Not available.                                                                                                                                                                        |
| Network Only                       |         | This flag indicates that the device can only be used in network displays (not reference displays).                                                                                    |
| Reference Only                     |         | This flag indicates that the device can only be used in reference-only displays (not network displays).                                                                               |
| Create DNOM Symbol                 |         |                                                                                                                                                                                       |
| DNOM Symbol Offset                 |         |                                                                                                                                                                                       |
| DNOM Symbol Rotation               |         |                                                                                                                                                                                       |
| Create DNOM Label                  |         |                                                                                                                                                                                       |
| DNOM Label Offset                  |         |                                                                                                                                                                                       |
| DNOM Label Rotation                |         |                                                                                                                                                                                       |
| MEAS Type                          |         |                                                                                                                                                                                       |
| MEAS Group                         |         |                                                                                                                                                                                       |
| Default Association                |         |                                                                                                                                                                                       |
| Default CAM                        |         |                                                                                                                                                                                       |
| Create Measurement                 |         |                                                                                                                                                                                       |
| Composite Key                      |         |                                                                                                                                                                                       |
| SCADA Data Source                  |         |                                                                                                                                                                                       |
| SCADA Model Name                   |         |                                                                                                                                                                                       |
| SCADA Area                         |         |                                                                                                                                                                                       |
| Controls                           |         |                                                                                                                                                                                       |
| Boundary Station                   |         | This field is the name of the adjacent station if the device connects two stations together. This field will be blank for most breaker objects.                                       |
| Maximum Interruption Rating (Amps) |         | This field is the maximum interruption rating in Amps.                                                                                                                                |
| Network Device ID                  |         | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |



| Element Name                       |                                                                        |                                                                                                                                                                |
|------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network Device Name                |                                                                        | This field is the device name. It is expected that this field should not be entered in the association (most devices within a station will have unique names). |
| Normally Open                      | Boolean                                                                | This field indicates the normal switching device status (true = normally open, false = normally closed).                                                       |
| Phase                              | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the device phasing. Most station internal devices will be three-phase (ABC).                                                              |
| Relay - Ground Trip Setting (Amps) |                                                                        | This field is the ground trip setting in Amps.                                                                                                                 |
| Relay - Overcurrent Relay Type     | Enumeration: Unknown, Electromechanical, or Electronic                 | This field is the time/overcurrent relay type.                                                                                                                 |
| Relay _Phase Trip Setting (Amps)   |                                                                        | This field is the phase trip setting in Amps.                                                                                                                  |
| Switch Model                       | Drop-down List                                                         | This field is the name of the associated switch model in the global resources file (global.csv).                                                               |

## 7.4.2. Substation Editor Configuration Associations - Bus

| Element Name         |       |                                                                                                         |
|----------------------|-------|---------------------------------------------------------------------------------------------------------|
| PARAMETER            | VALUE | COMMENTS                                                                                                |
| DNOM Type            | Bus   |                                                                                                         |
| Connection Pt 1:     |       | This field is the coordinate for the first connection point (format is X, Y).                           |
| Connection Pt 2:     |       | This field is the coordinate for the second connection point (format is X, Y).                          |
| Connection Pt 3:     |       | Not available.                                                                                          |
| Network Only         |       | This flag indicates that the device can only be used in network displays (not reference displays).      |
| Reference Only       |       | This flag indicates that the device can only be used in reference-only displays (not network displays). |
| Create DNOM Symbol   |       |                                                                                                         |
| DNOM Symbol Offset   |       |                                                                                                         |
| DNOM Symbol Rotation |       |                                                                                                         |
| Create DNOM Label    |       |                                                                                                         |
| DNOM Label Offset    |       |                                                                                                         |
| DNOM Label Rotation  |       |                                                                                                         |
| MEAS Type            |       |                                                                                                         |
| MEAS Group           |       |                                                                                                         |
| Default Association  |       |                                                                                                         |

| Element Name        |  |                                                                                                                                                                                       |
|---------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Default CAM         |  |                                                                                                                                                                                       |
| Create Measurement  |  |                                                                                                                                                                                       |
| Composite Key       |  |                                                                                                                                                                                       |
| SCADA Data Source   |  |                                                                                                                                                                                       |
| SCADA Model Name    |  |                                                                                                                                                                                       |
| SCADA Area          |  |                                                                                                                                                                                       |
| Controls            |  |                                                                                                                                                                                       |
| Network Device ID   |  | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name |  | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                   |
| Node                |  | This field is the associated node for the bus. This field is not enterable and should not be set in the association.                                                                  |

### 7.4.3. Substation Editor Configuration Associations - Connection Point

| Element Name         |                 |          |
|----------------------|-----------------|----------|
| PARAMETER            | VALUE           | COMMENTS |
| DNOM Type            | ConnectionPoint |          |
| Connection Pt 1:     |                 |          |
| Connection Pt 2:     |                 |          |
| Connection Pt 3:     |                 |          |
| Network Only         |                 |          |
| Reference Only       |                 |          |
| Create DNOM Symbol   |                 |          |
| DNOM Symbol Offset   |                 |          |
| DNOM Symbol Rotation |                 |          |
| Create DNOM Label    |                 |          |
| DNOM Label Offset    |                 |          |
| DNOM Label Rotation  |                 |          |
| MEAS Type            |                 |          |
| MEAS Group           |                 |          |
| Default Association  |                 |          |
| Default CAM          |                 |          |

| Element Name          |             |                                                                                                                                                                                                                                                          |
|-----------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Create Measurement    |             |                                                                                                                                                                                                                                                          |
| Composite Key         |             |                                                                                                                                                                                                                                                          |
| SCADA Data Source     |             |                                                                                                                                                                                                                                                          |
| SCADA Model Name      |             |                                                                                                                                                                                                                                                          |
| SCADA Area            |             |                                                                                                                                                                                                                                                          |
| Controls              |             |                                                                                                                                                                                                                                                          |
| Connection Node ID    |             | This field is the connection node (location) for connecting the station internal network with another station internal network or a station external network (commonly extracted from a GIS). Similar to other node fields, this field is not enterable. |
| External Station Name |             | This field is the name of the adjacent station if the connection is to network located inside another substation. For connections between station internal and station external networks, the external station name should not be filled in.             |
| Network Device ID     |             | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association.                                                                    |
| Network Device Name   |             | This field is the connection point name. Since a connection point is not a physical object in the topology model, the name should always be left blank (the user will never see the connection point).                                                   |
| Phase                 | Enumeration | This field is the phasing information. For a connection point, this value does not matter (leave at default value of ABC).                                                                                                                               |

## 7.4.4. Substation Editor Configuration Associations - Fault Detector

| Element Name       |                |                                                                                                         |
|--------------------|----------------|---------------------------------------------------------------------------------------------------------|
| PARAMETER          | VALUE          | COMMENTS                                                                                                |
| DNOM Type          | Fault_Detector |                                                                                                         |
| Connection Pt 1:   |                | This field is the coordinate for the first connection point (format is X, Y).                           |
| Connection Pt 2:   |                | Not available.                                                                                          |
| Connection Pt 3:   |                | Not available.                                                                                          |
| Network Only       |                | This flag indicates that the device can only be used in network displays (not reference displays).      |
| Reference Only     |                | This flag indicates that the device can only be used in reference-only displays (not network displays). |
| Create DNOM Symbol |                |                                                                                                         |

| Element Name         |                                                                        |                                                                                                                                                                                       |
|----------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DNOM Symbol Offset   |                                                                        |                                                                                                                                                                                       |
| DNOM Symbol Rotation |                                                                        |                                                                                                                                                                                       |
| Create DNOM Label    |                                                                        |                                                                                                                                                                                       |
| DNOM Label Offset    |                                                                        |                                                                                                                                                                                       |
| DNOM Label Rotation  |                                                                        |                                                                                                                                                                                       |
| MEAS Type            |                                                                        |                                                                                                                                                                                       |
| MEAS Group           |                                                                        |                                                                                                                                                                                       |
| Default Association  |                                                                        |                                                                                                                                                                                       |
| Default CAM          |                                                                        |                                                                                                                                                                                       |
| Create Measurement   |                                                                        |                                                                                                                                                                                       |
| Composite Key        |                                                                        |                                                                                                                                                                                       |
| SCADA Data Source    |                                                                        |                                                                                                                                                                                       |
| SCADA Model Name     |                                                                        |                                                                                                                                                                                       |
| SCADA Area           |                                                                        |                                                                                                                                                                                       |
| Controls             |                                                                        |                                                                                                                                                                                       |
| Network Device ID    |                                                                        | This field is the automatically created unique ID. Since each object in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name  |                                                                        | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                   |
| Node                 |                                                                        | This field is the associated node for the fault indicator. This field is not enterable and should not be set in the association.                                                      |
| Phase                | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the device phasing. Most station internal devices will be three-phase (ABC).                                                                                     |

## 7.4.5. Substation Editor Configuration Associations - Feeder

| Element Name     |        |                                                                                                         |
|------------------|--------|---------------------------------------------------------------------------------------------------------|
| PARAMETER        | VALUE  | COMMENTS                                                                                                |
| DNOM Type        | Feeder |                                                                                                         |
| Connection Pt 1: |        | This field is the coordinate for the first connection point (format is X, Y).                           |
| Connection Pt 2: |        | Not available.                                                                                          |
| Connection Pt 3: |        | Not available.                                                                                          |
| Network Only     |        | This flag indicates that the device can only be used in network displays (not reference-only displays). |

| Element Name                                                                                                                                                                                                                              |                 |                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reference Only                                                                                                                                                                                                                            |                 | This flag indicates that the device can only be used in reference-only displays (not network displays).                                                                               |
| Create DNOM Symbol                                                                                                                                                                                                                        |                 |                                                                                                                                                                                       |
| DNOM Symbol Offset                                                                                                                                                                                                                        |                 |                                                                                                                                                                                       |
| DNOM Symbol Rotation                                                                                                                                                                                                                      |                 |                                                                                                                                                                                       |
| Create DNOM Label                                                                                                                                                                                                                         |                 |                                                                                                                                                                                       |
| DNOM Label Offset                                                                                                                                                                                                                         |                 |                                                                                                                                                                                       |
| DNOM Label Rotation                                                                                                                                                                                                                       |                 |                                                                                                                                                                                       |
| MEAS Type                                                                                                                                                                                                                                 |                 |                                                                                                                                                                                       |
| MEAS Group                                                                                                                                                                                                                                |                 |                                                                                                                                                                                       |
| Default Association                                                                                                                                                                                                                       |                 |                                                                                                                                                                                       |
| Default CAM                                                                                                                                                                                                                               |                 |                                                                                                                                                                                       |
| Create Measurement                                                                                                                                                                                                                        |                 |                                                                                                                                                                                       |
| Composite Key                                                                                                                                                                                                                             |                 |                                                                                                                                                                                       |
| SCADA Data Source                                                                                                                                                                                                                         |                 |                                                                                                                                                                                       |
| SCADA Model Name                                                                                                                                                                                                                          |                 |                                                                                                                                                                                       |
| SCADA Area                                                                                                                                                                                                                                |                 |                                                                                                                                                                                       |
| Controls                                                                                                                                                                                                                                  |                 |                                                                                                                                                                                       |
| Feeder Color (1-24)                                                                                                                                                                                                                       |                 | This field is the feeder user interface color. The integer corresponds to a color defined in the station editor.                                                                      |
| Network Device ID                                                                                                                                                                                                                         |                 | This field is the automatically created unique ID. Since each object in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name                                                                                                                                                                                                                       |                 | This field is the feeder name This field should probably not be entered in the association (most devices within a station will have unique names).                                    |
| Node                                                                                                                                                                                                                                      |                 | This field is the associated node for the fault indicator. This field is not enterable and should not be set in the association.                                                      |
| PRV Evaluation                                                                                                                                                                                                                            | Enum:           | Specifies the fault impedance evaluation type that is applied by the PRV application.                                                                                                 |
|                                                                                                                                                                                                                                           | NotSet = 0      |                                                                                                                                                                                       |
|                                                                                                                                                                                                                                           | BoltedFault = 1 | When the value is selected and applied, either by saving to same editor file or exporting to an internal file, the value is updated as an integer value in respective XML file.       |
| If there is an invalid PRV evaluation type in the respective subeditor XML file, when a corresponding feeder is selected in Sub Station Editor UI a warning message appears and the PRV evaluation type is set to default value "NotSet". |                 |                                                                                                                                                                                       |

## 7.4.6. Substation Editor Configuration Associations - Fuse

| Element Name         |       |                                                                                                                                                                                       |
|----------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PARAMETER            | VALUE | COMMENTS                                                                                                                                                                              |
| DNOM Type            | Fuse  |                                                                                                                                                                                       |
| Connection Pt 1:     |       | This field is the coordinate for the first connection point (format is X, Y). A branch device has two connection points.                                                              |
| Connection Pt 2:     |       | This field is the coordinate for the second connection point (format is X, Y). A branch device has two connection points.                                                             |
| Connection Pt 3:     |       | Not available.                                                                                                                                                                        |
| Network Only         |       | This flag indicates that the device can only be used in network displays (not reference displays).                                                                                    |
| Reference Only       |       | This flag indicates that the device can only be used in reference-only displays (not network displays).                                                                               |
| Create DNOM Symbol   |       |                                                                                                                                                                                       |
| DNOM Symbol Offset   |       |                                                                                                                                                                                       |
| DNOM Symbol Rotation |       |                                                                                                                                                                                       |
| Create DNOM Label    |       |                                                                                                                                                                                       |
| DNOM Label Offset    |       |                                                                                                                                                                                       |
| DNOM Label Rotation  |       |                                                                                                                                                                                       |
| MEAS Type            |       |                                                                                                                                                                                       |
| MEAS Group           |       |                                                                                                                                                                                       |
| Default Association  |       |                                                                                                                                                                                       |
| Default CAM          |       |                                                                                                                                                                                       |
| Create Measurement   |       |                                                                                                                                                                                       |
| Composite Key        |       |                                                                                                                                                                                       |
| SCADA Data Source    |       |                                                                                                                                                                                       |
| SCADA Model Name     |       |                                                                                                                                                                                       |
| SCADA Area           |       |                                                                                                                                                                                       |
| Controls             |       |                                                                                                                                                                                       |
| Boundary Station     |       | This field is the name of the adjacent station if the device connects two stations together. This field will be blank for most breaker objects.                                       |
| Network Device ID    |       | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name  |       | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                   |

| Element Name  |                                                                        |                                                                                                          |
|---------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Normally Open |                                                                        | This field indicates the normal switching device status (true = normally open, false = normally closed). |
| Phase         | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the device phasing. Most station internal devices will be three-phase (ABC).        |
| Switch Model  | Drop-down List                                                         | This field is the name of the associated switch model in the global resources file (global.csv).         |

## 7.4.7. Substation Editor Configuration Associations - Generator

| Element Name         |           |                                                                                                                  |
|----------------------|-----------|------------------------------------------------------------------------------------------------------------------|
| PARAMETER            | VALUE     | COMMENTS                                                                                                         |
| DNOM Type            | Generator |                                                                                                                  |
| Connection Pt 1:     |           | This field is the coordinate for the connection point (format is X, Y). A shunt device has one connection point. |
| Connection Pt 2:     |           | Not available.                                                                                                   |
| Connection Pt 3:     |           | Not available.                                                                                                   |
| Network Only         |           | This flag indicates that the device can only be used in network displays (not reference displays).               |
| Reference Only       |           | This flag indicates that the device can only be used in reference-only displays (not network displays).          |
| Create DNOM Symbol   |           |                                                                                                                  |
| DNOM Symbol Offset   |           |                                                                                                                  |
| DNOM Symbol Rotation |           |                                                                                                                  |
| Create DNOM Label    |           |                                                                                                                  |
| DNOM Label Offset    |           |                                                                                                                  |
| DNOM Label Rotation  |           |                                                                                                                  |
| MEAS Type            |           |                                                                                                                  |
| MEAS Group           |           |                                                                                                                  |
| Default Association  |           |                                                                                                                  |
| Default CAM          |           |                                                                                                                  |
| Create Measurement   |           |                                                                                                                  |
| Composite Key        |           |                                                                                                                  |
| SCADA Data Source    |           |                                                                                                                  |
| SCADA Model Name     |           |                                                                                                                  |
| SCADA Area           |           |                                                                                                                  |
| Controls             |           |                                                                                                                  |

| Element Name                                 |                                                                        |                                                                                                                                                                                       |
|----------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Generator Model                              | Drop-down List                                                         | This field is the name of the associated generator model in the global resources file (global.csv).                                                                                   |
| Grounding Reactance (ohms)                   |                                                                        | This field is the grounding reactance (in ohms).                                                                                                                                      |
| Grounding Resistance (ohms)                  |                                                                        | This field is the grounding resistance (in ohms).                                                                                                                                     |
| Maximum Separation Voltage (per unit)        |                                                                        | This field is the maximum separation voltage (in per unit Volts).                                                                                                                     |
| Minimum Separation Voltage (per unit)        |                                                                        | This field is the minimum separation voltage (in per unit Volts).                                                                                                                     |
| Network Device ID                            |                                                                        | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name                          |                                                                        | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                   |
| Nominal Generator Voltage (kV L-L)           |                                                                        | This field is the nominal generator voltage (in kV line to line).                                                                                                                     |
| Phase                                        | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the device phasing. Generators should always be three-phase (ABC).                                                                                               |
| PV Generator Type (true = PV, false = slack) | Boolean                                                                | This field indicates the generator type (true = regulates power and voltage PV, false = slack).                                                                                       |

## 7.4.8. Substation Editor Configuration Associations - Ground

| Element Name         |        |                                                                                                                  |
|----------------------|--------|------------------------------------------------------------------------------------------------------------------|
| PARAMETER            | VALUE  | COMMENTS                                                                                                         |
| DNOM Type            | Ground |                                                                                                                  |
| Connection Pt 1:     |        | This field is the coordinate for the connection point (format is X, Y). A shunt device has one connection point. |
| Connection Pt 2:     |        | Not available.                                                                                                   |
| Connection Pt 3:     |        | Not available.                                                                                                   |
| Network Only         |        | This flag indicates that the device can only be used in network displays (not reference displays).               |
| Reference Only       |        | This flag indicates that the device can only be used in reference-only displays (not network displays).          |
| Create DNOM Symbol   |        |                                                                                                                  |
| DNOM Symbol Offset   |        |                                                                                                                  |
| DNOM Symbol Rotation |        |                                                                                                                  |
| Create DNOM Label    |        |                                                                                                                  |
| DNOM Label Offset    |        |                                                                                                                  |



| Element Name        |                                                                        |                                                                                                                                                                                       |
|---------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DNOM Label Rotation |                                                                        |                                                                                                                                                                                       |
| MEAS Type           |                                                                        |                                                                                                                                                                                       |
| MEAS Group          |                                                                        |                                                                                                                                                                                       |
| Default Association |                                                                        |                                                                                                                                                                                       |
| Default CAM         |                                                                        |                                                                                                                                                                                       |
| Create Measurement  |                                                                        |                                                                                                                                                                                       |
| Composite Key       |                                                                        |                                                                                                                                                                                       |
| SCADA Data Source   |                                                                        |                                                                                                                                                                                       |
| SCADA Model Name    |                                                                        |                                                                                                                                                                                       |
| SCADA Area          |                                                                        |                                                                                                                                                                                       |
| Controls            |                                                                        |                                                                                                                                                                                       |
| Network Device ID   |                                                                        | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name |                                                                        | This field is the device name This field should probably not be entered in the association (most devices within a station will have unique names).                                    |
| Phase               | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the device phasing. A ground will be typically be single-phase (A, B, C, or N).                                                                                  |

## 7.4.9. Substation Editor Configuration Associations - Jumper

| Element Name         |        |                                                                                                                           |
|----------------------|--------|---------------------------------------------------------------------------------------------------------------------------|
| PARAMETER            | VALUE  | COMMENTS                                                                                                                  |
| DNOM Type            | Jumper |                                                                                                                           |
| Connection Pt 1:     |        | This field is the coordinate for the first connection point (format is X, Y). A branch device has two connection points.  |
| Connection Pt 2:     |        | This field is the coordinate for the second connection point (format is X, Y). A branch device has two connection points. |
| Connection Pt 3:     |        | Not available.                                                                                                            |
| Network Only         |        | This flag indicates that the device can only be used in network displays (not reference displays).                        |
| Reference Only       |        | This flag indicates that the device can only be used in reference-only displays (not network displays).                   |
| Create DNOM Symbol   |        |                                                                                                                           |
| DNOM Symbol Offset   |        |                                                                                                                           |
| DNOM Symbol Rotation |        |                                                                                                                           |
| Create DNOM Label    |        |                                                                                                                           |

| Element Name        |                                                                        |                                                                                                                                                                                       |
|---------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DNOM Label Offset   |                                                                        |                                                                                                                                                                                       |
| DNOM Label Rotation |                                                                        |                                                                                                                                                                                       |
| MEAS Type           |                                                                        |                                                                                                                                                                                       |
| MEAS Group          |                                                                        |                                                                                                                                                                                       |
| Default Association |                                                                        |                                                                                                                                                                                       |
| Default CAM         |                                                                        |                                                                                                                                                                                       |
| Create Measurement  |                                                                        |                                                                                                                                                                                       |
| Composite Key       |                                                                        |                                                                                                                                                                                       |
| SCADA Data Source   |                                                                        |                                                                                                                                                                                       |
| SCADA Model Name    |                                                                        |                                                                                                                                                                                       |
| SCADA Area          |                                                                        |                                                                                                                                                                                       |
| Controls            |                                                                        |                                                                                                                                                                                       |
| Boundary Station    |                                                                        | This field is the name of the adjacent station if the device connects two stations together. This field will be blank for most jumper objects.                                        |
| Network Device ID   |                                                                        | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name |                                                                        | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                   |
| Normally Open       |                                                                        | This field indicates the normal switching device status (true = normally open, false = normally closed).                                                                              |
| Phase               | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the device phasing. Most station internal devices will be three-phase (ABC).                                                                                     |
| Switch Model        | Drop-down List                                                         | This field is the name of the associated switch model in the global resources file (global.csv).                                                                                      |

## 7.4.10. Substation Editor Configuration Associations - Line

| Element Name     |       |                                                                                                                           |
|------------------|-------|---------------------------------------------------------------------------------------------------------------------------|
| PARAMETER        | VALUE | COMMENTS                                                                                                                  |
| DNOM Type        | Line  |                                                                                                                           |
| Connection Pt 1: |       | This field is the coordinate for the first connection point (format is X, Y). A branch device has two connection points.  |
| Connection Pt 2: |       | This field is the coordinate for the second connection point (format is X, Y). A branch device has two connection points. |
| Connection Pt 3: |       | Not available.                                                                                                            |

| Element Name         |                                                                        |                                                                                                                                                                                       |
|----------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network Only         |                                                                        | This flag indicates that the device can only be used in network displays (not reference displays).                                                                                    |
| Reference Only       |                                                                        | This flag indicates that the device can only be used in reference-only displays (not network displays).                                                                               |
| Create DNOM Symbol   |                                                                        |                                                                                                                                                                                       |
| DNOM Symbol Offset   |                                                                        |                                                                                                                                                                                       |
| DNOM Symbol Rotation |                                                                        |                                                                                                                                                                                       |
| Create DNOM Label    |                                                                        |                                                                                                                                                                                       |
| DNOM Label Offset    |                                                                        |                                                                                                                                                                                       |
| DNOM Label Rotation  |                                                                        |                                                                                                                                                                                       |
| MEAS Type            |                                                                        |                                                                                                                                                                                       |
| MEAS Group           |                                                                        |                                                                                                                                                                                       |
| Default Association  |                                                                        |                                                                                                                                                                                       |
| Default CAM          |                                                                        |                                                                                                                                                                                       |
| Create Measurement   |                                                                        |                                                                                                                                                                                       |
| Composite Key        |                                                                        |                                                                                                                                                                                       |
| SCADA Data Source    |                                                                        |                                                                                                                                                                                       |
| SCADA Model Name     |                                                                        |                                                                                                                                                                                       |
| SCADA Area           |                                                                        |                                                                                                                                                                                       |
| Controls             |                                                                        |                                                                                                                                                                                       |
| Boundary Station     |                                                                        | This field is the name of the adjacent station if the device connects two stations together. This field will be blank for most jumper objects.                                        |
| Length (feet)        |                                                                        | This field is the line length (in feet).                                                                                                                                              |
| Limit 1 (amps)       |                                                                        | This field is the level 1 Current-carrying Capacity (ampacity) limit (Amps).                                                                                                          |
| Limit 2 (Amps)       |                                                                        | This field is the level 2 Current-carrying Capacity (ampacity) limit (Amps).                                                                                                          |
| Limit 3 (Amps)       |                                                                        | This field is the level 3 Current-carrying Capacity (ampacity) limit (Amps).                                                                                                          |
| Network Device ID    |                                                                        | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name  |                                                                        | This field is the device name. It is expected that this field should not be entered in the association (most devices within a station will have unique names).                        |
| Phase                | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the device phasing. Most station internal devices will be three-phase (ABC).                                                                                     |

| Element Name                                  |                                                                     |
|-----------------------------------------------|---------------------------------------------------------------------|
| Positive Sequence Charging Susceptance (MVAR) | This field is the positive sequence charging susceptance (in MVAR). |
| Positive Sequence Reactance (ohms)            | This field is the positive sequence reactance (in ohms).            |
| Positive Sequence Resistance (ohms)           | This field is the positive sequence resistance (in ohms).           |
| Zero Sequence Charging Susceptance (MVAR)     | This field is the zero sequence charging susceptance (in MVAR).     |
| Zero Sequence Reactance (ohms)                | This field is the zero sequence reactance (in ohms).                |
| Zero Sequence Resistance (ohms)               | This field is the zero sequence resistance (in ohms).               |

## 7.4.11. Substation Editor Configuration Associations - Load

| Element Name         |       |                                                                                                                  |
|----------------------|-------|------------------------------------------------------------------------------------------------------------------|
| PARAMETER            | VALUE | COMMENTS                                                                                                         |
| DNOM Type            | Load  |                                                                                                                  |
| Connection Pt 1:     |       | This field is the coordinate for the connection point (format is X, Y). A shunt device has one connection point. |
| Connection Pt 2:     |       | Not available.                                                                                                   |
| Connection Pt 3:     |       | Not available.                                                                                                   |
| Network Only         |       | This flag indicates that the device can only be used in network displays (not reference displays).               |
| Reference Only       |       | This flag indicates that the device can only be used in reference-only displays (not network displays).          |
| Create DNOM Symbol   |       |                                                                                                                  |
| DNOM Symbol Offset   |       |                                                                                                                  |
| DNOM Symbol Rotation |       |                                                                                                                  |
| Create DNOM Label    |       |                                                                                                                  |
| DNOM Label Offset    |       |                                                                                                                  |
| DNOM Label Rotation  |       |                                                                                                                  |
| MEAS Type            |       |                                                                                                                  |
| MEAS Group           |       |                                                                                                                  |
| Default Association  |       |                                                                                                                  |
| Default CAM          |       |                                                                                                                  |
| Create Measurement   |       |                                                                                                                  |
| Composite Key        |       |                                                                                                                  |
| SCADA Data Source    |       |                                                                                                                  |

| Element Name                                    |                                                                        |                                                                                                                                                                                       |
|-------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCADA Model Name                                |                                                                        |                                                                                                                                                                                       |
| SCADA Area                                      |                                                                        |                                                                                                                                                                                       |
| Controls                                        |                                                                        |                                                                                                                                                                                       |
| Network Device ID                               |                                                                        | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name                             |                                                                        | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                   |
| Phase                                           | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the load phasing.                                                                                                                                                |
| Phase A Nominal Reactive Power (kVAR) / PFactor |                                                                        | This field is the phase A nominal reactive power (in kVAR) or power factor.                                                                                                           |
| Phase A Nominal Real Power (kW)                 |                                                                        | This field is the phase A nominal real power (in kW).                                                                                                                                 |
| Phase B Nominal Reactive Power (kVAR) / PFactor |                                                                        | This field is the phase B nominal reactive power (in kVAR) or power factor.                                                                                                           |
| Phase B Nominal Real Power (kW)                 |                                                                        | This field is the phase B nominal real power (in kW).                                                                                                                                 |
| Phase C Nominal Reactive Power (kVAR) / PFactor |                                                                        | This field is the phase C nominal reactive power (in kVAR) or power factor.                                                                                                           |
| Phase C Nominal Real Power (kW)                 |                                                                        | This field is the phase C nominal real power (in kW).                                                                                                                                 |
| Wye Connected (Wye = true, Delta = false)       | Boolean                                                                | This field indicates whether the capacitor is connected in wye (true) or delta (line to line - false).                                                                                |

## 7.4.12. Substation Editor Configuration Associations - MeasObject

| Element Name       |            |          |
|--------------------|------------|----------|
| PARAMETER          | VALUE      | COMMENTS |
| DNOM Type          | MeasObject |          |
| Connection Pt 1:   |            |          |
| Connection Pt 2:   |            |          |
| Connection Pt 3:   |            |          |
| Network Only       |            |          |
| Reference Only     |            |          |
| Create DNOM Symbol |            |          |

| Element Name         |  |                                                                                                                                                                                       |
|----------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DNOM Symbol Offset   |  |                                                                                                                                                                                       |
| DNOM Symbol Rotation |  |                                                                                                                                                                                       |
| Create DNOM Label    |  |                                                                                                                                                                                       |
| DNOM Label Offset    |  |                                                                                                                                                                                       |
| DNOM Label Rotation  |  |                                                                                                                                                                                       |
| MEAS Type            |  |                                                                                                                                                                                       |
| MEAS Group           |  |                                                                                                                                                                                       |
| Default Association  |  |                                                                                                                                                                                       |
| Default CAM          |  |                                                                                                                                                                                       |
| Create Measurement   |  |                                                                                                                                                                                       |
| Composite Key        |  |                                                                                                                                                                                       |
| SCADA Data Source    |  |                                                                                                                                                                                       |
| SCADA Model Name     |  |                                                                                                                                                                                       |
| SCADA Area           |  |                                                                                                                                                                                       |
| Controls             |  |                                                                                                                                                                                       |
| Network Device ID    |  | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name  |  | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                   |

## 7.4.13. Substation Editor Configuration Associations - Measurement Point

| Element Name         |                  |          |
|----------------------|------------------|----------|
| PARAMETER            | VALUE            | COMMENTS |
| DNOM Type            | MeasurementPoint |          |
| Connection Pt 1:     |                  |          |
| Connection Pt 2:     |                  |          |
| Connection Pt 3:     |                  |          |
| Network Only         |                  |          |
| Reference Only       |                  |          |
| Create DNOM Symbol   |                  |          |
| DNOM Symbol Offset   |                  |          |
| DNOM Symbol Rotation |                  |          |

| Element Name         |             |                                                                                                                                                                                       |
|----------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Create DNOM Label    |             |                                                                                                                                                                                       |
| DNOM Label Offset    |             |                                                                                                                                                                                       |
| DNOM Label Rotation  |             |                                                                                                                                                                                       |
| MEAS Type            |             |                                                                                                                                                                                       |
| MEAS Group           |             |                                                                                                                                                                                       |
| Default Association  |             |                                                                                                                                                                                       |
| Default CAM          |             |                                                                                                                                                                                       |
| Create Measurement   |             |                                                                                                                                                                                       |
| Composite Key        |             |                                                                                                                                                                                       |
| SCADA Data Source    |             |                                                                                                                                                                                       |
| SCADA Model Name     |             |                                                                                                                                                                                       |
| SCADA Area           |             |                                                                                                                                                                                       |
| Controls             |             |                                                                                                                                                                                       |
| Associated Object ID |             | This field is the Network Device ID of the associated device. It is expected that this field will not be filled in from the associations.                                             |
| Network Device ID    |             | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name  |             | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                   |
| Phase                | Enumeration | This field is the phasing information. For a measurement point, this value does not matter (leave at default value of ABC).                                                           |

## 7.4.14. Substation Editor Configuration Associations - Recloser

| Element Name     |          |                                                                                                                           |
|------------------|----------|---------------------------------------------------------------------------------------------------------------------------|
| PARAMETER        | VALUE    | COMMENTS                                                                                                                  |
| DNOM Type        | Recloser |                                                                                                                           |
| Connection Pt 1: |          | This field is the coordinate for the first connection point (format is X, Y). A branch device has two connection points.  |
| Connection Pt 2: |          | This field is the coordinate for the second connection point (format is X, Y). A branch device has two connection points. |
| Connection Pt 3: |          | Not available.                                                                                                            |
| Network Only     |          | This flag indicates that the device can only be used in network displays (not reference displays).                        |

| Element Name                       |                                                                        |                                                                                                                                                                                       |
|------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reference Only                     |                                                                        | This flag indicates that the device can only be used in reference-only displays (not network displays).                                                                               |
| Create DNOM Symbol                 |                                                                        |                                                                                                                                                                                       |
| DNOM Symbol Offset                 |                                                                        |                                                                                                                                                                                       |
| DNOM Symbol Rotation               |                                                                        |                                                                                                                                                                                       |
| Create DNOM Label                  |                                                                        |                                                                                                                                                                                       |
| DNOM Label Offset                  |                                                                        |                                                                                                                                                                                       |
| DNOM Label Rotation                |                                                                        |                                                                                                                                                                                       |
| MEAS Type                          |                                                                        |                                                                                                                                                                                       |
| MEAS Group                         |                                                                        |                                                                                                                                                                                       |
| Default Association                |                                                                        |                                                                                                                                                                                       |
| Default CAM                        |                                                                        |                                                                                                                                                                                       |
| Create Measurement                 |                                                                        |                                                                                                                                                                                       |
| Composite Key                      |                                                                        |                                                                                                                                                                                       |
| SCADA Data Source                  |                                                                        |                                                                                                                                                                                       |
| SCADA Model Name                   |                                                                        |                                                                                                                                                                                       |
| SCADA Area                         |                                                                        |                                                                                                                                                                                       |
| Controls                           |                                                                        |                                                                                                                                                                                       |
| Boundary Station                   |                                                                        | This field is the name of the adjacent station if the device connects two stations together. This field will be blank for most breaker objects.                                       |
| Ground Trip Capable                | Boolean                                                                | This field indicates whether the recloser has ground trip capability (true = has ground trip capability, false = no ground trip capability).                                          |
| Ground Trip Setting (Amps)         |                                                                        | This field is the normal ground trip rating (in Amps).                                                                                                                                |
| Ground Trip Normally Enabled       | Boolean                                                                | This field is the normal ground trip status.                                                                                                                                          |
| Maximum Interruption Rating (Amps) |                                                                        | This field is the maximum interruption rating (in Amps).                                                                                                                              |
| Network Device ID                  |                                                                        | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name                |                                                                        | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                   |
| Normally Open                      | Boolean                                                                | This field indicates the normal switching device status (true = normally open, false = normally closed).                                                                              |
| Phase                              | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the device phasing. Most station internal devices will be three-phase (ABC).                                                                                     |
| Phase Trip Setting (Amps)          |                                                                        | This field is the phase trip setting (in Amps).                                                                                                                                       |



| Element Name |                |                                                                                                  |
|--------------|----------------|--------------------------------------------------------------------------------------------------|
| Switch Model | Drop-down List | This field is the name of the associated switch model in the global resources file (global.csv). |

## 7.4.15. Substation Editor Configuration Associations - Reference Bus Interface

| Element Name          |                 |                                                                                                                                                 |
|-----------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| PARAMETER             | VALUE           | COMMENTS                                                                                                                                        |
| DNOM Type             | RefBusInterface |                                                                                                                                                 |
| Connection Pt 1:      |                 | This field is the coordinate for the connection point (format is X, Y). A shunt device has one connection point.                                |
| Connection Pt 2:      |                 | Not available.                                                                                                                                  |
| Connection Pt 3:      |                 | Not available.                                                                                                                                  |
| Network Only          |                 | This flag indicates that the device can only be used in network displays (not reference displays).                                              |
| Reference Only        |                 | This flag indicates that the device can only be used in reference-only displays (not network displays).                                         |
| Study Only            |                 | This flag indicates that the device will be used for study purposes (not real time).                                                            |
| Interface with Switch |                 | This flag indicates that the switch having the measurement type SwitchStatusMeasurementType is modeled to determine the status of energization. |
| Create DNOM Symbol    |                 |                                                                                                                                                 |
| DNOM Symbol Offset    |                 |                                                                                                                                                 |
| DNOM Symbol Rotation  |                 |                                                                                                                                                 |
| Create DNOM Label     |                 |                                                                                                                                                 |
| DNOM Label Offset     |                 |                                                                                                                                                 |
| DNOM Label Rotation   |                 |                                                                                                                                                 |
| MEAS Type             |                 |                                                                                                                                                 |
| MEAS Group            |                 |                                                                                                                                                 |
| Default Association   |                 |                                                                                                                                                 |
| Default CAM           |                 |                                                                                                                                                 |
| Create Measurement    |                 |                                                                                                                                                 |
| Composite Key         |                 |                                                                                                                                                 |
| SCADA Data Source     |                 |                                                                                                                                                 |
| SCADA Model Name      |                 |                                                                                                                                                 |
| SCADA Area            |                 |                                                                                                                                                 |
| Controls              |                 |                                                                                                                                                 |

| Element Name        |                                                                        |                                                                                                                                                                                       |
|---------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network Device ID   |                                                                        | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name |                                                                        | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                   |
| Phase               | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the device phasing. Most station internal devices will be three-phase (ABC).                                                                                     |

## 7.4.16. Substation Editor Configuration Associations - Sectionalizer

| Element Name         |               |                                                                                                                           |
|----------------------|---------------|---------------------------------------------------------------------------------------------------------------------------|
| PARAMETER            | VALUE         | COMMENTS                                                                                                                  |
| DNOM Type            | Sectionalizer |                                                                                                                           |
| Connection Pt 1:     |               | This field is the coordinate for the first connection point (format is X, Y). A branch device has two connection points.  |
| Connection Pt 2:     |               | This field is the coordinate for the second connection point (format is X, Y). A branch device has two connection points. |
| Connection Pt 3:     |               | Not available.                                                                                                            |
| Network Only         |               | This flag indicates that the device can only be used in network displays (not reference displays).                        |
| Reference Only       |               | This flag indicates that the device can only be used in reference-only displays (not network displays).                   |
| Create DNOM Symbol   |               |                                                                                                                           |
| DNOM Symbol Offset   |               |                                                                                                                           |
| DNOM Symbol Rotation |               |                                                                                                                           |
| Create DNOM Label    |               |                                                                                                                           |
| DNOM Label Offset    |               |                                                                                                                           |
| DNOM Label Rotation  |               |                                                                                                                           |
| MEAS Type            |               |                                                                                                                           |
| MEAS Group           |               |                                                                                                                           |
| Default Association  |               |                                                                                                                           |
| Default CAM          |               |                                                                                                                           |
| Create Measurement   |               |                                                                                                                           |
| Composite Key        |               |                                                                                                                           |
| SCADA Data Source    |               |                                                                                                                           |

| Element Name        |                                                                        |                                                                                                                                                                                       |
|---------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCADA Model Name    |                                                                        |                                                                                                                                                                                       |
| SCADA Area          |                                                                        |                                                                                                                                                                                       |
| Controls            |                                                                        |                                                                                                                                                                                       |
| Boundary Station    |                                                                        | This field is the name of the adjacent station if the device connects two stations together. This field will be blank for most breaker objects.                                       |
| Network Device ID   |                                                                        | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name |                                                                        | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                   |
| Normally Open       | Boolean                                                                | This field indicates the normal switching device status (true = normally open, false = normally closed).                                                                              |
| Phase               | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the device phasing. Most station internal devices will be three-phase (ABC).                                                                                     |
| Switch Model        | Drop-down List                                                         | This field is the name of the associated switch model in the global resources file (global.csv).                                                                                      |

## 7.4.17. Substation Editor Configuration Associations - Series Reactor

| Element Name         |                |                                                                                                                           |
|----------------------|----------------|---------------------------------------------------------------------------------------------------------------------------|
| PARAMETER            | VALUE          | COMMENTS                                                                                                                  |
| DNOM Type            | Series_Reactor |                                                                                                                           |
| Connection Pt 1:     |                | This field is the coordinate for the first connection point (format is X, Y). A branch device has two connection points.  |
| Connection Pt 2:     |                | This field is the coordinate for the second connection point (format is X, Y). A branch device has two connection points. |
| Connection Pt 3:     |                | Not available.                                                                                                            |
| Network Only         |                | This flag indicates that the device can only be used in network displays (not reference displays).                        |
| Reference Only       |                | This flag indicates that the device can only be used in reference-only displays (not network displays).                   |
| Create DNOM Symbol   |                |                                                                                                                           |
| DNOM Symbol Offset   |                |                                                                                                                           |
| DNOM Symbol Rotation |                |                                                                                                                           |
| Create DNOM Label    |                |                                                                                                                           |
| DNOM Label Offset    |                |                                                                                                                           |

| Element Name         |                                                                        |                                                                                                                                                                                       |
|----------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DNOM Label Rotation  |                                                                        |                                                                                                                                                                                       |
| MEAS Type            |                                                                        |                                                                                                                                                                                       |
| MEAS Group           |                                                                        |                                                                                                                                                                                       |
| Default Association  |                                                                        |                                                                                                                                                                                       |
| Default CAM          |                                                                        |                                                                                                                                                                                       |
| Create Measurement   |                                                                        |                                                                                                                                                                                       |
| Composite Key        |                                                                        |                                                                                                                                                                                       |
| SCADA Data Source    |                                                                        |                                                                                                                                                                                       |
| SCADA Model Name     |                                                                        |                                                                                                                                                                                       |
| SCADA Area           |                                                                        |                                                                                                                                                                                       |
| Controls             |                                                                        |                                                                                                                                                                                       |
| Boundary Station     |                                                                        | This field is the name of the adjacent station if the device connects two stations together. This field will be blank for most breaker objects.                                       |
| Network Device ID    |                                                                        | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name  |                                                                        | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                   |
| Phase                | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the device phasing. Most station internal devices will be three-phase (ABC).                                                                                     |
| Series Reactor Model | Drop-down List                                                         | This field is the name of the associated series reactor model in the global resources file (global.csv).                                                                              |

## 7.4.18. Substation Editor Configuration Associations - Shunt Impedance Capacitor

| Element Name     |                           |                                                                                                                  |
|------------------|---------------------------|------------------------------------------------------------------------------------------------------------------|
| PARAMETER        | VALUE                     | COMMENTS                                                                                                         |
| DNOM Type        | Shunt_Impedance_Capacitor |                                                                                                                  |
| Connection Pt 1: |                           | This field is the coordinate for the connection point (format is X, Y). A shunt device has one connection point. |
| Connection Pt 2: |                           | Not available.                                                                                                   |
| Connection Pt 3: |                           | Not available.                                                                                                   |
| Network Only     |                           | This flag indicates that the device can only be used in network displays (not reference displays).               |

| Element Name                                                  |                                                              |                                                                                                                                                                                             |
|---------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reference Only                                                |                                                              | This flag indicates that the device can only be used in reference-only displays (not network displays).                                                                                     |
| Create DNOM Symbol                                            |                                                              |                                                                                                                                                                                             |
| DNOM Symbol Offset                                            |                                                              |                                                                                                                                                                                             |
| DNOM Symbol Rotation                                          |                                                              |                                                                                                                                                                                             |
| Create DNOM Label                                             |                                                              |                                                                                                                                                                                             |
| DNOM Label Offset                                             |                                                              |                                                                                                                                                                                             |
| DNOM Label Rotation                                           |                                                              |                                                                                                                                                                                             |
| MEAS Type                                                     |                                                              |                                                                                                                                                                                             |
| MEAS Group                                                    |                                                              |                                                                                                                                                                                             |
| Default Association                                           |                                                              |                                                                                                                                                                                             |
| Default CAM                                                   |                                                              |                                                                                                                                                                                             |
| Create Measurement                                            |                                                              |                                                                                                                                                                                             |
| Composite Key                                                 |                                                              |                                                                                                                                                                                             |
| SCADA Data Source                                             |                                                              |                                                                                                                                                                                             |
| SCADA Model Name                                              |                                                              |                                                                                                                                                                                             |
| SCADA Area                                                    |                                                              |                                                                                                                                                                                             |
| Controls                                                      |                                                              |                                                                                                                                                                                             |
| Control Type Mechanism                                        | Enumeration: Fixed, Local_Only, Remote_Only, or Remote_Local | This field is the control type mechanism for the capacitor. For locally controlled capacitors that can also be controlled remotely through Scada, this field should be set to Remote_Local. |
| Local Control - Associated Time Schedule                      | Drop-down List - check on this                               | This field is the name of an associated time schedule in the global resources file (global.csv). This schedule only needs to be filled in if the capacitor local control scheme is time.    |
| Local Control - Line to Line Voltage Regulation               | Boolean                                                      | This field indicates whether the voltage is measured line to line (true) or line to ground (false). This field only needs to be filled in for local voltage controlled capacitors.          |
| Local Control - Off Level (Amps, kVAR, Volt p.u., or Pfactor) |                                                              | This field is the local control off level setting for a flow regulating (Amp, VAR, or power factor) capacitor.                                                                              |
| Local Control - Off Level (Amps, kVAR, Volt p.u., or Pfactor) |                                                              | This field is the local control on level setting for a flow regulating (Amp, VAR, or power factor) capacitor.                                                                               |
| Local Control - Phases Sensed                                 | Enumeration: ABC, A, B, C, AB, BC, AC                        | This field is the local control on setting for a flow regulating (Amp, VAR, or power factor) or voltage regulating capacitor.                                                               |
| Local Control - Regulation Branch (Amps, kVAR, or Pfactor)    |                                                              | This field is the branch ID for a flow regulating (Amp, VAR, or power factor) locally controlled capacitor.                                                                                 |

| Element Name                                                    |                                                                        |                                                                                                                                                                                       |
|-----------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local Control - Voltage Override Capable                        |                                                                        | This field indicates whether a flow regulating (Amp, VAR, or power factor) locally controlled capacitor has voltage override capability.                                              |
| Local Control - Voltage Override Off Level (Voltage - per unit) |                                                                        | This field is the local control voltage override off level setting for a flow regulating (Amp, VAR, or power factor) capacitor that has voltage override capability.                  |
| Local Control - Voltage Override On Level (Voltage - per unit)  |                                                                        | This field is the local control voltage override on level setting for a flow regulating (Amp, VAR, or power factor) capacitor that has voltage override capability.                   |
| Local Control Type                                              | Enumeration: None, Power Factor, Time, VAR, Amp, or Voltage            | This field is the type of local control mechanism for capacitors that have local control capability (control type mechanism = Local_Only or Remote_Local).                            |
| Maximum Number of Daily Switching Operations                    |                                                                        | This field is the maximum number of daily switching operations.                                                                                                                       |
| Minimum Time Between Switching Operations (hours)               |                                                                        | This field is the minimum time interval between switching operations (hours).                                                                                                         |
| Network Device ID                                               |                                                                        | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name                                             |                                                                        | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                   |
| Nominal Voltage (kV line to line)                               |                                                                        | This field is the nominal voltage (in kV line to line).                                                                                                                               |
| Number of Steps                                                 |                                                                        | This field is the number of steps. For a capacitor that has all VARs on or off, the number of steps is 1.                                                                             |
| Phase                                                           | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the device phasing. Most station internal devices will be three-phase (ABC).                                                                                     |
| Total Three-Phase kVAR                                          |                                                                        | This field is the total three-phase VAR rating (in kVAR).                                                                                                                             |
| Wye Connected (Wye = true, Delta = false)                       | Boolean                                                                | This field indicates whether the capacitor is connected in wye (true) or delta (line to line - false).                                                                                |

## 7.4.19. Substation Editor Configuration Associations - Substation

| Element Name     |            |          |
|------------------|------------|----------|
| PARAMETER        | VALUE      | COMMENTS |
| DNOM Type        | Substation |          |
| Connection Pt 1: |            |          |
| Connection Pt 2: |            |          |
| Connection Pt 3: |            |          |

| Element Name         |  |                                                                                                                                                                                       |
|----------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network Only         |  |                                                                                                                                                                                       |
| Reference Only       |  |                                                                                                                                                                                       |
| Create DNOM Symbol   |  |                                                                                                                                                                                       |
| DNOM Symbol Offset   |  |                                                                                                                                                                                       |
| DNOM Symbol Rotation |  |                                                                                                                                                                                       |
| Create DNOM Label    |  |                                                                                                                                                                                       |
| DNOM Label Offset    |  |                                                                                                                                                                                       |
| DNOM Label Rotation  |  |                                                                                                                                                                                       |
| MEAS Type            |  |                                                                                                                                                                                       |
| MEAS Group           |  |                                                                                                                                                                                       |
| Default Association  |  |                                                                                                                                                                                       |
| Default CAM          |  |                                                                                                                                                                                       |
| Create Measurement   |  |                                                                                                                                                                                       |
| Composite Key        |  |                                                                                                                                                                                       |
| SCADA Data Source    |  |                                                                                                                                                                                       |
| SCADA Model Name     |  |                                                                                                                                                                                       |
| SCADA Area           |  |                                                                                                                                                                                       |
| Controls             |  |                                                                                                                                                                                       |
| Network Device ID    |  | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name  |  | This field is the station name.                                                                                                                                                       |

## 7.4.20. Substation Editor Configuration Associations - Switch

| Element Name     |        |                                                                                                                           |
|------------------|--------|---------------------------------------------------------------------------------------------------------------------------|
| PARAMETER        | VALUE  | COMMENTS                                                                                                                  |
| DNOM Type        | Switch |                                                                                                                           |
| Connection Pt 1: |        | This field is the coordinate for the first connection point (format is X, Y). A branch device has two connection points.  |
| Connection Pt 2: |        | This field is the coordinate for the second connection point (format is X, Y). A branch device has two connection points. |
| Connection Pt 3: |        | Not available.                                                                                                            |
| Network Only     |        | This flag indicates that the device can only be used in network displays (not reference displays).                        |
| Reference Only   |        | This flag indicates that the device can only be used in reference-only displays (not network displays).                   |

| Element Name         |                                                                        |                                                                                                                                                                                       |
|----------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Create DNOM Symbol   |                                                                        |                                                                                                                                                                                       |
| DNOM Symbol Offset   |                                                                        |                                                                                                                                                                                       |
| DNOM Symbol Rotation |                                                                        |                                                                                                                                                                                       |
| Create DNOM Label    |                                                                        |                                                                                                                                                                                       |
| DNOM Label Offset    |                                                                        |                                                                                                                                                                                       |
| DNOM Label Rotation  |                                                                        |                                                                                                                                                                                       |
| MEAS Type            |                                                                        |                                                                                                                                                                                       |
| MEAS Group           |                                                                        |                                                                                                                                                                                       |
| Default Association  |                                                                        |                                                                                                                                                                                       |
| Default CAM          |                                                                        |                                                                                                                                                                                       |
| Create Measurement   |                                                                        |                                                                                                                                                                                       |
| Composite Key        |                                                                        |                                                                                                                                                                                       |
| SCADA Data Source    |                                                                        |                                                                                                                                                                                       |
| SCADA Model Name     |                                                                        |                                                                                                                                                                                       |
| SCADA Area           |                                                                        |                                                                                                                                                                                       |
| Controls             |                                                                        |                                                                                                                                                                                       |
| Boundary Station     |                                                                        | This field is the name of the adjacent station if the device connects two stations together. This field will be blank for most breaker objects.                                       |
| Network Device ID    |                                                                        | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association. |
| Network Device Name  |                                                                        | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                   |
| Normally Open        | Boolean                                                                | This field indicates the normal switching device status (true = normally open, false = normally closed).                                                                              |
| Phase                | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the device phasing. Most station internal devices will be three-phase (ABC).                                                                                     |
| Switch Model         | Drop-down List                                                         | This field is the name of the associated switch model in the global resources file (global.csv).                                                                                      |

## 7.4.21. Substation Editor Configuration Associations - Transformer

| Element Name |       |          |
|--------------|-------|----------|
| PARAMETER    | VALUE | COMMENTS |



| Element Name                 |             |                                                                                                                                                                                                                                                                        |
|------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DNOM Type                    | Transformer |                                                                                                                                                                                                                                                                        |
| Connection Pt 1:             |             | This field is the coordinate for the first connection point (format is X, Y). A branch device has two connection points.                                                                                                                                               |
| Connection Pt 2:             |             | This field is the coordinate for the second connection point (format is X, Y). A branch device has two connection points.                                                                                                                                              |
| Connection Pt 3:             |             | This field is the coordinate for the third connection point (format is X, Y). A three winding transformer (tertiary winding) is the only device that has a third connection point.                                                                                     |
| Network Only                 |             | This flag indicates that the device can only be used in network displays (not reference displays).                                                                                                                                                                     |
| Reference Only               |             | This flag indicates that the device can only be used in reference-only displays (not network displays).                                                                                                                                                                |
| Create DNOM Symbol           |             |                                                                                                                                                                                                                                                                        |
| DNOM Symbol Offset           |             |                                                                                                                                                                                                                                                                        |
| DNOM Symbol Rotation         |             |                                                                                                                                                                                                                                                                        |
| Create DNOM Label            |             |                                                                                                                                                                                                                                                                        |
| DNOM Label Offset            |             |                                                                                                                                                                                                                                                                        |
| DNOM Label Rotation          |             |                                                                                                                                                                                                                                                                        |
| MEAS Type                    |             |                                                                                                                                                                                                                                                                        |
| MEAS Group                   |             |                                                                                                                                                                                                                                                                        |
| Default Association          |             |                                                                                                                                                                                                                                                                        |
| Default CAM                  |             |                                                                                                                                                                                                                                                                        |
| Create Measurement           |             |                                                                                                                                                                                                                                                                        |
| Composite Key                |             |                                                                                                                                                                                                                                                                        |
| SCADA Data Source            |             |                                                                                                                                                                                                                                                                        |
| SCADA Model Name             |             |                                                                                                                                                                                                                                                                        |
| SCADA Area                   |             |                                                                                                                                                                                                                                                                        |
| Controls                     |             |                                                                                                                                                                                                                                                                        |
| Associated Transformer Model |             | This field is the name of the associated transformer model in the global resources file (global.csv). If this model is not filled in (which is probably the case for most power transformers), then the field names that start with "Model" will need to be filled in. |
| Boundary Station             |             | This field is the name of the adjacent station if the device connects two stations together. This field will be blank for most breaker objects.                                                                                                                        |
| Model - Core Loss (MW)       |             | This field is the core loss (in MW). This field does not need to be filled in if an associated transformer model is assigned.                                                                                                                                          |

| Element Name                                         |                                                                                                                                                                                                                                                                   |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model - Excitation Current (%)                       | This field is the excitation current percentage. If a value is not provided here, then an internal Converter default will be used to estimate the magnetizing reactance. This field does not need to be filled in if an associated transformer model is assigned. |
| Model - Limit 1 (Amps)                               | This field is the first current-carrying capacity limit value for regulators. A regulator may have a variable limit depending on the tap step. This field does not need to be filled in if an associated transformer model is assigned.                           |
| Model - Limit 2 (Amps)                               | This field is the second current-carrying capacity limit value for regulators. A regulator may have a variable limit depending on the tap step. This field does not need to be filled in if an associated transformer model is assigned.                          |
| Model - Limit 3 (Amps)                               | This field is the third current-carrying capacity limit value for regulators. A regulator may have a variable limit depending on the tap step. This field does not need to be filled in if an associated transformer model is assigned.                           |
| Model - Limit 4 (Amps)                               | This field is the fourth current-carrying capacity limit value for regulators. A regulator may have a variable limit depending on the tap step. This field does not need to be filled in if an associated transformer model is assigned.                          |
| Model - Limit 5 (Amps)                               | This field is the fifth current-carrying capacity limit value for regulators. A regulator may have a variable limit depending on the tap step. This field does not need to be filled in if an associated transformer model is assigned.                           |
| Model - Operational Limit 1 (MVA)                    | This field is the level 1 limit (in MVA). This field does not need to be filled in if an associated transformer model is assigned.                                                                                                                                |
| Model - Operational Limit 2 (MVA)                    | This field is the level 2 limit (in MVA). This field does not need to be filled in if an associated transformer model is assigned.                                                                                                                                |
| Model - Operational Limit 3 (MVA)                    | This field is the level 3 limit (in MVA). This field does not need to be filled in if an associated transformer model is assigned.                                                                                                                                |
| Model - Operational Limit(Secondary Winding) 1 (MVA) | This field is the level 1 limit (in MVA) for the transformer's secondary winding. This field does not need to be filled in if an associated transformer model is assigned.                                                                                        |
| Model - Operational Limit(Secondary Winding) 2 (MVA) | This field is the level 2 limit (in MVA) for the transformer's secondary winding. This field does not need to be filled in if an associated transformer model is assigned.                                                                                        |
| Model - Operational Limit(Secondary Winding) 3 (MVA) | This field is the level 3 limit (in MVA) for the transformer's secondary winding. This field does not need to be filled in if an associated transformer model is assigned.                                                                                        |
| Model - Operational Limit(Tertiary Winding)1 (MVA)   | This field is the level 1 limit (in MVA) for the transformer's tertiary winding. This field does not need to be filled in if an associated transformer model is assigned.                                                                                         |
| Model - Operational Limit (Tertiary Winding) 2 (MVA) | This field is the level 2 limit (in MVA) for the transformer's tertiary winding. This field does not need to be filled in if an associated transformer model is assigned.                                                                                         |

| Element Name                                        |                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model - Operational Limit(Tertiary Winding) 3 (MVA) | This field is the level 3 limit (in MVA) for the transformer's tertiary winding. This field does not need to be filled in if an associated transformer model is assigned.                                                                                                                       |
| Model - Primary-Secondary Reactance (percent)       | This field is for three-winding transformers. It is the leakage reactance (%) measured in the primary with the secondary short circuited and the tertiary open. The impedance value is referred to the transformer power base, and it is used to determine the total winding impedance values.  |
| Model - Primary-Secondary Resistance (percent)      | This field is for three-winding transformers. It is the leakage resistance (%) measured in the primary with the secondary short circuited and the tertiary open. The impedance value is referred to the transformer power base, and it is used to determine the total winding impedance values. |
| Model - Primary-Tertiary Reactance (percent)        | This field is for three-winding transformers. It is the leakage reactance (%) measured in the primary with the tertiary short circuited and the secondary open. The impedance value is referred to the transformer power base, and it is used to determine the total winding impedance values.  |
| Model - Primary-Tertiary Resistance (percent)       | This field is for three-winding transformers. It is the leakage resistance (%) measured in the primary with the tertiary short circuited and the secondary open. The impedance value is referred to the transformer power base, and it is used to determine the total winding impedance values. |
| Model - Reactance (per unit)                        | This field is the per unit reactance (on transformer power rating and nominal voltage rating). This field does not need to be filled in if an associated transformer model is assigned.                                                                                                         |
| Model - Regulation Step 1 (%)                       | This field is the first current-carrying capacity limit step value ( $\pm 5\%$ for example) for regulators. A regulator may have a variable limit depending on the tap step. This field does not need to be filled in if an associated transformer model is assigned.                           |
| Model - Regulation Step 2 (%)                       | This field is the second current-carrying capacity limit step value ( $\pm 6.25\%$ for example) for regulators. A regulator may have a variable limit depending on the tap step. This field does not need to be filled in if an associated transformer model is assigned.                       |
| Model - Regulation Step 3 (%)                       | This field is the third current-carrying capacity limit step value ( $\pm 7.5\%$ for example) for regulators. A regulator may have a variable limit depending on the tap step. This field does not need to be filled in if an associated transformer model is assigned.                         |
| Model - Regulation Step 4 (%)                       | This field is the fourth current-carrying capacity limit step value ( $\pm 8.75\%$ for example) for regulators. A regulator may have a variable limit depending on the tap step. This field does not need to be filled in if an associated transformer model is assigned.                       |

| Element Name                                       |                                                              |                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model - Regulation Step 5 (%)                      |                                                              | This field is the fifth current-carrying capacity limit step value ( $\pm 10.0\%$ for example) for regulators. A regulator may have a variable limit depending on the tap step. This field does not need to be filled in if an associated transformer model is assigned.                        |
| Model - Resistance (per unit)                      |                                                              | This field is the per unit resistance (on transformer power rating and nominal voltage rating). This field does not need to be filled in if an associated transformer model is assigned.                                                                                                        |
| Model - Secondary-Tertiary Reactance (percent)     |                                                              | This field is for three-winding transformers. It is the leakage reactance (%) measured in the secondary with the tertiary short circuited and the primary open. The impedance value is referred to the transformer power base, and it is used to determine the total winding impedance values.  |
| Model - Secondary-Tertiary Resistance (percent)    |                                                              | This field is for three-winding transformers. It is the leakage resistance (%) measured in the secondary with the tertiary short circuited and the primary open. The impedance value is referred to the transformer power base, and it is used to determine the total winding impedance values. |
| Model - Size Units                                 | Enumeration: MVA, kVA, or Regulated_kVA                      | This field lists the units for specifying the size of the transformer. This field does not need to be filled in if an associated transformer model is assigned.                                                                                                                                 |
| Model - Transformer/Regulator Size                 |                                                              | This field is the size of the transformer. This field does not need to be filled in if an associated transformer model is assigned.                                                                                                                                                             |
| Network Device ID                                  |                                                              | This field is the automatically created unique ID. Since each device in the DMS model must have a unique ID, it is expected that this field should not be entered in the association.                                                                                                           |
| Network Device Name                                |                                                              | This field is the device name. This field should probably not be entered in the association (most devices within a station will have unique names).                                                                                                                                             |
| Primary - Associated Tap Model Name                | Drop-down List                                               | This field is the name of the primary winding tap changer model (if the winding has a tap) in the global resources file (global.csv).                                                                                                                                                           |
| Primary - Control Type                             | Enumeration: Fixed, Local_Only, Remote_Only, or Remote_Local | This field is the control type mechanism for the primary winding tap changer (if the winding has a tap).                                                                                                                                                                                        |
| Primary - CT Full Secondary Amps                   |                                                              | This field is the primary winding tap changer current transformer full load secondary amp value (this field may be defaulted in the DMS_Defaults.csv file). This field is only needed for voltage regulating tap changers.                                                                      |
| Primary - CT Ratio                                 |                                                              | This field is the primary winding tap changer current transformer ratio. This field is only needed for voltage regulating tap changers.                                                                                                                                                         |
| Primary - Line Drop Compensation Reactance (Volts) |                                                              | This field is the primary winding tap changer line drop compensator reactance setting (in Volts). This field is only needed for voltage regulating tap changers.                                                                                                                                |

| Element Name                                         |                                                                        |                                                                                                                                                                                                          |
|------------------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary - Line Drop Compensation Resistance (Volts)  |                                                                        | This field is the primary winding tap changer line drop compensator resistance setting (in Volts). This field is only needed for voltage regulating tap changers.                                        |
| Primary - Maximum Tap Step                           |                                                                        | This field is the primary winding tap changer maximum tap position (maybe +16). This field does not need to be filled in if the tap model name is filled in.                                             |
| Primary - Minimum Tap Step                           |                                                                        | This field is the primary winding tap changer minimum tap position (maybe -16). This field does not need to be filled in if the tap model name is filled in.                                             |
| Primary - Nominal Tap Position                       |                                                                        | This field is the primary winding tap changer nominal tap position. This field does not need to be filled in if the tap model name is filled in.                                                         |
| Primary - Normal Tap Position                        |                                                                        | This field is the primary winding tap changer normal tap position.                                                                                                                                       |
| Primary - Phase(s) Measured                          | Enumeration: A, B, C, AB,BC, AC, ABC                                   | This field is the primary winding tap changer phases that are measured. This field is only needed for voltage regulating tap changers.                                                                   |
| Primary - PT Ratio                                   |                                                                        | This field is the primary winding tap changer potential transformer (PT) ratio. This field is only needed for voltage regulating tap changers.                                                           |
| Primary - Tap Step Increment (%)                     |                                                                        | This field is the primary winding tap changer tap step increment (%). This field does not need to be filled in if the tap model name is filled in.                                                       |
| Primary - Two Sided Bandwidth (120.0 Volt Base)      |                                                                        | This field is the primary winding tap changer two sided bandwidth (in Volts on 120.0 Volt base). This field is only needed for voltage regulating tap changers.                                          |
| Primary - Voltage Measured Line to Line (true = L-L) | Boolean                                                                | This field is the primary winding tap changer flag indicating that the voltage is measured line to line (true) or line to ground (false). This field is only needed for voltage regulating tap changers. |
| Primary - Voltage Reduction (%)                      |                                                                        | This field is the primary winding tap changer voltage reduction setting (in %). This field is only needed for tap changers that have voltage reduction capability.                                       |
| Primary - Voltage Target (120.0 Volt Base)           |                                                                        | This field is the primary winding tap changer voltage target setting (in Volts on a 120.0 volt base). This field is only needed for voltage regulating tap changers.                                     |
| Primary Phase Connection                             | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the primary (high voltage side) phasing. For a power transformer (connected delta/grounded wye), the typical primary phasing will be ABC.                                           |
| Primary Tap Type                                     | Enumeration: None, Fixed, or Regulating                                | This field is the primary tap changer type. For a power transformer, it is common that the primary winding has a fixed tap (or possibly no tap).                                                         |
| Primary Voltage (kV)                                 |                                                                        | This field is the primary rated voltage (kV, line to line).                                                                                                                                              |
| Primary Winding Type                                 | Enumeration: None, Delta, Grounded_Wye, Wye, ZigZag                    | This field is the primary winding type. For a power transformer (connected delta/grounded wye), the primary winding type is Delta.                                                                       |

| Element Name                                           |                                                              |                                                                                                                                                                                                                              |
|--------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Secondary - Associated Tap Model Name                  |                                                              | This field is the name of the secondary winding tap changer model in the global resources file (global.csv).                                                                                                                 |
| Secondary - Control Type                               | Enumeration: Fixed, Local_Only, Remote_Only, or Remote_Local | This field is the control type mechanism for the secondary winding tap changer (if the winding has a tap).                                                                                                                   |
| Secondary - CT Full Secondary Amps                     |                                                              | This field is the secondary winding tap changer current transformer full load secondary amp value (this field may be defaulted in the DMS_Defaults.csv file). This field is only needed for voltage regulating tap changers. |
| Secondary - CT Ratio                                   |                                                              | This field is the secondary winding tap changer current transformer ratio. This field is only needed for voltage regulating tap changers.                                                                                    |
| Secondary - Line Drop Compensation Reactance (Volts)   |                                                              | This field is the secondary winding tap changer line drop compensator reactance setting (in Volts). This field is only needed for voltage regulating tap changers.                                                           |
| Secondary - Line Drop Compensation Resistance (Volts)  |                                                              | This field is the secondary winding tap changer line drop compensator resistance setting (in Volts). This field is only needed for voltage regulating tap changers.                                                          |
| Secondary - Maximum Tap Step                           |                                                              | This field is the secondary winding tap changer maximum tap position (maybe +16). This field does not need to be filled in if the tap model name is filled in.                                                               |
| Secondary - Minimum Tap Step                           |                                                              | This field is the secondary winding tap changer minimum tap position (maybe -16). This field does not need to be filled in if the tap model name is filled in.                                                               |
| Secondary - Nominal Tap Position                       |                                                              | This field is the secondary winding tap changer nominal tap position. This field does not need to be filled in if the tap model name is filled in.                                                                           |
| Secondary - Normal Tap Position                        |                                                              | This field is the secondary winding tap changer normal tap position.                                                                                                                                                         |
| Secondary - Phase(s) Measured                          | Enumeration: A, B, C, AB,BC, AC, ABC                         | This field is the primary winding tap changer phases that are measured. This field is only needed for voltage regulating tap changers.                                                                                       |
| Secondary - PT Ratio                                   |                                                              | This field is the secondary winding tap changer potential transformer (PT) ratio. This field is only needed for voltage regulating tap changers.                                                                             |
| Secondary - Tap Step Increment (%)                     |                                                              | This field is the secondary winding tap changer tap step increment (%). This field does not need to be filled in if the tap model name is filled in.                                                                         |
| Secondary - Two Sided Bandwidth (120.0 Volt Base)      |                                                              | This field is the primary winding tap changer two sided bandwidth (in Volts on 120.0 Volt base). This field is only needed for voltage regulating tap changers.                                                              |
| Secondary - Voltage Measured Line to Line (true = L-L) | Boolean                                                      | This field is the secondary winding tap changer flag indicating that the voltage is measured line to line (true) or line to ground (false). This field is only needed for voltage regulating tap changers.                   |

| Element Name                                         |                                                                        |                                                                                                                                                                                                                             |
|------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Secondary - Voltage Reduction (%)                    |                                                                        | This field is the secondary winding tap changer voltage reduction setting (in %). This field is only needed for tap changers that have voltage reduction capability.                                                        |
| Secondary - Voltage Target (120.0 Volt Base)         |                                                                        | This field is the secondary winding tap changer voltage target setting (in Volts on a 120.0 volt base). This field is only needed for voltage regulating tap changers.                                                      |
| Secondary Phase Connection                           | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the secondary (low voltage side) phasing. For a power transformer (connected delta/grounded wye), the typical secondary phasing will be ABCN.                                                          |
| Secondary Tap Type                                   | Enumeration: None, Fixed, or Regulating                                | This field is the primary tap changer type. For a power transformer, it is common that the secondary winding has a regulating tap (if it is a TCUL). A fixed tap may exist if the transformer is not a TCUL.                |
| Secondary Voltage - Nominal (kV)                     |                                                                        | This field is the primary nominal voltage (kV, line to line). It is possible to have a transformer with a rated voltage that is slightly higher or lower than the system nominal voltage.                                   |
| Secondary Voltage - Rated (kV)                       |                                                                        | This field is the secondary rated voltage (kV, line to line).                                                                                                                                                               |
| Secondary Winding Type                               | Enumeration: None, Delta, Grounded_Wye, Wye, ZigZag                    | This field is the primary winding type. For a power transformer (connected delta/grounded wye), the primary winding type is Grounded Wye.                                                                                   |
| Tertiary - Associated Tap Model Name                 |                                                                        | This field is the name of the Tertiary winding tap changer model in the global resources file (global.csv).                                                                                                                 |
| Tertiary - Control Type                              | Enumeration: Fixed, Local_Only, Remote_Only, or Remote_Local           | This field is the control type mechanism for the Tertiary winding tap changer (if the winding has a tap).                                                                                                                   |
| Tertiary - CT Full Secondary Amps                    |                                                                        | This field is the Tertiary winding tap changer current transformer full load secondary amp value (this field may be defaulted in the DMS_Defaults.csv file). This field is only needed for voltage regulating tap changers. |
| Tertiary - CT Ratio                                  |                                                                        | This field is the Tertiary winding tap changer current transformer ratio. This field is only needed for voltage regulating tap changers.                                                                                    |
| Tertiary - Line Drop Compensation Reactance (Volts)  |                                                                        | This field is the Tertiary winding tap changer line drop compensator reactance setting (in Volts). This field is only needed for voltage regulating tap changers.                                                           |
| Tertiary - Line Drop Compensation Resistance (Volts) |                                                                        | This field is the Tertiary winding tap changer line drop compensator resistance setting (in Volts). This field is only needed for voltage regulating tap changers.                                                          |
| Tertiary - Maximum Tap Step                          |                                                                        | This field is the Tertiary winding tap changer maximum tap position (maybe +16). This field does not need to be filled in if the tap model name is filled in.                                                               |
| Tertiary - Minimum Tap Step                          |                                                                        | This field is the Tertiary winding tap changer minimum tap position (maybe -16). This field does not need to be filled in if the tap model name is filled in.                                                               |
| Tertiary - Nominal Tap Position                      |                                                                        | This field is the Tertiary winding tap changer nominal tap position. This field does not need to be filled in if the tap model name is filled in.                                                                           |

| Element Name                                          |                                                                        |                                                                                                                                                                                                           |
|-------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tertiary - Normal Tap Position                        |                                                                        | This field is the Tertiary winding tap changer normal tap position.                                                                                                                                       |
| Tertiary - Phase(s) Measured                          | Enumeration: A, B, C, AB,BC, AC, ABC                                   | This field is the Tertiary winding tap changer phases that are measured. This field is only needed for voltage regulating tap changers.                                                                   |
| Tertiary - PT Ratio                                   |                                                                        | This field is the Tertiary winding tap changer potential transformer (PT) ratio. This field is only needed for voltage regulating tap changers.                                                           |
| Tertiary - Tap Step Increment (%)                     |                                                                        | This field is the Tertiary winding tap changer tap step increment (%). This field does not need to be filled in if the tap model name is filled in.                                                       |
| Tertiary - Two Sided Bandwidth (120.0 Volt Base)      |                                                                        | This field is the Tertiary winding tap changer two sided bandwidth (in Volts on 120.0 Volt base). This field is only needed for voltage regulating tap changers.                                          |
| Tertiary - Voltage Measured Line to Line (true = L-L) | Boolean                                                                | This field is the Tertiary winding tap changer flag indicating that the voltage is measured line to line (true) or line to ground (false). This field is only needed for voltage regulating tap changers. |
| Tertiary - Voltage Reduction (%)                      |                                                                        | This field is the Tertiary winding tap changer voltage reduction setting (in %). This field is only needed for tap changers that have voltage reduction capability.                                       |
| Tertiary - Voltage Target (120.0 Volt Base)           |                                                                        | This field is the Tertiary winding tap changer voltage target setting (in Volts on a 120.0 volt base). This field is only needed for voltage regulating tap changers.                                     |
| Tertiary Phase Connection                             | Enumeration: ABC, ABCN, A, AN, B, BN, C, CN, AB, ABN, BC, BCN, AC, ACN | This field indicates the Tertiary (low voltage side) phasing. For a power transformer (connected delta/grounded wye), the typical Tertiary phasing will be ABCN.                                          |
| Tertiary Tap Type                                     | Enumeration: None, Fixed, or Regulating                                | This field is the Tertiary tap changer type.                                                                                                                                                              |
| Tertiary Voltage - Nominal (kV)                       |                                                                        | This field is the Tertiary nominal voltage (kV, line to line). It is possible to have a transformer with a rated voltage that is slightly higher or lower than the system nominal voltage.                |
| Tertiary Voltage - Rated (kV)                         |                                                                        | This field is the Tertiary rated voltage (kV, line to line).                                                                                                                                              |
| Tertiary Winding Type                                 | Enumeration: None, Delta, Grounded_Wye, Wye, ZigZag                    | This field is the Tertiary winding type. For a power transformer (connected delta/grounded wye), the Tertiary winding type is Grounded Wye.                                                               |
| Transformer Type                                      | Enumeration: Power, Regulator, StepUpStepDown, Auto, Distribution      | This field indicates the type of transformer. Most of the time, the transformer will be either a Power transformer or a Regulator.                                                                        |



## 8. <UIContainer> Postage Stamp Display

Substation Editor is able to create full graphics displays and enhancements to the DNOM model for devices that are outside the substation fence. This combination of displays and models is known as a Postage Stamp display.

The enhancements to the model consist of new Measurements and Measurement Points that are attached to nodes and devices defined in the station externals file.

Substation Editor is able to create Postage Stamp Displays both for devices that are wholly defined in a single station, and for boundary devices, which are partially defined in different stations.

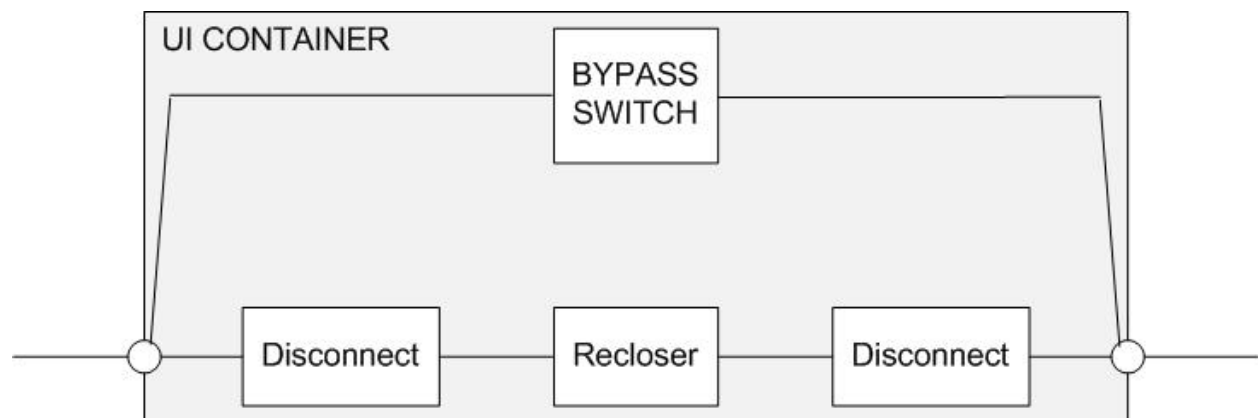
The supported device types for Postage Stamp displays are:

- Capacitors
- Monitoring Points (Sensor Locations, Fault Detectors, and others)
- Regulators
- Switches and Reclosers
- Topology Device Groups (currently limited to UI Containers)

Each device, when traced electrically, has a circuit pattern such as:

Line - Capacitor - Line

Or:



A Postage Stamp Display can only be constructed from a complete circuit pattern. For example, if the UI Container definition does not include one of the Disconnect Switches that is positioned between two included line segments, Substation Editor is not able to trace the circuit.

This chapter limits its discussion to the process of creating a Postage Stamp Display for a UI Container, although much of the material is applicable to the other devices.

## 8.1. Externals File Prerequisites

The Externals view (from Substation Editor's File menu > Import Station Externals operation) is expected to define every line segment, node, and device required to trace the circuit that is to be drawn by the Create Postage Stamp Display function. For example, the (concatenated, if boundary) UI Container requires<sup>[1]</sup>:

```
<UIContainer>
  <Name>UI_78372</Name>
  <Objects><ID>78372</ID>  <Type>Recloser  </Type></Objects>
  <Objects><ID>78387</ID>  <Type>Switch    </Type></Objects>
  <Objects><ID>78390</ID>  <Type>Switch    </Type></Objects>
  <Objects><ID>78379</ID>  <Type>Line      </Type></Objects>
  <Objects><ID>78376</ID>  <Type>Line      </Type></Objects>
  <Objects><ID>78381</ID>  <Type>Line      </Type></Objects>
  <Objects><ID>78378</ID>  <Type>Line      </Type></Objects>
  <Objects><ID>78392</ID>  <Type>Switch    </Type></Objects>
  <Objects><ID>78377</ID>  <Type>Line      </Type></Objects>
  <Objects><ID>78382</ID>  <Type>Line      </Type></Objects>
  <Objects><ID>78383</ID>  <Type>Line      </Type></Objects>
  <Objects><ID>78375</ID>  <Type>Line      </Type></Objects>
</UIContainer>
```

The order of Objects within the UI Container is important. However, the order of an object's definitions, shown below, is not important. They are listed here in UIContainer-order:

| Type       | <Id>  | <FromNode> | <ToNode> | <ToStation> |
|------------|-------|------------|----------|-------------|
| <Recloser> | 78372 | 78372      | 78372.1  | STATIONB    |
| <Switch>   | 78387 | 78387      | 78387.1  | STATIONB    |
| <Switch>   | 78390 | 78390      | 78390.1  |             |
| <Line>     | 78379 | 78395      | 78390    |             |
| <Line>     | 78376 | 78394      | 78395    |             |
| <Line>     | 78381 | 78390.1    | 78372.1  |             |
| <Line>     | 78378 | 78395      | 78387.1  |             |
| <Switch>   | 78392 | 78392      | 78392.1  |             |
| <Line>     | 78377 | 78396      | 78387    |             |
| <Line>     | 78382 | 78392.1    | 78372    |             |
| <Line>     | 78383 | 78396      | 78392    |             |
| <Line>     | 78375 | 78397      | 78396    |             |

For a Boundary Device, only the devices that belong to a station are defined in that station's externals file. Those which straddle the boundary are defined in both stations' externals file. For example:

| STATIONA Externals                                                                                                                                                                                                                                                                                                                                                                                                                                              | STATIONB Externals                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> &lt;UIContainer&gt;   &lt;Name&gt;UI_78372&lt;/Name&gt;   &lt;Objects&gt;&lt;ID&gt;78372 &lt;Type&gt;Recloser   &lt;Objects&gt;&lt;ID&gt;78387 &lt;Type&gt;Switch   &lt;Objects&gt;&lt;ID&gt;78390 &lt;Type&gt;Switch   &lt;Objects&gt;&lt;ID&gt;78377 &lt;Type&gt;Line   &lt;Objects&gt;&lt;ID&gt;78382 &lt;Type&gt;Line   &lt;Objects&gt;&lt;ID&gt;78383 &lt;Type&gt;Line   &lt;Objects&gt;&lt;ID&gt;78375 &lt;Type&gt;Line &lt;/UIContainer&gt; </pre> | <pre> &lt;UIContainer&gt;   &lt;Name&gt;UI_78372&lt;/Name&gt;   &lt;Objects&gt;&lt;ID&gt;78372 &lt;Type&gt;Recloser   &lt;Objects&gt;&lt;ID&gt;78387 &lt;Type&gt;Switch   &lt;Objects&gt;&lt;ID&gt;78392 &lt;Type&gt;Switch   &lt;Objects&gt;&lt;ID&gt;78379 &lt;Type&gt;Line   &lt;Objects&gt;&lt;ID&gt;78381 &lt;Type&gt;Line   &lt;Objects&gt;&lt;ID&gt;78378 &lt;Type&gt;Line   &lt;Objects&gt;&lt;ID&gt;78376 &lt;Type&gt;Line &lt;/UIContainer&gt; </pre> |

When multiple station externals files are loaded into Substation Editor (File > Import Station Externals > Multiple Stations), they are loaded in alphabetical order, regardless of the order in which they are selected. Depending upon how the station externals files are produced, this may affect which Object is the "first" one. It is important that the same Object is listed first in both station externals files.

### 8.1.1. Rules for a Valid UIContainer

For the purposes of constructing a Postage Stamp Display, a valid UIContainer has two requirements:

1. Every node on every (non-line) device listed within the container is connected to a line segment listed within the container.
  - A UIContainer must contain at least: -Line-nonLine-Line-
2. Exactly two line segments each have exactly one unmatched node.
  - There are exactly two external connections from the UIContainer.
  - Exactly two devices each have exactly one unmatched node, and those two devices are line segments.
  - All devices have at least two terminals.

## 8.2. Substation Editor Configuration and ESETs

The naming convention for ESET pictures differs according to the Create Postage Stamp Displays pop-up tab on which the equipment is listed. For example:

| Tab               | Equipment Type    | Circuit Pattern | Picture Name         |
|-------------------|-------------------|-----------------|----------------------|
| Capacitors        | Capacitor         | CLL             | CAPACITOR_CLL        |
| Regulators        | Transformer       | <TTT>LL         | REGULATOR_<TTT>LL    |
| Monitoring Points | Fault Detector    | FLS             | MONITORING_POINT_FLS |
| Monitoring Points | Measurement Point | MLL             | MONITORING_POINT_MLL |
| Switches          | Switch            | SLL             | SWITCH_SLL           |

| Tab                    | Equipment Type | Circuit Pattern | Picture Name         |
|------------------------|----------------|-----------------|----------------------|
| Switches               | Recloser       | RLL             | RECLOSER_RLL         |
| Topology Device Groups | UI Container   | FLXLSRLRLXL     | RECLOSER_FLXLSRLRLXL |

Note that parallel devices are grouped using left and right carets, and that these carets must appear in the picture name (for example, <TTT>LL).

Note that even though the Circuit Pattern of the <UIContainer> starts with the Fuse ("F"), the prefix of the picture is "RECLOSER" because that's the first device in the container.

The following is the easiest way to determine what SSE expects the picture name to be:

1. Open the Tools > Configuration > General tab.
2. Set Log Level to Debug3 or greater.
3. Open File > Import Station Externals > Single Station (or Multiple Stations).
4. Open Tools > Create Postage Stamp Displays.
5. Select the appropriate tab and click Populate Grid.
6. Select the equipment you need information about and click Create Views.

The name of the non-existent picture appears in the grid list of missing pictures. The circuit pattern of each device traced by Populate Grid is also logged in the trace log. Search the trace log for the device name, or the externals view name, bracketed by colons, for example:

```
-D3-found:Switch:MULTIPLE_EXTERNALS::SLS
-D3-found:Switch:STATIONB_EXTERNALS:CXC-63869:SL
```

For UI Containers, search for "UicTrace for <UIContainer>":

```
UicTrace for <UIContainer>=R758_UIC (R758) with Circuit Pattern = FLXLSRLRLXL
```

The way Substation Editor constructs both the picture name and the proxies is described below.

## 8.2.1. Tracing the Circuit

The following example shows an abstract form of Substation Editor's loop through a list of Objects in the UIContainer. Error checking is not shown.

From a very high level, the algorithm is:

```
Start with the first <Objects> in the <UIContainer>
Until all <Objects> are matched,
    Try to find a match for the <ToNode>.
    Else try to find a match for the <FromNode>
    Else take the first <Objects> on the "Roads Not Taken" list.
```

When reading the following pseudo-code, note that:

- Once an Object is added to the CircuitTrace, it is removed from consideration (that is, a device appears once and only once in the circuit pattern).
- The "Roads Not Taken" list is examined first-in-first-out. This means that after the path originating from the first Object's ToNode completes, the first Object's FromNode is then traced.
- Line segments that connect outside the UIContainer (i.e., external lines) are inserted into the circuit pattern at the very first opportunity. This means, using the UI\_78372 example, that Line 78376 (external) is inserted after Line 78379 (from the Disconnect switch) and before Line 78378 (to the Bypass switch).

```
currentDevice = first <Objects> in <UIContainer>
for each device which has a node matching currentDevice's <FromNode>,
    add that device to the "Roads Not Taken" list
TopOfLoop:
    add currentDevice to the CircuitTrace
    if currentDevice is in a set of parallel devices,
        add each device in the set to the CircuitTrace
    if an external line attaches to currentDevice,
        add the external line to the CircuitTrace

    // Find the next device in the circuit
    for every remaining device which has a node matching currentDevice's <ToNode>,
        the first such device becomes the next currentDevice;
        all the others are added to the "Roads Not Taken" list;
        goto TopOfLoop;

    // Here when no match was found for the <ToNode>
    for every remaining device which has a node matching currentDevice's <FromNode>,
        the first such device becomes the next currentDevice;
        all the others are added to the "Roads Not Taken" list;
        goto TopOfLoop;

    // Here when no match was found for either node
    if the "Roads Not Taken" list is not empty,
        the first device on the "Roads Not Taken" list becomes currentDevice
        goto TopOfLoop;

    // Here when we have nothing else to look at: we're done.
```

Using the UI\_78372 example, the Circuit Trace is built like this:

It is easiest, before beginning the loop, to identify the external lines:

| ID    | Internal Node | External Node |
|-------|---------------|---------------|
| 78376 | 78395         | 78394         |
| 78375 | 78396         | 78397         |

Start with Recloser 78372, and add it to the Circuit Trace. Find all the devices that attach to its <FromNode>, and add them to the "Roads Not Taken" list:

| Circuit Trace  | Roads Not Taken |
|----------------|-----------------|
| Recloser 78372 | Line 78382      |

Does any external line attach to either of the Recloser's nodes? No.

Follow the Recloser's <ToNode> to Line 78381, and add it to the Circuit Trace. Find all the devices that attach to its other node (which happens to be its <FromNode>) and add them to the "Roads Not Taken" list.

| Circuit Trace  | Roads Not Taken |
|----------------|-----------------|
| Recloser 78372 | Line 78382      |
| Line 78381     | Switch 78390    |

Does any external line attach to either of the Line's nodes? No.

Follow Line 78381's <ToNode> back to the Recloser already in the Circuit Trace. There are no other matches.

Try Line 78381's <FromNode>. There is a match with Switch 78390. Add it to the Circuit Trace (removing it from "Roads Not Taken") and add all the devices that attach to its other node (which again happens to be its <FromNode>).

| Circuit Trace  | Roads Not Taken |
|----------------|-----------------|
| Recloser 78372 | Line 78382      |
| Line 78381     |                 |
| Switch 78390   | Line 78379      |

Does any external line attach to either of the Switch's nodes? No.

Does any other untraced device attach to the Switch's <ToNode>? No.

Does any other untraced device attach to the Switch's <FromNode>? Yes, Line 78379. Process it.

| Circuit Trace  | Roads Not Taken |
|----------------|-----------------|
| Recloser 78372 | Line 78382      |
| Line 78381     |                 |
| Switch 78390   |                 |
| Line 78379     | Line 78378      |

Does any external line attach to either of the Line's nodes? Yes, add it to the trace, and then continue to check for other attachments.

| Circuit Trace  | Roads Not Taken |
|----------------|-----------------|
| Recloser 78372 | Line 78382      |
| Line 78381     |                 |
| Switch 78390   |                 |

| Circuit Trace | Roads Not Taken |
|---------------|-----------------|
| Line 78379    |                 |
| Line 78376    |                 |
| Line 78378    |                 |

The process continues until it has no more Roads Not Taken.

| Circuit Trace  | Roads Not Taken |
|----------------|-----------------|
| Recloser 78372 |                 |
| Line 78381     |                 |
| Switch 78390   |                 |
| Line 78379     |                 |
| Line 78376     |                 |
| Line 78378     |                 |
| Switch 78387   |                 |
| Line 78377     |                 |
| Line 78375     |                 |
| Line 78383     |                 |
| Switch 78392   |                 |
| Line 78382     |                 |

## 8.2.2. Determining the Sorted Circuit Pattern

The unsorted circuit pattern is a simple concatenation of the initial letters of the Object's Types, with two substitutions:

- "X" is used for an external line.
- "Z" is used for a serializer.

For UI\_78372, the unsorted circuit pattern is:

- RLSLXLXLXL

If there are any parallel devices in the circuit, they are grouped using left and right angle brackets (for example, RL<SSS>LXLXL<SSS>L).

If you continually move the first letter to the last position, you will eventually arrive at the alphabetically lowest ordering of the circuit pattern:

RLSLXLXLXL ⇒ LSLXLXLXSLR ⇒ ... ⇒ LRLSLXLXL

### Note

Parallel devices are moved as a whole: <SSS>LXLXL<SSS>L RL becomes

LXLSLXL<SSS>LRL<SSS>.

The alphabetically lowest ordering of the circuit pattern is also known as the sorted Circuit Pattern. It appears as the Type on the Postage Stamp pop-up's Topological Device Groups tab.

### 8.2.3. Naming the Picture

The ESET picture used to draw the Postage Stamp Display must be named according to the first Object (device) listed in the UIContainer, followed by an underscore, followed by the sorted Circuit Pattern. For UI\_78372, that is:

RECLOSER\_ LRLSLXLXLXS

## 8.3. How Proxies for Lines and Devices are Named

Six proxies are generated for each device (including line segments) in the Circuit Pattern, formatted as:

|                          |                |              |
|--------------------------|----------------|--------------|
| %/device/number/%        | %LINE3%-       | 114639483    |
| %/device/number/NAME%    | %LINE3NAME%    | 1075-32998   |
| %/device/number/STATION% | %LINE3STATION% | PEBBLE BEACH |
| %/device/number/FRNODE%  | %LINE3FRNODE%  | 84N32SE      |
| %/device/number/TONODE%  | %LINE3TONODE%  | 16SW499E     |
| %.device/number/PHASE%   | %LINE3PHASE%   | ABC          |

The numbers are assigned sequentially for each equipment type in the sorted Circuit Pattern starting from the left:

For example, using the second Switch listed in the UI\_78372 UI Container:

|                  |          |
|------------------|----------|
| %SWITCH2%        | 78387    |
| %SWITCH2NAME%    | X587     |
| %SWITCH2FRNODE%  | 78387    |
| %SWITCH2TONODE%  | 78387.1  |
| %SWITCH2STATION% | STATIONA |

### 8.3.1. How Proxies Can Be Used to Set Properties

Proxies can be used to create templates for properties in three places:

1. On the Properties tab

For creating displays in bulk, having individual entries on the Properties tab is not very useful,



but it is possible. Once the Postage Stamp display is created, any of the editable properties exposed on the Properties tab can be modified, and a proxy string can be used to do so. A value entered on the Properties tab overwrites any value that may have been set by PictureNetworkAssociation or by the Picture/Symbol Placement Properties.

Proxies entered on the Measurements tab and the Equipment tab are not resolved.

## 2. In the Properties grid of the Tools > Configuration > Associations tab

Base default values for properties may be set using the properties grid. Proxies may be used in defining those base default values.

For example, suppose the placement of an ALG\_KV\_S picture were configured to create (Create Meas = checked) an Analog (MEAS Group = AnalogMeasurementGroup). One of the properties of an Analog Measurement is Measurement: Composite ID. You could define the default Composite ID to be, for example,  
%SUBSTN%.RECLSR.%RECLOSER1NAME%.SOURCE\_KV.

These base default values are preserved in PictureNetworkAssociation. They are applied to every placement of the Element (picture or symbol).

## 3. On the Define Link tab of the Picture Placement Properties or Symbol Placement Properties pop-up from a Nested Picture in FG Display Builder.

Values that overwrite those defined in PictureNetworkAssociation may be defined for each placement of a picture or symbol within a Nested Picture, as described in [Modeling Equipment Groups](#).

Continuing the example of defining the Measurement: Composite ID, the proxy string might be set for a specific placement of the ALG\_KV\_S picture to be:  
%SUBSTN%.RECLSR.%RECLOSER1NAME%.SRC\_A\_KV.

In Display Builder, the syntax would look like:

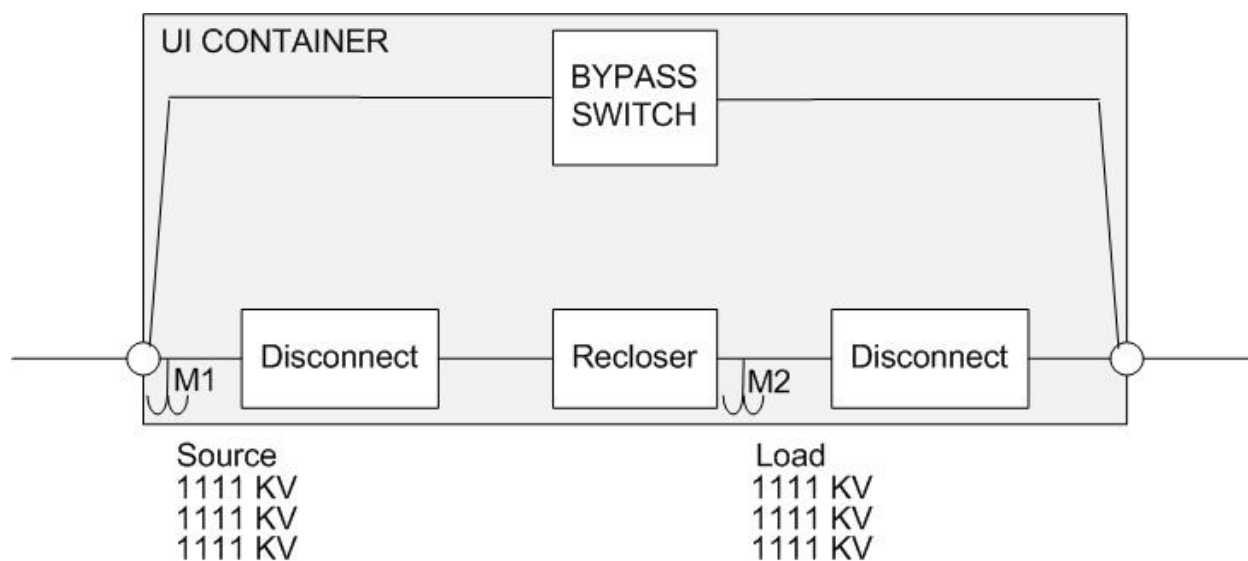
```
MEAS[MEASUREMENT:COMPOSITEID = %SUBSTN%.RECLSR.%RECLOSER1NAME%.SRC_A_KV
```

Examples of how this looks when preserved in the ESET can be found in [Associating Analogs to MeasurementPoints](#). Notice that the value can be different for each placement. The per-placement defaults overwrite anything defined in PictureNetworkAssociation.

## 8.3.2. Associating Analogs to MeasurementPoints

For each Analog MEAS Type-to-specific-MeasurementPoint pairing, you need to create a PictureNetworkAssociation (defined on the Tools > Configuration > Associations tab) between a unique Analog FG picture and a unique MeasurementPoint FG picture.

For example, consider the following two measurement points, M1 and M2, each with three phase-specific voltage measurements:



For the above Analogs and MeasurementPoints, you need four pictures (each Analog associated to a given MeasurementPoint can use the same picture definition):

| Source-Side         |                 | Load-Side           |                 |
|---------------------|-----------------|---------------------|-----------------|
| Object              | Example Picture | Object              | Example Picture |
| MeasurementPoint M1 | MEASPT_SRC      | MeasurementPoint M2 | MEASPT_LOD      |
| Analog for Phase A  | ALG_KV_S        | Analog for Phase A  | ALG_KV_L        |
| Analog for Phase B  | ALG_KV_S        | Analog for Phase B  | ALG_KV_L        |
| Analog for Phase C  | ALG_KV_S        | Analog for Phase C  | ALG_KV_L        |

And two PictureNetworkAssociations:

| <PictureNetworkAssociation> property | Label              | PNA #1 value           | PNA #2 value           |
|--------------------------------------|--------------------|------------------------|------------------------|
| <ElementName>                        | Element:           | ALG_KV_S               | ALG_KV_L               |
| <DnomObjectname>                     | DNOM Type:         | <null>                 | <null>                 |
| <CreateDnomSymbol>                   | Create DNOM Symbol | true                   | true                   |
| <pMeas><MeasType>                    | MEAS Type          | MeasVolts              | MeasVolts              |
| <pMeas><MeasGroup>                   | MEAS Group         | AnalogMeasurementGroup | AnalogMeasurementGroup |
| <MeasDefaultAssoc>                   | Default Assoc      | MEASPT_SRC             | MEASPT_LOD             |
| <CreateMeasObject>                   | Create Meas        | true                   | true                   |

The table above shows a minimum set of properties that must be configured to make your Postage Stamp display work. Many other properties are available, and you would want to change some properties for different readings. For example, "MeasVolts" is not the correct MEAS Type for an amps reading.

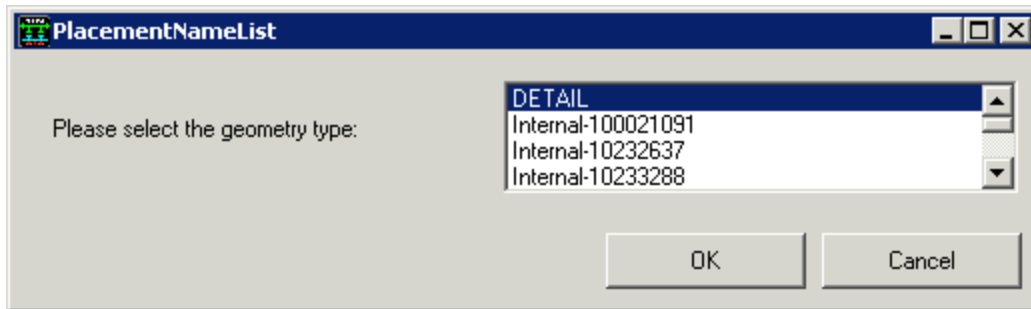
You can also configure properties in the ESET as described in [How Proxies Can Be Used to Set](#)

**Properties**, especially those properties that vary from one placement to the next. For example, the Source voltages in the example could have their Composite ID and Phase properties tailored to suit your needs:

```
picture_in_picture "ALG_KV_S"
(
  origin(39 458)
  composite_key
  (
    record("MEAS[MEASUREMENT:COMPOSITEID]") record_key("%RECLOSER1NAME%.RECLSR.SRC_A_KV")
    record("MEAS[MEASUREMENT:PHASE]") record_key("A")
  )
)
picture_in_picture "ALG_KV_S"
(
  origin(39 480)
  composite_key
  (
    record("MEAS[MEASUREMENT:COMPOSITEID]") record_key("%RECLOSER1NAME%.RECLSR.SRC_B_KV")
    record("MEAS[MEASUREMENT:PHASE]") record_key("B")
  )
)
picture_in_picture "ALG_KV_S"
(
  origin(39 502)
  composite_key
  (
    record("MEAS[MEASUREMENT:COMPOSITEID]") record_key("%RECLOSER1NAME%.RECLSR.SRC_C_KV")
    record("MEAS[MEASUREMENT:PHASE]") record_key("C")
  )
)
```

## 8.4. Creating a Postage Stamp Display

1. Open the SUBEDITOR file (from the File > Open menu) into which you want to save the created view(s).
2. Import the Station Externals file of a single station, or of both stations that share a boundary.  
  
From the File menu, choose Import Station Externals, and select either Single Station or Multiple Stations.
3. Select a file and click Open for as many files as you need to import. To end the selection, click Cancel.
4. Substation Editor opens a list of geometry types that are available in the selected files. Select a geometry type and click OK. Click Cancel to cancel the entire import process.



5. To find the desired Postage Stamp Equipment, from the Tools > Create Postage Stamp Displays pop-up, choose the desired equipment tab and click Populate Grid.

The grid is filled with the equipment that could be traced successfully, and a warning message indicates how many could not be traced (if any).

More information about why the equipment could not be traced is available in the Messages tab of the main Substation Editor form.

6. Select the desired boundary equipment and click Create Views. If the required picture is available in the ESET, the new view(s) appear.

## 8.5. Processing a Different Boundary

If a boundary involves a station that has not yet been imported, you must exit Substation Editor (or open a different SUBEDITOR file) to clear the Multiple Externals view before importing any more externals.

[1] More is required for correct ADMS modeling. Only what is required for Postage Stamp modeling is listed here.

## 9. SSE Tools

SSE Tools are used for batch processing common modeling tasks. They are installed in the \Tools\bin directory and are expected to run in the same environment as the Substation Editor (although they may use different directories).

### 9.1. About Logging

There are three significant differences between Substation Editor and the SSE Tools in the way messages are output.

#### 9.1.1. Where Messages Are Sent

Messages from Substation Editor may appear:

- On the Messages tab of the main SSE form
- On the Messages tab of the Postage Stamp Displays window
- In Message Box pop-ups
- In the trace log

Messages from SSE Tools are only output to the console and to the trace log. Every message sent to the console is also captured in the trace log.

Messages that are sent by SSE code to one of the Messages tabs, or to a Message Box pop-up, are instead captured by SSE Tools in the trace log. When SSE Tools can detect that a message is an Error or Warning, it also attempts to send that message to the console (depending upon the current severity filter).

#### 9.1.2. How Filtering is Handled

Substation Editor does not allow filtering of Messages tabs and Message Box pop-ups. You are allowed to control the size of the trace log by filtering out messages whose severity is less than the SubstationEditorConfiguration <LogLevel> property. <LogLevel> can be set from the SSE Tools > Configuration pop-up, on the General tab.

The SSE Tools do not allow filtering messages sent to the trace log. You are allowed to control which messages appear on the console by filtering out messages whose severity is less than the SubstationEditorConfiguration <LogLevel> property.

Because the SSE Tools do not have a graphical user interface, you may either use Substation Editor to set the filter, or you may edit the SubstationEditorConfiguration file directly. For a list of appropriate values, refer to the *Substation Editor Installation Guide*.



## Note

Fatal messages cannot be filtered out.

### 9.1.3. Where to Find the Trace Log

Substation Editor allows you to specify the location of the trace log using the SubstationEditorConfiguration <LogFilePath> property. If that is not defined, SSE attempts to open the file in %IDMS\_ROOT%\log. If it cannot do that, it attempts to open the file in SSE's working directory.

The SSE Tools, on the other hand, pay no attention to the <LogFilePath> property. The SSE Tools attempt to open the file in %IDMS\_ROOT%\log, and if unsuccessful, opens the trace log file in the SSE Tool's working directory.

### 9.1.4. About Reporting Problems

Note that the trace log file is GE's best source of information about any problem that you may encounter. When you report a problem, please include:

- Step-by-step instructions for reproducing the problem.
- All the materials necessary to follow your instructions to see what you saw.
- The trace logs from all relevant runs.

## 9.2. BatchPostageStamps

### 9.2.1. BatchPostageStamps.exe

The BatchPostageStamps tool is used to create or re-create large numbers of Postage Stamp Displays (and their measurements) in batch mode. It is invoked from the command line with or without the following arguments. If an argument is not given on the command line, you are prompted to provide it. For example:

```
>BatchPostageStamps.exe -inputdir %BPS_IN2%
                        -include %BPS_INC%\20131125_testINCLUDE_SWITCHES.csv
                        -out %BPS_OUT2%
...-E-Filename for SubstationEditorConfig file not given
Please tell me where to find the current Substation Editor Configuration file:
%BPS_CFG%\SubstationEditorConfiguration_standard.xml
...-E-Directory for station externals files not given
Please tell me where to find the newest Station Externals files: %BPS_EXT4%
```

The arguments are:

- **INPUTDIR:** The location where to find the SUBEDITOR files to be processed (for example,

\\SubstationEditor\\data). Each of the SUBEDITOR files is examined.

- If it contains a view whose name is included for removal, the view is deleted in the OUT copy of the SUBEDITOR file. All of the DNOM objects referenced (via <dnomRef>) by the placements in the view are also deleted.
- If a piece of equipment, whose view is included for creation, is found in the station's externals file, the corresponding view and DNOM objects are created in the OUT copy of the SUBEDITOR file using SSE's Create Postage Stamp Display functionality.

If a piece of equipment has changed owning stations, every SUBEDITOR file involved in the move must be found in INPUTDIR. For example, if equipment Z used to be a boundary device between stations A and B, but is now wholly owned by station C, all three A\_SUBEDITOR.xml, B\_SUBEDITOR.xml, and C\_SUBEDITOR.xml files must be in INPUTDIR.

- **INCLUDE:** Specifies the file containing the list of included equipment and views. For format and contents, see [The INCLUDE List](#).
- **OUT:** The directory in which the saved SUBEDITOR files are deposited.

Even if no changes are made to a particular SUBEDITOR file, it is re-saved to the OUT directory.

- **CONFIG:** Specifies the SubstationEditorConfiguration.xml file.

When creating a view, the tool uses the Element Set and Menu Element Set specified by the CONFIG file exactly as Substation Editor does.

This tool also uses the logging parameters specified by the CONFIG file.

- **EXTDIR:** The location where to find the station externals files to be processed.

It must contain all the station externals files that define the equipment whose views are included for creation in the INCLUDE file.

Continuing our example with equipment Z, C.xml must be in EXTDIR, but A.xml and B.xml are not necessary.

- **DUMP:** When present, this argument causes two results files to be created, in addition to the trace log. Both are CSV files named *BPSIncludeList\_dumpAllEquipment<timestamp>.csv\_* and are deposited in %IDMS\_ROOT%\log. The columns vary according to the BatchPostageStamps version, but the most common are:
  - **rowNumber:** The number of the row in the INCLUDE list which produced this results line.
  - **eqpPSTYPE:** The name of the Postage Stamp tab to which this piece of equipment belongs.
  - **eqpID:** The <Id> of the equipment.
  - **eqpVIEWNAME:** The name of the view to be deleted or created.
  - **eqpSUBSTN:** The value of the %SUBSTN% proxy to be used in that view.
  - **eqpDEVTYP:** The value of the %DEVTYP% proxy to be used in that view.
  - **eqpDEVICE:** The value of the %DEVICE% proxy to be used in that view.
  - **eqpPICTURE:** The name of the picture used to create the view.



- **eqpDELETE\_VIEW:** True or false: from the INCLUDE list.
- **eqpCREATE\_VIEW:** True or false: from the INCLUDE list.

The first results file is dumped after the INCLUDE list has been loaded, expanded, and validated.

The second results file is dumped after all processing is complete.

- **SEVERITY:** Controls the number of messages output to the console; messages of lower severity are suppressed. (Fatal is the most severe; Debug10 is the least severe.)
- **TRACE:** Controls the number of messages output to the log file; messages of lower severity are suppressed. (Fatal is the most severe; Debug10 is the least severe.) The default is Debug10, which can produce a large file; but when a hard-to-reproduce problem occurs, GE will greatly prefer to have a complete log.
- **WAIT:** Sets the number of seconds to wait for the user to respond to a question.

If BatchPostageStamps finds non-Fatal errors before beginning to loop through the SUBEDITOR files, it asks the user whether to continue. This argument determines how long BPS waits for an answer.

If no answer is provided, the answer is assumed to be "Yes, continue".

If the answer starts with "Y" (case insensitive), BPS continues. Any other answer causes BPS to abort the run.

- **LOGFILEPREFIX:** Specifies the string which prepends the created BPS log file name. If this value is "BPSTrace\_3LIONS" for example, all created logs will be called "BPSTrace\_3LIONS\*.log". If not specified, the prepended value is by default "BatchPostageStampsTrace".
- **MAXLOGFILESIZE:** Controls the maximum size of the BPS log file in MB. Once the size limit is reached, a new log is created. If not provided the default size is 10, meaning the maximum log size will be 10 MB.
- **LOGDIRECTORY:** Indicates what directory the log files should be placed in. If not provided, the default value is "%IDMS\_ROOT%\Log".

In addition to the saved SUBEDITOR files in the OUT directory, the tool outputs status messages to the console, and preserves audit trail and diagnostic messages in the trace log file as described above.

## 9.2.2. PrepBPSIncludeList.exe

Occasionally, modelers may not know that a station boundary has moved until CONVERTER starts warning that it cannot find the equipment to which a measurement is attached (the FWF172 error). The PrepBPSIncludeList tool uses those errors to construct a list of moved equipment, with all the information that will allow a BatchPostageStamps run to remove the affected views and measurements out of the old "owning" station and into the new one.

PrepBPSIncludeList is run from the command line with the following arguments:

- **CNVDIR:** The location where to find the CONVERTER log files containing FWF172 error messages.
- **OUT:** The name of the INCLUDE file to be produced.

This file is a CSV file whose first (left-most) columns have the format of the INCLUDE list. This file has four additional columns as follows:

| Column Header        | Data Value                                                                                                                                                                                                                                                                                                                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>OLD_STATION_1</b> | This column contains the name of the station for which the CONVERTER error occurred. This value is taken from the name of the CONVERTER log file.                                                                                                                                                                                       |
| <b>OLD_STATION_2</b> | If the moved equipment was a boundary device, and if it is no longer in either of its former stations, CONVERTER produces error messages for both of the formerly "owning" stations. This value in this column is taken from the name of the second CONVERTER log file in which the tool found an FWF172 error for the moved equipment. |
| <b>NEW_STATION_1</b> | If EXTDIR is specified, this column contains the name of the station externals file in which the equipment was found.                                                                                                                                                                                                                   |
| <b>NEW_STATION_2</b> | If EXTDIR is specified, and if the moved equipment is currently a boundary device, this column contains the name of the second externals file in which the equipment was found.                                                                                                                                                         |

When running BatchPostageStamps using the new INCLUDE file, the externals files for each of the listed stations must be in the EXTDIR directory, and the SUBEDITOR files for each of the listed stations must be in the INPUT directory.

- **INCLUDE:** Specifies the Master INCLUDE file, if any.

This argument is optional. Because BatchPostageStamps allows the modeler more options than Substation Editor does (for ViewName, SUBSTN, DEVICE, and Picture), the modeler may choose to construct a master INCLUDE file that preserves non-default values for specific pieces of equipment.

If a master INCLUDE file is given to PrepBPSIncludeList, it will search that file for each piece of moved equipment. If a matching ID is found, the tool uses the listed values for PSTYPE, ViewName, SUBSTN, DEVICE, and Picture in the new (OUT) INCLUDE file.

- **EXTDIR:** The location where to find the latest station externals files.

This argument is optional. If you run PrepBSPIncludeList with externals files, the tool searches them for the ID of the moved equipment. If found, the tool:

- Ensures that the PSTYPE is correct. Note that any value taken from a Master INCLUDE list is overwritten.
- If the ViewName is null, fills the ViewName per Substation Editor rules:

If the equipment <Name> is specified, ViewName is set to <Name>. Otherwise, ViewName is set to <Id>.

Note that any ViewName taken from a Master INCLUDE list is **not** overwritten.

- Fills the NEW\_STATION\_1 column.
- Fills the NEW\_STATION\_2 column if the moved equipment is a boundary device.

## About Logging

PrepBPSIncludeList behaves as described in [About Logging](#). However, PrepBPSIncludeList does not take a configuration parameter, so it is not possible to adjust the filtering of console messages.

### 9.2.3. PurgePlacements.exe

After a piece of equipment changes its owning station, its measurements and view are still modeled in the previously owning station. BatchPostageStamps.exe takes care of the SUBEDITOR file, and SubstationEditor is used to generate a new INTERNALS and set of PLACEMENTS files, but the old PLACEMENTS file is not deleted.

PurgePlacements is used to facilitate the cleanup of the CONVERTER input directory so that obsolete PLACEMENTS files do not cause an error. PurgePlacements produces batch files containing deletion commands for each of the obsolete files as follows:

- **PURGEPLACEMENTS\_ALL.BAT**

For each station that is found to have obsolete PLACEMENTS files in the input directory, one line is placed in this batch file, invoking the station-specific batch file to clean them up.

The names of the station-specific batch files are enclosed in quotes so that there are no issues with embedded spaces.

This batch file takes one parameter, the specification of the directory in which the target files are found. That parameter is passed to each of the station-specific batch files.

- **PURGEPLACEMENTS\_station.BAT**

For each station that is found to have obsolete PLACEMENTS files in the input directory, one batch file is created and named according to the station. For each PLACEMENTS file that is obsolete, one line is placed in its station-specific batch file, containing a delete command for that file.

This batch file takes one parameter, the specification of the directory in which the target files are found. That parameter is included in the specification of each of the files.

Each file specification is enclosed in double quotes, so that there are no issues with embedded spaces.

A PLACEMENTS file is determined to be obsolete when

- It was serialized by SubstationEditor.
- Its serialization timestamp exists but does not match the serialization timestamp of the INTERNALS file for the same station.

This means that it contains:

```
<SerializationInfo> <ModelingApplication>SubstationEditor</ModelingApplication>
<Timestamp>yyyymmddhhmmss</Timestamp> </SerializationInfo>
```

where the value of "yyyymmddhhmmss" is different than the same element in the station's INTERNALS file.

Each batch file may contain a timestamp in its name, if that option was chosen when PurgePlacements was run. The timestamp is of the time at which PurgePlacements was run.

PurgePlacements is run from the command line with the following arguments:

- INPUTDIR

The location where to find the INTERNALS and PLACEMENTS files to be examined.

- OUTPUTDIR

The location to write the batch files.

- TIMESTAMP

If given, causes a timestamp to be added to the names of the batch files.

## About Logging

PurgePlacements behaves as described in [About Logging](#). However, PrepBPSIncludeList does not take a configuration parameter, so it is not possible to adjust the filtering of console messages.

## 9.2.4. The INCLUDE List

The INCLUDE list takes the form of a CSV file where each line (row), other than the header row, provides information for one piece of PS equipment. The values in the header row are as follows:

| Column Header   | Data Value                                                                                                                                                                                                                               |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PSTYPE</b>   | The Postage Stamp equipment type. There is no default value.<br><br>For more information, see <a href="#">About the PSTYPE Column</a> .                                                                                                  |
| <b>Id</b>       | The DNOM <Id> of one particular piece of PS equipment, or an asterisk (*) to indicate "all instances of this PSTYPE". There is no default value.<br><br>For more information, see <a href="#">About the ID Column</a> .                  |
| <b>ViewName</b> | The <EditorView><Name> of the view, or null.<br><br>The default value is described in <a href="#">About the ViewName Column</a> .                                                                                                        |
| <b>SUBSTN</b>   | The value to use for the %SUBSTN% proxy, which is often used as the ID_SUBSTN in the SCADA Composite ID for all devices and measurements in the view.<br><br>The default value is described in <a href="#">About the SUBSTN Column</a> . |
| <b>DEVTYP</b>   | The value to use for the %DEVTYP% proxy, which is often used as the ID_DEVTYP in the SCADA Composite ID for all devices and measurements in the view.<br><br>The default value is described in <a href="#">About the DEVTYP Column</a> . |

| Column Header         | Data Value                                                                                                                                                                                                              |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DEVICE</b>         | The value to use for the %DEVICE% proxy, which is often used as the ID_DEVICE in the SCADA Composite ID for all devices and measurements in the view.<br><br>The default value is described in About the DEVICE Column. |
| <b>Picture</b>        | The name of the ESET picture to be used for drawing the view for this particular piece of PS equipment, or null.<br><br>The default value is described in About the Picture Column.                                     |
| <b>DELETE_VIEW</b>    | True if an existing view and its DNOM objects should be deleted.<br><br>For more information, see About the DELETE_VIEW Column.                                                                                         |
| <b>CREATE_VIEW</b>    | True if a non-existent (or deleted) view and its DNOM objects should be created.<br><br>For more information, see About the CREATE_VIEW Column.                                                                         |
| <b>%user_defined%</b> | Please see the About User-Defined Proxies paragraph below.                                                                                                                                                              |

The first line of the INCLUDE file must be a header row. Only the PSTYPE and the ID are required, strictly speaking, but if both DELETE\_VIEW and CREATE\_VIEW are missing, BPS will have no work to do.

The columns may be in any order, and other user-defined columns may be interspersed. However, BPS only uses columns whose headers exactly match the "Column Header" given above.

## About the PSTYPE Column

| PSTYPE                                                    | Corresponding Create Postage Stamp Display Tab |
|-----------------------------------------------------------|------------------------------------------------|
| Capacitor<br>Cap*                                         | Capacitors                                     |
| MonitoringPoint<br>Mon*                                   | Monitoring Points                              |
| Regulator<br>Reg*                                         | Regulators                                     |
| Switch<br>SWT<br>Sw*                                      | Switches                                       |
| UIContainer<br>TopologyDeviceGroup<br>UIC<br>UIC*<br>Top* | Topology Device Groups                         |

As you can see by the variations listed in the PSTYPE column, the BatchPostageStamps tool is lenient in its interpretation of the input value.

- It recognizes and accepts anything beginning with the letters given in the asterisk-wildcard strings shown above.
- It ignores leading whitespace.
- Its comparisons are case-insensitive.

## About the ID Column

Either this column value must be an asterisk, or it must match the <Id> of an instance of the PSTYPE. The comparison is case-insensitive.

## About the ViewName Column

When the ID column value is not an asterisk, and this column value is not null, this is the value compared against <EditorView><Name> when the BatchPostageStamps tool is searching for existing views that need to be deleted. The comparison against <EditorView><Name> is case-insensitive.

This is also the value (converted to uppercase) given to <EditorView><Name> when the view is created.

When Id=\*, or when this column value is null, BatchPostageStamps uses the Substation Editor algorithm to determine the <EditorView><Name>. Currently, the algorithm is:

- If the PS Equipment's <Name> is provided, it is used.
- Otherwise, the PS Equipment's <Id> is used.

## About the SUBSTN Column

This column specifies the value given to the %SUBSTN% proxy, which is often used in ESET pictures to define the SCADA Composite IDs of elements. The value may be:

- **EMSName**

When the tool sees this literal in the SUBSTN column, it sets the %SUBSTN% proxy to the value of the current station's <Substation><pSubName><EMSName> property.

- **DeviceName**

When the tool sees this literal in the SUBSTN column, it sets the %SUBSTN% proxy to the value of the Name property of the current PS equipment.

This is the equivalent of clicking the "Set %SUBSTN% to Device Name" option on the Substation Editor's Create Postage Stamp Displays window.

- **DeviceId**

When the tool sees this literal in the SUBSTN column, it sets the %SUBSTN% proxy to the value of the Id property of the current PS equipment.

- **DoNotSet**

When the tool sees this literal in the SUBSTN column, it does not set a value for the %SUBSTN% proxy for the current view.

- **Another string**

This is almost the equivalent of clicking the "Set %SUBSTN% to:" option on the Substation Editor's Create Postage Stamp Displays window and typing a string in the text box. Currently,

for the SSE text box:

- The maximum length is eight characters
- Leading and trailing spaces are not trimmed
- Case is preserved
- Punctuation characters are accepted

In contrast, the BatchPostageStamps tool requires that the string comply with the rules for a valid SCADA ID as described in the Source *EMS Modeling Guide*, in the "Comm Modeling" chapter, under the "Building ISD Names" section. Although BatchPostageStamps does not check, this string must match an ID\_SUBSTN in the SCADA database.

- **Null**

The default value is DeviceName.

### **Note**

The %/device/number/STATION% proxy value (see [Postage Stamp Proxies](#)) is still set to the name of the station that owns the equipment, regardless of what value is given to the proxy %SUBSTN%.

## About the DEVTYP Column

This column specifies the value given to the %DEVTYPE% proxy, which is often used in ESET pictures to define the SCADA Composite IDs of elements. The value may be:

- **DoNotSet**

When the tool sees this literal in the DEVICE column, it does not set a value for the %DEVTYPE% proxy for the current view.

- **Another string**

The given string must comply with the rules for a valid SCADA ID as described in the **Source** *EMS Modeling Guide*, in the "Comm Modeling" chapter, under the "Building ISD Names" section. Although BatchPostageStamps does not check, this string must match an ID\_DEVTYP in the SCADA database.

- **Null**

The default value is DoNotSet.

The value from this column overrides any value specified by other Substation Editor mechanisms, such as the picture's Define Link tab (for information about the Define Link tab, see [Modeling Equipment Groups](#)).

## About the DEVICE Column

This column specifies the value given to the %DEVICE% proxy, which is often used in ESET pictures to define the SCADA Composite IDs of elements. The value may be:

- **DeviceName**

When the tool sees this literal in the DEVICE column, it sets the %DEVICE% proxy to the value of the Name property of the current PS equipment.

- **DeviceId**

When the tool sees this literal in the SUBSTN column, it sets the %SUBSTN% proxy to the value of the Id property of the current PS equipment.

- **DoNotSet**

When the tool sees this literal in the DEVICE column, it does not set a value for the %DEVICE% proxy for the current view.

- **Another string**

The given string must comply with the rules for a valid SCADA ID as described in the **Source EMS Modeling Guide**, in the "Comm Modeling" chapter, under the "Building ISD Names" section. Although BatchPostageStamps does not check, this string must match an ID\_DEVICE in the SCADA database.

- **Null**

The default value is DeviceName.

The value from this column overrides any value specified by other Substation Editor mechanisms, such as the picture's Define Link tab (for information about the Define Link tab, see [Modeling Equipment Groups](#)).

## About the Picture Column

This column specifies the name of a picture element in the ESET file. It may be:

- **A string**

The given string must exactly match the name of an appropriate picture element in the ESET file.

- **Null**

When the picture value is null, the tool constructs a picture name as described in section 2.4.1 Circuit Pattern.

The value in this column must match the name of the picture in the ESET in all respects, including case, or the SSE Tool will not be able to find it.

## About the DELETE\_VIEW Column

To set the DELETE\_VIEW value to true, this column must contain a string beginning with the letter "T", in either upper or lower case. Anything else - including null - sets the DELETE\_VIEW value to false.

Setting DELETE\_VIEW=true can facilitate moving a boundary piece of PS equipment from one



station to another, as long as the ViewName to be deleted is the same as the ViewName to be created.

## About the CREATE\_VIEW Column

To set the CREATE\_VIEW value true, this column must contain a string beginning with the letter "T", in either upper or lower case. Anything else - including null - sets the CREATE\_VIEW value to false.

### Note

A number of other conditions must be met before a view is successfully created. However, if CREATE\_VIEW is not true, its creation is not attempted.

## About User-Defined Proxies

BatchPostageStamps supports the ability to define a new proxy ("Template Name" on the SSE Properties tab) as an INCLUDE List column, and to provide individual values for each row.

- The column header must have the form "%proxy%".
- The 'proxy' string must only have characters chosen from A through Z, 0 through 9, and underscore.
- The entire length of the column header string, including the percent signs, must be between 3 and 30, inclusive.
- Leading and trailing white space is ignored.
- The 'proxy' string may be provided in any case, but is forced to uppercase.

The column values:

- May contain any inkable UTF-8 culture invariant characters, including spaces and horizontal tabs
- May be zero-length (which is different from DoNotSet; see below)
- Must not exceed 255 characters
- Are trimmed of leading and trailing white space and quotes (both single and double)
- Remain in the case found in the INCLUDE list
- May specify DoNotSet (case insensitive), in which case the proxy will not be defined for that row's View.

Note that "DoNotSet" is the flag: "Do Not Set" (with embedded spaces) is interpreted as a literal value.

The modeler may define as many new proxies as she wishes from within the INCLUDE list.

## 9.3. BulkEdit

The BulkEdit tools are used to extract electrical properties of targeted equipment from SUBEDITOR

files, then to merge modified properties back into SUBEDITOR files. The prototypical use case is:

1. Run BulkExtract to create CSV files of properties extracted from all the stations in the model, producing one CSV file per equipment type.
2. Edit the CSV files using Microsoft Excel or any other tool of the user's choice.
3. Run BulkMerge to overwrite the properties in the SUBEDITOR files with the modified values from the CSV files.

#### Note

The values are transferred from-and-to the SUBEDITOR files, without modification from any Associate Models (as found in GLOBALS.CSV or GLOBAL\_INTERNALS.CSV).

### 9.3.1. BulkExtract.exe

BulkExtract is invoked from the command line with or without the following arguments. If an argument is not given on the command line, you are prompted to provide it. For example:

```
>BulkExtract.exe -I %bps_inp% -o %bps_out% -config %bps_cfg%  
...  
...-E-Filename for extract request file not given  
Please tell me where to find the extract request file: %BULK_QRY%\BulkEdit_MixedQueries_001-002a.txt
```

The arguments are:

- **INPUTDIR:** The directory where the SUBEDITOR files are to be processed (for example, \SubstationEditor\data). Each of the SUBEDITOR files is examined for instances of the equipment targeted by a REQUEST.
- **CONFIG:** Specifies the SubstationEditorConfiguration.xml file.

The tool uses the Element Set and Menu Element Set specified by the CONFIG file exactly as Substation Editor does.

This tool also uses the <LogLevel> parameter specified by the CONFIG file.

- **OUT:** The directory where the CSV files are created.
- **REQUEST:** The specification of a plain-text file in which the SQL-like extraction queries are found. A quick-guide to the syntax of a request is found in [The REQUEST Syntax \(aka Query Syntax\)](#).

In addition to the CSV files in the OUT directory, the tool outputs status messages to the console, and preserves audit trail and diagnostic messages in the trace log file as described above.

### 9.3.2. BulkMerge.exe

BulkMerge is run from the command line with the following arguments:

- **INPUTDIR:** The directory where the SUBEDITOR files are to be processed (for example, \SubstationEditor\data). Each of the SUBEDITOR files are examined for equipment identified in the CSV files.
- **CONFIG:** Specifies the SubstationEditorConfiguration.xml file.

The tool uses the Element Set and Menu Element Set specified by the CONFIG file exactly as Substation Editor does.

This tool also uses the logging parameters specified by the CONFIG file.

- **CSV:** Where to find the CSV files that will be used to modify the SUBEDITOR file properties.

### 9.3.3. The REQUEST Syntax (aka Query Syntax)

The complete specification of the request syntax is in the *SSE Bulk Edit Query Syntax* document, which is available from GE upon request.

Multiple requests can be provided in a single -REQUEST file. Each request (aka query) must begin on a new line, and must be terminated by a semicolon.

#### Query Format

The format of a Bulk Edit Query comprises four clauses:

```
SELECT <select-list>INTO <user phrase>
FROM <equipment>
[WHERE <selection expression>];
```

Where:

- A <select-list> is a comma-separated list of DNOM property identifiers.
- A <user phrase> is a string that enables the user to distinguish the output files and is also used by Bulk Edit to identify the extraction request being processed.
- <equipment> is the DNOM identifier of a single equipment type.
- A <selection expression> specifies the criteria by which instances of the <equipment> are selected.

Notes:

- Every clause, except the WHERE clause, is required.
- The order of the clauses must be SELECT, INTO, FROM, then WHERE (if provided).
- A semicolon (;) must be present at the end of the Query.

#### <select-list>

The <select-list> of properties specifies both which properties are output, and the order in which they are output. It is a list of qualified property identifiers.

Except for the *Station* property identifier, the property identifiers are specified as the names of the corresponding elements in DNOMConverter.xsd.

A qualified property identifier is one in which the identifiers of referenced objects other than the `<equipment>` are explicit. The explicitly referenced object must be a child of the selected `<equipment>`. For example,

```
SELECT Station,Id,CID,pMeas.MeasType,pMeas.MeasGroup
INTO Orphans FROM Analog WHERE (Device = null)
```

The order of the properties in the `<select-list>` determines the order of the columns in the output CSV file.

Not every property is supported for extraction. In particular, graphic elements and infrastructure elements are not supported. Refer to the *SSE Bulk Edit Query Syntax* document for more detail, which is available from GE upon request.

### About *Station* and *Id*

The property identifier for the SUBEDITOR *Station* (for example, *Station*), and the property identifier for the customer's equipment unique identifier (for example, *Id*) are required. If they are not in the `<select-list>`, they are output anyway because they are required if a subsequent BulkMerge is to be successful.

The value of the *Station* is taken from the `<Substation><Name>` property in the SUBEDITOR file.

The value of the *Id* is taken from the `<equipment>` object's `<Id>` property.

### `<equipment>`

As a simple rule, the supported values for `<equipment>` are the names of electrical devices such as Line, Switch, and Transformer. The exact rules for which elements listed in DnomConverter.xsd are supported by BulkExtract can be found in the *SSE Bulk Edit Query Syntax* document, which is available from GE upon request.

### `<user-phrase>`

This may be any string composed of any of the following characters:

- Upper- or lower-case A through Z
- Digits 0 through 9
- Underscore (" \_ ")

BulkExtract does not explicitly limit the length of the string. However, it becomes part of the output filename, so if you provide a string that forces the filename to violate a file-system limitation, the results are unpredictable. See more about the output filename below.

### `<selection-expression>`

The `<selection-expression>` is expected to be a fairly simple, fully-parenthesized conjunction of

comparisons.

The following are examples of valid WHERE clauses:

```
WHERE ((Phase = "ABCN") and (Length > 250))
where (ConstructionModel = Null)
where (NormallyLooped = true)
Where ((Repair = 2) or ((Repair = 3) and (NumCircuits > 2)))
where (not ((Repair = 3) and (NumCircuits > 2)))
WHERE ((Phase <> "A") and (phase <> "B") and (Phase <> 'c'))
```

For the examples of invalid WHERE clauses, and for a full specification of the WHERE clause syntax, refer to the *SSE Bulk Edit Query Syntax* document, which is available from GE upon request.

### 9.3.4. About the Output Filename

The name of the output file is

<user-phrase>\_<equipment>\_<timestamp>.csv

Where:

- <userphrase> is as specified above.
- <equipment> is as specified above.
- <timestamp> is the year, month, day, hour, and minute the query was processed, in the format "yyyymmddhhnn" in which:
  - "yyyy" is the four-digit year
  - "mm" is the two-digit (leading zero) month
  - "dd" is the two-digit (leading zero) day of month
  - "hh" is the two-digit (leading zero) hour of the 24-hour clock
  - "nn" is the two-digit (leading zero) minute

For example:

- SWupdate\_SWITCH\_201403051123.csv
- SWupdate\_LINE\_201403051126.csv